REEPHAM HIGH SCHOOL & COLLEGE

OCR H446/01

Computer Systems
Visual revision guide

UNDERSTAND > MODEL > TRY > CHECK

Five chapters, one consistent revision journey

1.1	Processors and storage	30 pp
1.2	Software development	58 pp
1.3	Exchanging data	82 pp
1.4	Data types, structures and logic	98 pp
1.5	Ethical and legal issues	58 pp

REVIEW EDITION / NOT THE PUBLISHED GUIDE

Includes a legacy supplementary recall appendix. Subject accuracy and retained artwork require final teacher sign-off.

Choose your chapter

Start with the linked unit below. Each chapter has its own clickable specification checklist, followed by four-step learning cycles.

1.1 Processors, input/output and storage

Pages 4-33 | Chapter checklist: page 5

1.2 Software and software development

Pages 34-91 | Chapter checklist: page 35

1.3 Exchanging data

Pages 92-173 | Chapter checklist: page 93

1.4 Data types, structures and Boolean algebra

Pages 174-271 | Chapter checklist: page 175

1.5 Legal, moral, cultural and ethical issues

Pages 272-329 | Chapter checklist: page 273

Additional recall: original memory banks

Pages 330-337 | Archived material included for comparison

Navigation tip

Use the PDF contents panel (bookmarks) to jump directly to any of the 79 learning cycles, or click a chapter row above.

How to use each four-page cycle

01 **UNDERSTAND**

Read the explanation; follow the diagram and highlighted vocabulary.

02 **MODEL**

Trace the fully worked example and explain why each step happens.

03 **TRY**

Complete the independent exam-style questions before checking.

04 **CHECK**

Use the next page to self-mark in green, correct errors and explain why.

Teachers: pre-release checks

Every chapter is independently drafted. The 264-page published PDF, the revision question banks and existing page-link map have not been replaced. Full content parity with the published edition is still to be checked.

1.1 / WORKING CHAPTER / TEACHER REVIEW

Understand it. Model it. Try it. Check it.

An expanded, student-facing draft for OCR H446/01 specification 1.1. It supplements the existing RHSC guide; it has not been integrated or published.

VISUAL ROUTE THROUGH SECTION 1.1

1.1.1
Processor function, throughput and architectures

1.1.2
RISC/CISC, GPUs and parallel systems

1.1.3
Choosing devices, memory and virtual storage

ONE CONSISTENT REVISION ROUTINE

01 UNDERSTAND
A visual model connected to precise A Level terms.

02 MODEL
A complete worked example with every step visible.

03 TRY
Original exam-style questions and writing room.

04 CHECK
Indicative marking points and green-pen corrections.

WHAT IS PRESERVED

The published 264-page H446 Paper 1 guide, original diagrams, existing exam questions, quiz IDs and all student website links remain unchanged. This chapter is a separate candidate addition.

Release condition: check terminology, examples, layout, duplication against the current PDF, and every stable revision-quiz page key before integration.

1.1 / SPECIFICATION ALIGNMENT

Coverage before page count

The official OCR H446 specification identifies the requirements below. Seven four-page learning cycles cover all 12 sub-points in section 1.1.

SPEC	SPECIFICATION CONTENT	THIS DRAFT
1.1.1(a,b)	ALU, CU, PC/ACC/MAR/MDR/CIR, buses and the fetch-decode-execute cycle	pp. 3-6
1.1.1(c,d)	Clock speed, cores, cache and pipelining	pp. 7-10
1.1.1(e)	Von Neumann, Harvard and contemporary architecture	pp. 11-14
1.1.2(a)	RISC and CISC differences and uses	pp. 15-18
1.1.2(b,c)	GPUs, multicore and parallel systems	pp. 19-22
1.1.3(a,b)	Selecting I/O, magnetic, flash and optical devices	pp. 23-26
1.1.3(c,d)	RAM, ROM and virtual storage	pp. 27-30

REVIEW PRIORITY

Check every model answer against the OCR wording and the existing RHSC lessons. The question/mark-point sets here are original revision questions, not official past-paper questions.

Source: OCR H446 A Level Computer Science specification, section 2c; existing RHSC H446 Paper 1 guide and the approved 18-page visual revision pilot. This draft uses the same four-stage pattern.

1.1.1(a,b) / SEE IT / REGISTER ROLES

Follow the instruction, not just the arrow

A register stores a particular kind of information. Track the location, transferred value, instruction and calculated result separately.

FOUR VALUES ARE NOT INTERCHANGEABLE

PC / NEXT

Holds address of next instruction in a normal sequential trace.

MAR / ADDRESS

Holds address currently being accessed in main memory.

MDR / TRANSFER

Holds data or instruction being transferred to/from memory.

CIR / CURRENT

Holds instruction being decoded or executed.

TRACE ONE INSTRUCTION: RAM[40] = ADD 85

PC	MAR	MDR	CIR
40	40	ADD 85	ADD 85

FROM DIAGRAM TO EXAM LANGUAGE

Fetch: the PC address is copied to MAR; memory is read via MDR into CIR; PC normally advances. Decode: the CU interprets the instruction. Execute: the ALU or another unit performs the operation. ACC can retain arithmetic results.

WORDS THAT EARN MARKS

- **PC** Next instruction
- **MAR** Accessed address
- **MDR** Transferred value
- **CIR** Current instruction

1.1.1(a,b) / WORKED EXAMPLE / TEACHING CPU

ADD 85: a full fetch-and-execute trace

At the start, PC = 40, ACC = 9, RAM[40] = ADD 85, RAM[85] = 6. Here the operand 85 is a memory address.

PHASE	WHAT CHANGES	WHY IT CHANGES
Fetch 1	MAR ← PC = 40	Select RAM address of instruction.
Fetch 2	MDR ← RAM[40] = ADD 85	Transfer instruction on data bus.
Fetch 3	CIR ← MDR; PC ← 41	Store current instruction; move on.
Decode	CU identifies ADD and operand 85	Determine operation and operand address.
Execute 1	MAR ← 85; MDR ← RAM[85] = 6	Read the operand, not address 85 as a value.
Execute 2	ACC ← 9 + 6 = 15	ALU performs arithmetic; ACC stores result.

END STATE

PC = 41; CIR = ADD 85; ACC = 15; MAR = 85; MDR = 6 (in this simplified teaching trace).

Where the model stops: a real CPU can have additional registers, different buses and instruction formats. A jump/branch can change the PC rather than allowing sequential advancement.

1.1.1(a,b) / INDEPENDENT EXAM PRACTICE

Trace, label and explain

Assume a simplified accumulator CPU unless stated otherwise.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 3 MARKS

State the different roles of PC, MAR and MDR in fetching an instruction.

QUESTION 2 | 5 MARKS

PC = 72, ACC = 4, RAM[72] = ADD 90 and RAM[90] = 8. State the final PC, CIR and ACC, showing the operand read.

QUESTION 3 | 6 MARKS

Explain how the address, data and control buses contribute to fetching an instruction and then obtaining an operand.

1.1.1(a,b) / SELF-MARKING / INDICATIVE ANSWERS

Check each register transfer

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 3 MARKS

- PC holds the address of the next instruction [1].
- MAR holds the memory location being accessed [1].
- MDR holds data or instruction transferred to/from memory [1].

Q2 / 5 MARKS

- MAR receives 72 [1]; RAM[72] supplies ADD 90 through MDR [1].
- CIR becomes ADD 90 [1]; PC normally advances to 73 [1].
- CPU reads RAM[90] = 8 and ACC becomes 4 + 8 = 12 [1].

Q3 / 6 MARKS

- Address bus selects a memory location [1]; control bus signals READ [1].
- Data bus carries fetched instruction towards CPU [1]; MDR/CIR retain it [1].
- An operand memory address is subsequently placed on address bus [1]; data bus returns operand for ALU operation [1].

COMMON MISCONCEPTION

ADD 90 means add the value stored at memory address 90. It does not necessarily mean add the literal number 90.

Green pen: Choose the least precise of your bus or register explanations and rewrite it as a full cause-and-effect sentence.

1.1.1(c,d) / SEE IT / BOTTLENECKS

A faster clock is not the whole story

Speed depends on the workload and implementation. Clock frequency, cache access, core count and pipeline efficiency influence different bottlenecks.

THREE PROCESSOR LEVERS: THREE DIFFERENT QUESTIONS

CLOCK SPEED

How many clock cycles occur each second? Work per cycle still matters.

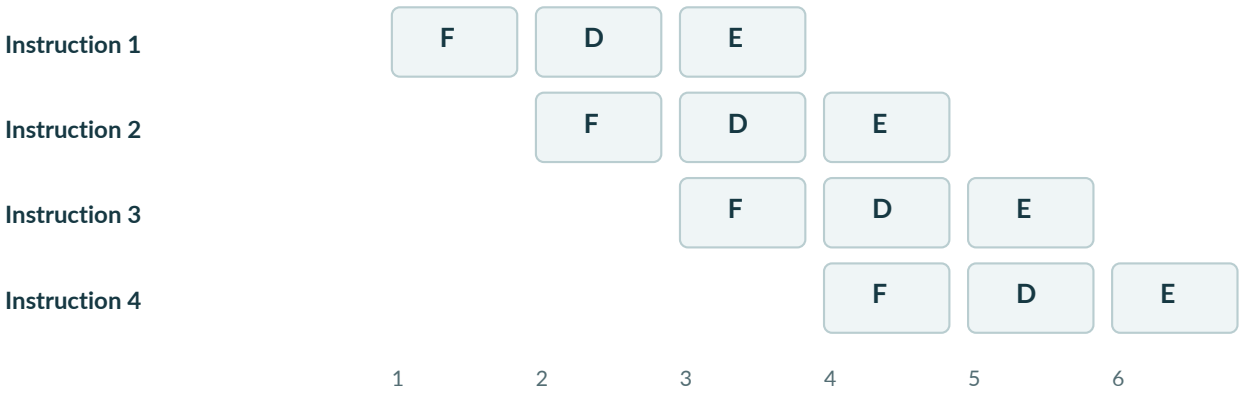
CACHE

Can frequently used instructions/data avoid slower main-memory access?

NUMBER OF CORES

Can multiple independent threads/tasks actually run in parallel?

PIPELINE: OVERLAP INSTRUCTION STAGES



One simplified cycle per stage: after the pipeline fills, a result may finish each cycle, provided there are no stalls. The latency of a single instruction need not become shorter.

WHERE THE ANALOGY STOPS

Data dependencies, branches and shared resources can stall or flush work. Real pipelines have different stage timings and overhead; ideal timing is a model, not a guarantee.

WORDS THAT EARN MARKS

- **Clock** Cycles/sec
- **Cache** Fast nearby data
- **Pipeline** Overlap stages
- **Stall** Delay a stage

1.1.1(c,d) / WORKED EXAMPLE / THREE STAGES

Compare serial and pipelined execution

Four independent instructions use fetch (2 ns), decode (2 ns) and execute (2 ns). Assume an ideal pipeline, no overhead or hazards.

WITHOUT OVERLAP / 4 INSTRUCTIONS

INSTR. 1	INSTR. 2	INSTR. 3	INSTR. 4
F 2 + D 2 + E 2	F 2 + D 2 + E 2	F 2 + D 2 + E 2	F 2 + D 2 + E 2

4 × (2 + 2 + 2) = **24 ns** total

WITH OVERLAP / COMPLETE A NEW INSTRUCTION EVERY 2 NS (IDEAL)

INTERVAL	STAGE F	STAGE D	STAGE E
0-2 ns	Instruction 1	-	-
2-4 ns	Instruction 2	Instruction 1	-
4-6 ns	Instruction 3	Instruction 2	Instruction 1
6-8 ns	Instruction 4	Instruction 3	Instruction 2
8-10 ns	-	Instruction 4	Instruction 3
10-12 ns	-	-	Instruction 4

EXPLAIN THE RESULT

The first instruction still takes 6 ns; total time is 6 + 3 × 2 = **12 ns**. Under these ideal assumptions the four-instruction batch completes in half the non-pipelined time.

Important: this example teaches increased **throughput**, not a shorter single-instruction **latency**. Actual speed-up can be lower if dependencies or branches introduce stalls.

1.1.1(c,d) / INDEPENDENT EXAM PRACTICE

Prove, calculate, evaluate

Show the units and state the ideal assumptions in numerical answers.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 3 MARKS

Explain why a CPU with a higher clock frequency might not always complete a program sooner.

QUESTION 2 | 5 MARKS

Five independent instructions each require F, D and E stages of 2 ns. Calculate the non-pipelined and ideal pipelined completion times, then the speed-up.

QUESTION 3 | 6 MARKS

Evaluate how cache, more cores and pipelining could affect a CPU running a program with some branches and limited parallel work.

1.1.1(c,d) / SELF-MARKING / INDICATIVE ANSWERS

What improves speed, and when?

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 3 MARKS

- Clock speed is cycles per second [1].
- Different CPUs can do different work per cycle [1].
- Cache misses, instruction mix or stalls may outweigh frequency advantage [1].

Q2 / 5 MARKS

- One instruction uses 3 × 2 = 6 ns [1].
- Non-pipelined five-instruction time = 5 × 6 = 30 ns [1].
- Ideal pipelined time = 6 + 4 × 2 = 14 ns [1].
- Speed-up = 30 / 14 ≈ 2.14× [1]; no stalls/overhead assumed [1].

Q3 / 6 MARKS

- Cache can reduce average main-memory access time on hits [1]; effectiveness depends on locality/hit rate [1].
- Multiple cores can run independent threads concurrently [1], but sequential code cannot benefit in the same way [1].
- Pipelining overlaps stages to raise throughput [1]; branches/data dependencies may stall or flush it [1].

COMMON MISCONCEPTION

Pipelining improves throughput in a well-filled pipeline. It does not simply divide the latency of one instruction by the stage count.

Green pen: Show your 14 ns calculation as a timeline rather than only writing a speed-up number.

1.1.1(e) / SEE IT / ARCHITECTURES

Two routes, one critical difference

Von Neumann and Harvard describe memory organisation. Contemporary processors can combine ideas at different levels of the memory hierarchy.

MEMORY ACCESS PATHS

VON NEUMANN

One shared memory / pathway
Instructions and data share an address space and access path in the simplified model.

Potential limit: instruction fetch and data access can contend.

HARVARD

Separated instruction and data stores/paths
Fetching instructions and accessing data can use separate routes.

Potential benefit: concurrent access to the two.

A CONTEMPORARY DESIGN CAN BE MIXED

INSTRUCTION CACHE

Dedicated nearby instruction path.

DATA CACHE

Dedicated nearby data path.

SHARED LOWER-LEVEL MEMORY

Eventually accesses shared wider memory.

Separate L1 instruction/data caches can resemble Harvard close to the CPU while lower-level memory still uses a unified address space. Architecture labels describe organisation, **not a guaranteed performance ranking**.

EXAM LANGUAGE

Shared instruction/data route can create a Von Neumann bottleneck. Separate routes can allow simultaneous accesses. Actual performance also depends on cache, bandwidth, hazards and implementation.

1.1.1(e) / WORKED EXAMPLE / ACCESS DEMAND

Predict a memory-access bottleneck

A simplified CPU needs an instruction fetch and one data read during the same clock interval.

SCENARIO A / SINGLE SHARED MEMORY ACCESS PORT

CLOCK SLOT	REQUEST	OUTCOME
Slot 1	Fetch next instruction	One use of the shared port.
Slot 2	Read operand	Must use the port later if both conflict.

SCENARIO B / SEPARATE PATHS AVAILABLE

CLOCK SLOT	INSTRUCTION PATH	DATA PATH
Slot 1	Fetch next instruction	Read operand at the same time

SO WHAT?

For this simplified pair of memory operations, separate paths avoid the shared-port conflict. This alone does **not** prove the whole program finishes twice as quickly.

TURN OBSERVATION INTO A COMPARISON

Weak: “Harvard is faster.”
Stronger: “Separate instruction and data pathways may allow concurrent fetch and data access, reducing contention where both are needed; other bottlenecks can remain.”

1.1.1(e) / INDEPENDENT EXAM PRACTICE

Compare architectures precisely

Avoid unsupported claims about guaranteed speed.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 3 MARKS

State three structural differences or similarities between simplified Von Neumann and Harvard architectures.

QUESTION 2 | 5 MARKS

A processor often reads data while fetching upcoming instructions. Explain why a shared memory pathway may reduce throughput and how separate pathways help.

QUESTION 3 | 6 MARKS

A modern CPU has separate instruction and data L1 caches but uses shared main memory. Explain why this is a hybrid arrangement and evaluate the claim that it eliminates memory bottlenecks.

1.1.1(e) / SELF-MARKING / INDICATIVE ANSWERS

Use architecture vocabulary

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 3 MARKS

- Von Neumann model uses a shared instruction/data memory space [1].
- The access path may be shared between fetching and data transfers [1].
- Harvard uses distinct instruction and data storage/access paths [1].

Q2 / 5 MARKS

- Both instruction fetch and data read demand memory access [1].
- Shared port/path cannot serve conflicting demands simultaneously [1].
- One request may wait, reducing available throughput [1].
- Separate instruction/data paths can permit the two accesses in parallel [1].
- Benefit is conditional on implementation and workload [1].

Q3 / 6 MARKS

- Separate L1 caches provide distinct nearby instruction/data paths [1].
- Main memory still uses a shared system/address space [1].
- Hence it combines Harvard-like cache separation with a unified wider memory model [1].
- A cache miss can still access slower shared memory [1].
- Bandwidth/latency and contention remain relevant [1].
- Therefore it can reduce some conflicts but cannot eliminate all bottlenecks [1].

COMMON MISCONCEPTION

Separate caches alone do not mean that the whole machine has fully separate instruction and data address spaces.

Green pen: Rewrite any unqualified “always faster” statement as a conditional explanation.

1.1.2(a) / SEE IT / TWO INSTRUCTION-SET IDEAS

Instruction count is not execution time

RISC and CISC are instruction-set design approaches. Both can be implemented in sophisticated modern processors.

ONE CALCULATION, TWO ILLUSTRATIVE ENCODINGS

RISC-LIKE SEQUENCE

```
LOAD R1, [A]
LOAD R2, [B]
ADD R3, R1, R2
STORE [C], R3
```

The operations are simpler and load/store-style. This is teaching pseudocode, not a particular ISA.

CISC-LIKE EXAMPLE

```
ADD_MEMORY C, A, B
```

One **hypothetical** compound instruction expresses more work; decoding and execution may still involve several internal steps.

COMPARE CHARACTERISTICS WITHOUT CLAIMING ABSOLUTES

RISC / TYPICAL	CISC / TYPICAL
More regular, simpler instructions	Richer, sometimes variable-length instructions
Load/store style is common	More addressing modes may be available
Decoding and pipelining can be simpler	Complex instructions may improve code density
May need more instructions for same job	Fewer instructions does not imply fewer cycles

WHY THIS MATTERS

Performance depends on implementation, clock rate, caches, compiler and workload. Instruction-set category alone is not a reliable speed comparison.

1.1.2(a) / WORKED EXAMPLE / NO MADE-UP TIMINGS

Compare the cost of two sequences

Assume A holds 3 and B holds 5. Both illustrative sequences compute C = A + B.

STEP	LOAD / STORE APPROACH	COMPOUND APPROACH
1	LOAD R1, [A] -> R1 = 3	ADD_MEMORY C, A, B
2	LOAD R2, [B] -> R2 = 5	Meaning: read A and B; compute 8; write C
3	ADD R3, R1, R2 -> R3 = 8	Implementation and timing unspecified
4	STORE [C], R3 -> C = 8	Possible internal micro-operations

MODEL ANSWER: WHAT CAN WE CONCLUDE?

Both approaches produce C = 8. The compound form uses fewer **visible instructions**. We cannot claim it runs faster: each instruction may take different cycles and incur different memory/decode work.

HOW TO STRUCTURE A SIX-MARK COMPARISON

CHARACTERISE

Discuss typical instruction complexity and code density.

EXPLAIN TRADE-OFF

Link decoding / pipelines / memory accesses to possible benefits or costs.

EVALUATE

Conclude performance depends on actual hardware and task.

1.1.2(a) / INDEPENDENT EXAM PRACTICE

Compare instead of repeating slogans

Use typical characteristics and explain why exceptions exist.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 3 MARKS

State one characteristic of RISC, one of CISC, and the meaning of code density.

QUESTION 2 | 5 MARKS

Compare LOAD/LOAD/ADD/STORE with one hypothetical complex instruction to calculate C = A + B. Explain why fewer instructions need not mean faster execution.

QUESTION 3 | 6 MARKS

Evaluate “RISC processors are always faster than CISC processors” with reference to decoding, pipelining, instruction count and memory accesses.

1.1.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Distinguish ISA from implementation

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 3 MARKS

- RISC typically uses simpler/more regular instructions [1].
- CISC can provide richer/compound operations [1].
- Code density is functionality encoded per unit of machine-code size [1].

Q2 / 5 MARKS

- LOAD reads A [1] and another LOAD reads B [1].
- ADD combines register values [1]; STORE writes result C [1].
- Compound instructions can have more internal operations/cycles; counting instructions alone does not establish execution time [1].

Q3 / 6 MARKS

- RISC regularity can simplify decoding [1] and help efficient pipelining [1].
- But a task may require more RISC instructions, creating fetch/code-size costs [1].
- CISC can express some work with fewer instructions / greater code density [1].
- CISC decoding may be complex, though implementation can mitigate it [1].
- Caches, compiler, microarchitecture and workload determine actual speed; the absolute claim is false [1].

COMMON MISCONCEPTION

RISC does not mean that every instruction completes in one cycle; CISC does not mean a complex instruction is automatically slower.

Green pen: Turn a simple “RISC is fast” line into a comparison of at least two technical factors.

1.1.2(b,c) / SEE IT / DATA PARALLELISM

Match the hardware to the work

A CPU can combine relatively few powerful cores with flexible control flow. GPUs offer many execution units suited to repeated work on large data sets.

DIFFERENT SHAPE OF TASKS

MULTICORE CPU

Core 1 Core 2 Core 3 Core 4

Different threads can make progress in parallel, but shared data may need synchronisation.

GPU WORKLOAD

Many image pixels can receive similar calculations concurrently. Branch-heavy, tiny sequential tasks may use GPU resources poorly.

A VISUAL OF 16 INDEPENDENT ITEMS

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16

For a data-parallel operation, items 1–16 can be divided among execution units. The **serial part** (loading, control, final aggregation) still costs time, and communication may add overhead.

WHERE IT IS USEFUL

GPU applications extend beyond drawing graphics: machine-learning tensor operations, scientific simulations and parallel image processing can use many similar independent calculations.

WORDS THAT EARN MARKS

- **Core** CPU executor
- **Thread** Execution flow
- **GPU** Data parallel
- **Sync** Coordinate tasks

1.1.2(b,c) / WORKED EXAMPLE / SERIAL FRACTION

Why four cores do not give 4× speed

A job takes 16 time units on one core: 4 units are inherently sequential and 12 can be divided perfectly over four cores.

ONE CORE / BASELINE

Serial: 4

Parallel: 12

FOUR CORES / BEST-CASE CALCULATION

First: serial stage = 4 units

THEN / FOUR CORES WORK SIMULTANEOUSLY

Core 1: 3 units

Core 2: 3 units

Core 3: 3 units

Core 4: 3 units

WORKED CALCULATION

Sequential 4 + parallel 12 ÷ 4 = **7 units**. Ideal speed-up = 16 ÷ 7 ≈ **2.29×**, not 4×. Synchronisation, memory bandwidth or imbalance could make real results worse.

Transfer this reasoning: more cores or a GPU only accelerate work they can execute concurrently. Always identify serial work before predicting speed-up.

1.1.2(b,c) / INDEPENDENT EXAM PRACTICE

Apply parallel thinking

Show how a task is divided and any unavoidable sequential stage.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 3 MARKS

Give one graphics and one non-graphics GPU application and explain one feature that makes both suitable.

QUESTION 2 | 5 MARKS

A job takes 16 units on one core: 4 are serial and 12 can be split over four cores. Calculate the ideal four-core completion time and speed-up, and identify one reason reality may be slower.

QUESTION 3 | 6 MARKS

Evaluate whether a GPU or a multicore CPU is the better choice for a workload that transforms millions of pixels but also performs some sequential file handling.

1.1.2(b,c) / SELF-MARKING / INDICATIVE ANSWERS

Find the serial limit

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 3 MARKS

- Graphics example: applying an image filter/rendering [1].
- Non-graphics example: matrix/tensor computations in machine learning [1].
- Many data elements use similar independent operations [1].

Q2 / 5 MARKS

- Serial portion remains 4 units [1].
- Parallel portion per core is 12/4 = 3 units [1].
- Total ideal time is 4 + 3 = 7 units [1].
- Speed-up is 16/7 ≈ 2.29× [1].
- Synchronization/data-transfer/memory contention/imbalance can reduce gain [1].

Q3 / 6 MARKS

- Many pixel transformations can be performed independently [1]; GPU excels at this data-parallel section [1].
- Sequential file handling may suit a CPU core [1]; it does not scale across GPU units in the same way [1].
- Transfers to/from GPU or memory bandwidth may cost time [1].
- A hybrid CPU/GPU design is justified; overall speed depends on the serial portion and overhead [1].

COMMON MISCONCEPTION

The maximum speed-up is not the number of cores when serial code remains. GPU and multicore CPU performance both depend on task structure.

Green pen: Circle each claim about speed and add the relevant assumption or limit.

1.1.3(a,b) / SEE IT / PRACTICAL CONSTRAINTS

Choose by the problem, not the product

A strong H446 answer names a suitable device and links a property of that device directly to the scenario.

THE SCENARIO DECIDES THE CRITERIA

INPUT

Barcode reader: fast, consistent item capture at a stockroom checkout.

OUTPUT

E-ink screen: low power for infrequently changing text, not rapid animation.

STORAGE

SSD: fast random access and no moving read/write mechanism for a moving device.

COMPARE STORAGE MEDIA USING NEEDS

MEDIUM	ADVANTAGE	LIMIT / TYPICAL MATCH
Magnetic disk / tape	Often economical for large capacity	Disk has moving parts; tape sequential; suitable bulk archives
Flash / SSD	Fast access, resistant to movement	Cost/GB can be higher; suitable active portable work
Optical disc	Physical, inexpensive distribution of a fixed copy	Capacity/speed limits; suitable offline distribution

EXAM METHOD / CHOOSE -> PROPERTY -> BECAUSE

"Use flash storage **because** the data logger moves over rough ground and the device has no moving read/write head." This earns more than "flash is good."

1.1.3(a,b) / WORKED EXAMPLE / THREE SCENARIOS

Make a defensible recommendation

A wildlife monitoring project has moving data loggers; an office has live shared files; a school archives infrequently accessed historical footage.

NEED	CHOICE	REASON LINKED TO NEED
Moving logger	Flash storage	No moving read/write parts; more tolerant of vibration.
Active office work	SSD in server	Fast random access suits repeated small-file reads and writes.
Large cold archive	Magnetic tape	High capacity and economical archival media; sequential access acceptable.

NOW ASSESS A COMPETING OPTION

WHY NOT OPTICAL?

Optical disc media can distribute an offline fixed copy, but may be inconvenient for large, frequently updated, high-volume archives.

WHY NOT SSD FOR EVERYTHING?

SSD is responsive, but the cheaper cost per capacity of an alternative may matter when retrieval is rare and large volumes dominate.

EVALUATIVE CONCLUSION

Select by capacity, cost, speed, portability, durability, access pattern and maintenance. No medium is universally best; the priorities decide.

1.1.3(a,b) / INDEPENDENT EXAM PRACTICE

Justify each technology choice

State the device, then give a property and a consequence in the scenario.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 3 MARKS

A museum needs visitors to hear recorded descriptions at exhibits, while preventing the sound disturbing nearby areas. Recommend an output device and explain two reasons.

QUESTION 2 | 5 MARKS

A school needs fast storage for working files and low-cost storage for a rarely accessed video archive. Recommend a medium for each and explain why it fits.

QUESTION 3 | 6 MARKS

Evaluate magnetic, flash and optical storage for a portable field recorder that is frequently carried, must write data rapidly and occasionally shares completed recordings offline.

1.1.3(a,b) / SELF-MARKING / INDICATIVE ANSWERS

Select, justify, qualify

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 3 MARKS

- Headphones/earphones are suitable [1].
- Direct sound to the visitor without broadcasting to the room [1].
- Other visitors are less disturbed; suitable for simultaneous nearby exhibits [1].

Q2 / 5 MARKS

- Flash SSD is appropriate for active working files [1].
- It provides responsive random access for many reads/writes [1].
- Magnetic tape or suitably priced magnetic storage can serve a cold archive [1].
- It offers economical large capacity [1]; slower/sequential retrieval is acceptable because files are rarely requested [1].

Q3 / 6 MARKS

- Flash has no moving read/write parts and tolerates carrying/vibration [1].
- It typically provides fast writes/access for regular recording [1].
- Magnetic disk may offer economical capacity but mechanical vulnerability in a field device [1].
- Optical media can provide removable offline copies [1].
- Optical capacity/write speed/rewrite convenience may limit primary recording use [1].
- A justified combination is flash for recording plus appropriate removable/export format for distribution [1].

COMMON MISCONCEPTION

“Magnetic is slow” is too vague: hard disks and tape have different access patterns. Explain the medium and the specific workload.

Green pen: Underline each “because” in your answers. Add one wherever a choice has no scenario-specific justification.

1.1.3(c,d) / SEE IT / MEMORY VERSUS STORAGE

What disappears when power is off?

RAM, ROM and virtual storage solve different problems. In this specification point, virtual storage is not virtual memory.

THREE DIFFERENT FUNCTIONS

RAM

Volatile working memory for instructions and data in active use; normally read/write.

ROM / FIRMWARE

Non-volatile startup / embedded instructions; contents remain without power.

VIRTUAL STORAGE

Remote/network-backed files presented as a convenient logical store.

WHERE THE FILES REALLY LIVE

USER DEVICE

Opens a familiar drive/folder.

›

NETWORK

Carries requests and returned data.

›

REMOTE STORE

Physical disks/SSDs hold files.

CRITICAL DISTINCTION

Virtual storage (1.1.3d) makes remote or pooled secondary storage appear accessible as a logical drive. **Virtual memory (1.2.1b)** uses secondary storage to support a process address space when RAM is constrained.

Access and backup: remote access can help sharing and off-site recovery, but synchronised deletion can propagate. Separate retained backups and tested restoration are still needed.

WORDS THAT EARN MARKS

RAM Volatile working

ROM Non-volatile code

Virtual Remote/logical

Backup Recover earlier

1.1.3(c,d) / WORKED EXAMPLE / JUSTIFY AND WARN

Choose storage for two offices

A design team has offices in two towns. Staff need to work on shared files, and storage requirements are expected to grow.

STEP	OBSERVATION	EXPLANATION
1	Files appear on a shared drive	The interface is logical; data lives on remote servers.
2	Both offices use the same project store	Network access makes one managed set of files available.
3	The provider expands capacity	Physical disks/SSDs can be added without changing user paths.
4	A link or service fails	Uncached remote files may be temporarily unavailable.
5	A deletion synchronises	Independent backup/version retention is required to recover.

TURN TO EXAM WORDING

Virtual storage is useful here because staff in separate sites can access centrally held files and the logical capacity can expand. It depends on the network/provider, and the data must still be stored physically somewhere.

DO NOT CONFUSE THE TERMS

RAM: fast, volatile working store. **ROM:** non-volatile startup/firmware. **Remote storage:** files behind a logical drive. **Virtual memory:** address-space management using secondary storage, covered separately under 1.2.1.

1.1.3(c,d) / INDEPENDENT EXAM PRACTICE

Tell memory and storage apart

The final question requires both benefit and limitation.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 3 MARKS

Explain one role and one property of RAM, and contrast this with one role of ROM in a computer.

QUESTION 2 | 5 MARKS

Explain how a shared network drive can be virtual storage and distinguish it from virtual memory.

QUESTION 3 | 6 MARKS

A school stores staff files in a synchronised cloud drive. Evaluate two benefits and two risks, and explain why an independent backup is still needed.

1.1.3(c,d) / SELF-MARKING / INDICATIVE ANSWERS

Remote files are still physical

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 3 MARKS

- RAM holds instructions/data in active use [1].
- RAM is volatile (normally read/write) [1].
- ROM is non-volatile and can hold startup/firmware code [1].

Q2 / 5 MARKS

- Virtual storage presents a logical drive/folder to the user [1].
- Files are physically kept on network/remote/cloud storage [1].
- It requires network access for uncached files [1].
- Virtual memory uses secondary storage as part of address-space/RAM management [1].
- The two uses of “virtual” refer to different abstractions [1].

Q3 / 6 MARKS

- Remote files can be accessed from different locations [1]; capacity can expand centrally [1].
- Network/provider unavailability can block access [1]; third-party security/privacy and bandwidth are risks [1].
- Synchronization can propagate accidental deletion/corruption [1].
- Independent versioned backups and restore testing enable recovery of prior states [1].

COMMON MISCONCEPTION

Virtual storage does not eliminate physical storage. A synchronised cloud folder is not automatically a reliable, independent backup.

Green pen: Correct any use of “virtual memory” when you actually mean remote/cloud storage.

1.2 / TEACHER-REVIEW WORKING CHAPTER

Understand it. Model it. Try it. Check it.

This is a standalone candidate chapter covering H446/01 section 1.2. It is not integrated into the published 264-page guide.

FOUR SPECIFICATION FAMILIES

1.2.1 SYSTEMS SOFTWARE
OS, memory, interrupts, scheduling, firmware and VMs

1.2.2 APPLICATIONS
Utilities, translators, linking and licensing

1.2.3 DEVELOPMENT
Methodologies and algorithm tracing

1.2.4 LANGUAGES
Paradigms, LMC, addressing and OOP

FOUR PASSES FOR EVERY TOPIC

01 UNDERSTAND
Build a precise conceptual picture.

02 MODEL
Follow a fully worked trace or choice.

03 TRY
Answer original exam-style questions.

04 CHECK
Self-mark with separate indicative points.

RELEASE SAFETY

The existing student guide, original artwork, live revision-quiz page references and H446 lessons are not modified by this draft. Topic alignment and mark points require teacher review before integration.

Scope: 14 learning cycles × 4 pages plus 2 orientation pages = 58 pages. Total is not 58 pages added to the published edition; duplicated material must be reconciled first.

1.2 / SPECIFICATION CHECKLIST

Every examined 1.2 strand has a working example

Use the linked rows to jump to a topic. Content follows the accredited OCR H446 section 1.2 requirements, including all named scheduling methods and language paradigms.

SPEC	LEARNING CYCLE	PAGES
1.2.1(a)	Operating-system responsibilities	3–6
1.2.1(b)	Paging, segmentation and virtual memory	7–10
1.2.1(c)	Interrupts and ISRs	11–14
1.2.1(d)	FCFS, SJF and round robin	15–18
1.2.1(d)	SRT and feedback queues	19–22
1.2.1(e-h)	OS types, firmware, drivers and VMs	23–26
1.2.2(a-c)	Applications, utilities and licensing	27–30
1.2.2(d,e)	Translators and compilation	31–34
1.2.2(f)	Linkers, loaders and libraries	35–38
1.2.3(a,b)	Software-development methodologies	39–42
1.2.3(c)	Write and follow algorithms	43–46
1.2.4(a,b)	Paradigms and procedural code	47–50
1.2.4(c,d)	Little Man Computer and addressing	51–54
1.2.4(e)	Object-oriented programming	55–58

SCOPE BOUNDARY

1.2.4(c) LMC instructions use the standard OCR/LMC model; immediate, indirect and indexed examples are explicitly distinguished as generic pseudo-assembly.

Specification source: OCR H446 A Level Computer Science, version 2.7 (2024), section 2c, 1.2. The matching chapter in the earlier RHSC guide was reviewed for coverage; current final-edition visual comparison remains part of release QA.

1.2.1(a) / BUILD THE MENTAL MODEL

The OS is the coordinator, not the application

A running program requests resources; the operating system coordinates who gets them and when.

THREE PIECES OF THE IDEA

PROCESS CONTROL

Allocate CPU time, change the running process and handle interrupts.

MEMORY CONTROL

Allocate and protect memory so processes do not overwrite one another.

FILES + DEVICES

Manage file permissions, file systems, I/O requests and hardware access.

MODEL / FOLLOW THE FLOW

Application → **System call** → **OS / scheduler** → **Hardware**

An application requests a file read; the OS mediates the operation rather than the application controlling the disk directly.

EXAM-READY STATEMENT

The OS is system software; the browser and backup tool remain separate programs using its services.

WORDS THAT EARN MARKS

- **Kernel** Core OS code
- **Scheduler** Chooses next task
- **System call** OS service request
- **Permissions** Restrict access

1.2.1(a) / STEP-BY-STEP WORKED EXAMPLE

Browser, backup and printer compete for one computer

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Request	Browser asks to load a file	OS checks permissions and file-system metadata.
2. Schedule	A backup runs in background	Scheduler allocates time so the browser remains responsive.
3. Memory	Each program needs RAM	Memory manager allocates/protects address spaces.
4. Device	Printer needs a particular driver	OS forwards request via driver and tracks its completion.
5. Return	OS reports outcome	Application receives data or an error, not raw hardware control.

TRANSFER IT TO AN EXAM ANSWER

The OS is system software; the browser and backup tool remain separate programs using its services.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.1(a) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Questions are original teacher-written practice, not OCR past-paper questions. Work in black pen; check the next page only afterwards.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Explain two ways an operating system permits a browser and a background backup to run concurrently.

QUESTION 2 | 5 MARKS

A student application needs to print a confidential report. Explain the roles of the operating system, driver and file permissions.

1.2.1(a) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Scheduler shares CPU time between processes and permits progress [2].
- Memory management allocates/protects their memory spaces [2].

Q2 / 5 MARKS

- OS receives and manages the print request, possibly spooling it.
- Printer driver translates OS requests into hardware-specific commands.
- File/print access permissions restrict unauthorised users.
- The OS tracks I/O completion and reports failure to the application.

COMMON MISCONCEPTION

A multitasking OS does not require two CPU cores. Even one core can interleave tasks.

Green pen: Explain one OS function using a real request, action and outcome.

1.2.1(b) / BUILD THE MENTAL MODEL

A page is a fixed-size block; a segment has meaning

An address seen by a process need not equal a physical RAM location. The memory manager performs the mapping.

THREE PIECES OF THE IDEA

PAGING

Split virtual memory into equal-sized pages and physical RAM into equal-sized frames.

SEGMENTATION

Group a program into logical, variable-sized segments such as code, stack and data.

VIRTUAL MEMORY

Move pages/regions between RAM and secondary storage when needed; avoid frequent page faults.

MODEL / FOLLOW THE FLOW

Virtual page 3 → **Page table** → **Physical frame 7** → **RAM access**

Example: 1 KB pages; virtual address 3,100 uses page 3 plus offset 28. If page 3 maps to frame 7, physical address is 7,196.

EXAM-READY STATEMENT

Virtual memory can extend available address space, but access to secondary storage is much slower than RAM.

WORDS THAT EARN MARKS

Page	Fixed-size block	**Frame**	RAM block
Page fault	Page not resident	**Thrashing**	Excessive swapping

1.2.1(b) / STEP-BY-STEP WORKED EXAMPLE

Translate 3,100 with 1,024-byte pages

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Divide	3,100 ÷ 1,024 = 3 rem 28	Virtual page number = 3, offset = 28.
2. Look up	Page table: page 3 → frame 7	Page table supplies the RAM frame number.
3. Multiply	7 × 1,024 = 7,168	Frame 7 starts at byte address 7,168.
4. Add offset	7,168 + 28 = 7,196	Physical address = 7,196; offset is unchanged.
5. Alternative	Page not resident?	Page fault: OS loads it from secondary storage, then retries.

TRANSFER IT TO AN EXAM ANSWER

Virtual memory can extend available address space, but access to secondary storage is much slower than RAM.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.1(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Questions are original teacher-written practice, not OCR past-paper questions. Work in black pen; check the next page only afterwards.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Explain the difference between paging, segmentation and virtual memory. Refer to page tables or logical units.

QUESTION 2 | 5 MARKS

A system uses 512-byte pages. Virtual address 1,700 is on a page mapped to frame 9. Calculate the page, offset and physical address.

1.2.1(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Paging uses fixed-size pages and frames, linked by page tables.
- Segmentation uses variable-size logical program regions.
- Virtual memory allows non-resident data/pages to reside on secondary storage; access may trigger page faults.

Q2 / 5 MARKS

- 1,700 = 3 × 512 + 164: page 3, offset 164.
- Frame 9 base = 9 × 512 = 4,608.
- Physical address = 4,608 + 164 = 4,772.
- The offset is preserved through address translation.

COMMON MISCONCEPTION

OCR 1.2.1(b) virtual memory is NOT 1.1.3(d) virtual storage (network/cloud-backed files).

Green pen: Redo the address translation without looking, then explain the role of the page table.

1.2.1(c) / BUILD THE MENTAL MODEL

An interrupt changes control, not the entire program

Hardware or software signals an event. The processor completes an appropriate instruction boundary, saves state and services it.

THREE PIECES OF THE IDEA

INTERRUPT

A signal that requests processor attention, such as I/O completion.

CONTEXT

Preserve program counter, registers and relevant processor state.

ISR

Interrupt service routine handles the event and returns control afterwards.

MODEL / FOLLOW THE FLOW



Simplified end-of-instruction model: an interrupt is recognised, the handler runs, then the original process resumes.

EXAM-READY STATEMENT

An interrupt is not the same as polling: polling repeatedly checks device status, while an interrupt signals attention.

WORDS THAT EARN MARKS

- **Interrupt** Attention signal
- **ISR** Event handler
- **Context** Saved CPU state
- **Priority** Service urgency

1.2.1(c) / STEP-BY-STEP WORKED EXAMPLE

Keyboard input arrives during an ADD instruction

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Normal	CPU fetches, decodes and executes	The running process retains its ordinary sequence.
2. Event	Keyboard controller raises interrupt	CPU detects a permitted pending request.
3. Preserve	Save PC/status/needed registers	Protect the interrupted process state.
4. Service	Run keyboard ISR	Read/acknowledge device data and record event.
5. Return	Restore state and resume	Next instruction continues as if the ISR had not altered it.

TRANSFER IT TO AN EXAM ANSWER

An interrupt is not the same as polling: polling repeatedly checks device status, while an interrupt signals attention.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.1(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Questions are original teacher-written practice, not OCR past-paper questions. Work in black pen; check the next page only afterwards.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Describe the sequence of actions from an I/O interrupt to resuming the interrupted program.

QUESTION 2 | 4 MARKS

Explain why an interrupt-driven input device may be preferable to continuous polling.

1.2.1(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Interrupt is recognised by the processor at a suitable point.
- Current process context, such as PC and registers, is saved.
- Control transfers to the relevant ISR, which handles/acknowledges the event.
- Saved context is restored and the interrupted process resumes.

Q2 / 4 MARKS

- Polling uses repeated CPU checks even when no input is ready.
- Interrupt-based handling allows useful work between events.
- Interrupt dispatch/context saving has overhead; it is not always better for extremely frequent events.

COMMON MISCONCEPTION

The ISR is code; the interrupt is the event signal. A process is normally resumed, not started from its first instruction.

Green pen: Write the entire interrupt-handling sequence in numbered steps from memory.

1.2.1(d) / BUILD THE MENTAL MODEL

Same tasks, different waiting times

A scheduling timeline must state arrival times, burst lengths and whether pre-emption is permitted.

THREE PIECES OF THE IDEA

FCFS

Run in arrival order; simple but a long task can delay later jobs.

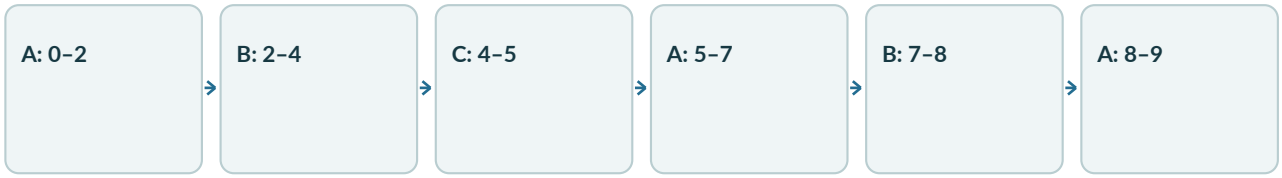
SJF

Choose shortest available total job; may leave large jobs waiting.

ROUND ROBIN

Use a time quantum, then rotate unfinished tasks to ready-queue tail.

MODEL / FOLLOW THE FLOW



Round robin example (all ready at t=0, quantum = 2): A needs 5, B needs 3 and C needs 1 time units.

EXAM-READY STATEMENT

For non-preemptive SJF, choose the shortest *available* task only when the CPU becomes free.

WORDS THAT EARN MARKS

Pre-emption Pause running job

Quantum CPU time slice

Waiting Ready but not run

Turnaround Finish minus arrival

1.2.1(d) / STEP-BY-STEP WORKED EXAMPLE

Calculate the round-robin completion and waiting times

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Input	A:5; B:3; C:1; all arrive t=0	Ready queue begins A, B, C; quantum = 2.
2. Slice 1	A 0–2; B 2–4; C 4–5	A has 3 left; B 1; C completes at t=5.
3. Slice 2	A 5–7; B 7–8	A has 1 left; B completes at t=8.
4. Last	A 8–9	A completes at t=9.
5. Waiting	Completion − arrival − CPU time	A = 4; B = 5; C = 4; mean = 13/3 = 4.33.

TRANSFER IT TO AN EXAM ANSWER

For non-preemptive SJF, choose the shortest *available* task only when the CPU becomes free.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.1(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Questions are original teacher-written practice, not OCR past-paper questions. Work in black pen; check the next page only afterwards.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Using A=5, B=3, C=1 (all arriving at time 0), draw a round-robin trace with time quantum 2.

QUESTION 2 | 5 MARKS

Explain a strength and a weakness of each of FCFS and SJF compared with round robin.

1.2.1(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- A 0–2, B 2–4, C 4–5, A 5–7, B 7–8, A 8–9.
- Completion times: A=9, B=8, C=5.
- Waiting times: A=4, B=5, C=4.

Q2 / 5 MARKS

- FCFS is simple with little scheduling overhead; a long first job causes convoy delays.
- SJF can reduce average waiting time when burst lengths are known; long tasks may starve.
- Round robin is responsive/fair to interactive jobs, but frequent switches add overhead.

COMMON MISCONCEPTION

Waiting time is not the same as turnaround time. Use waiting = completion − arrival − CPU burst.

Green pen: Draw a numbered timeline, check that the total span equals the sum of CPU bursts, and compute each completion.

1.2.1(d) / BUILD THE MENTAL MODEL

Pre-emption changes a running decision

Shortest remaining time (SRT) can replace the running task on a new arrival; multilevel feedback queues change priority based on behaviour.

THREE PIECES OF THE IDEA

SRT

At each arrival/completion, choose task with least remaining CPU time.

MLFQ

Several priority queues with differing time quanta; priorities can change as tasks use CPU.

TRADE-OFF

Excellent short-job response can cause starvation without suitable ageing or promotion.

MODEL / FOLLOW THE FLOW



SRT: A arrives at 0 (8 units), B at 1 (4 units), C at 2 (2 units). Each new arrival is shorter than the remaining job.

EXAM-READY STATEMENT

A scheduling simulation needs a clear rule for ties; these jobs avoid ties at decision points.

WORDS THAT EARN MARKS

- **SRT** Least time remaining
- **MLFQ** Feedback priorities
- **Ageing** Raise waiting task
- **Starvation** Waits indefinitely

1.2.1(d) / STEP-BY-STEP WORKED EXAMPLE

SRT: finish times for staggered arrivals

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
t=0	Run A for 1 unit	A has 7 remaining when B arrives.
t=1	Run B for 1 unit	B has 3 remaining when C arrives.
t=2	Run C for 2 units	C completes at t=4; B is now shortest.
t=4	Run B for 3 units	B completes at t=7; only A remains.
t=7	Run A for 7 units	A completes at t=14.
Waiting	A=14−0−8=6; B=7−1−4=2; C=4−2−2=0	Mean waiting time = 8/3 ≈ 2.67.

TRANSFER IT TO AN EXAM ANSWER

A scheduling simulation needs a clear rule for ties; these jobs avoid ties at decision points.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.1(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Questions are original teacher-written practice, not OCR past-paper questions. Work in black pen; check the next page only afterwards.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Draw the SRT timeline for A(0,8), B(1,4), C(2,2), where pairs represent (arrival, CPU time).

QUESTION 2 | 5 MARKS

Explain how a multilevel feedback queue might help short interactive jobs while risking starvation of a CPU-bound job.

1.2.1(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- A 0–1; B 1–2; C 2–4; B 4–7; A 7–14.
- Completion A=14, B=7, C=4.
- Waiting A=6, B=2, C=0.

Q2 / 5 MARKS

- High-priority queue permits quicker response/shorter quantum for interactive bursts.
- CPU-intensive tasks may be demoted to lower-priority queues.
- A stream of high-priority jobs can delay low-priority tasks indefinitely.
- Ageing or periodic priority boosts can mitigate starvation.

COMMON MISCONCEPTION

Shortest remaining time compares *remaining* burst length, not original burst length or job ID.

Green pen: At each new arrival, annotate the remaining burst time beside every ready task.

1.2.1(e-h) / BUILD THE MENTAL MODEL

Know whether the task is time-critical or hardware-specific

Choose the operating-system model by what the system must guarantee, then separate firmware, device-driver and virtual-machine roles.

THREE PIECES OF THE IDEA

OS TYPES

Distributed, embedded, multitasking, multi-user and real-time describe different requirements.

BOOT + DEVICE

BIOS/UEFI initialises hardware; device drivers expose device-specific functionality.

VIRTUAL MACHINE

Software emulates a whole system or runs intermediate code within a managed execution environment.

MODEL / FOLLOW THE FLOW

Power on → **BIOS / UEFI** → **Boot loader / OS** → **Driver / service**

At startup, firmware performs initialisation and begins booting; it does not replace the OS or a printer driver.

EXAM-READY STATEMENT

A Java-style bytecode virtual machine and an OS guest virtual machine are two different examples of software providing machine functionality.

WORDS THAT EARN MARKS

- **Real-time** Deadline matters
- **BIOS** Startup firmware
- **Driver** Device interface
- **VM** Software machine

1.2.1(e-h) / STEP-BY-STEP WORKED EXAMPLE

Choose software for three real requirements

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
Vehicle	Airbag response has hard deadline	Real-time system; predictability is crucial, not just average speed.
School	Many users share a server	Multi-user OS offers isolation, permissions and concurrent service.
Cluster	Machines cooperate over a network	Distributed OS/software coordinates work across multiple computers.
Sensor	Single-purpose environmental logger	Embedded OS may be compact and dedicated to the device.
Start-up	Firmware initialises and checks hardware	BIOS/UEFI transfers control towards boot-loading OS.
Printer	OS asks driver to print a page	Driver translates generic request to device protocol.
Testing	Run guest OS on host hypervisor	VM isolates a different system; separate from bytecode VM.

TRANSFER IT TO AN EXAM ANSWER

A Java-style bytecode virtual machine and an OS guest virtual machine are two different examples of software providing machine functionality.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.1(e-h) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Questions are original teacher-written practice, not OCR past-paper questions. Work in black pen; check the next page only afterwards.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Justify a suitable OS classification for an airbag controller and a school file server.

QUESTION 2 | 6 MARKS

Distinguish BIOS/UEFI, a device driver, an OS-hosting virtual machine and a bytecode virtual machine.

1.2.1(e-h) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Airbag: real-time/embedded; deadline and predictable response are essential.
- School server: multi-user; must manage accounts, file permissions and simultaneous access.
- Other OS descriptors can overlap; justify the actual requirement.

Q2 / 6 MARKS

- BIOS/UEFI: firmware initialises hardware and starts boot process.
- Driver: device-specific software translates I/O requests.
- System VM: guest OS executes on virtualised hardware via hypervisor.
- Bytecode VM: interprets/executes intermediate code for a runtime environment.

COMMON MISCONCEPTION

Real-time does not mean fastest possible; it means meeting specified timing constraints reliably.

Green pen: Replace a vague phrase like “faster” with an explicit requirement such as response deadline or isolation.

1.2.2(a-c) / BUILD THE MENTAL MODEL

Select software because its properties meet the problem

A good justification begins with the user need, then links a specific product property to an outcome.

THREE PIECES OF THE IDEA

APPLICATION

Performs a user task: bespoke or off-the-shelf; assess fitness, support and cost.

UTILITY

Maintains/protects/optimises a system: backup, compression, anti-malware, encryption.

SOURCE MODEL

Open source offers source access under licence; closed source withholds source. Neither guarantees zero cost.

MODEL / FOLLOW THE FLOW

User requirement → **Required features** → **Constraints** → **Justified choice**

A school needs an accessible timetable tool: choose by features, data protection, support and maintenance, not by label alone.

EXAM-READY STATEMENT

Open source and proprietary describe licensing/source availability. They are not synonyms for secure/insecure or free/paid.

WORDS THAT EARN MARKS

Bespoke	Custom-made	**Utility**	Maintains system
Open source	Source available	**Licence**	Usage conditions

1.2.2(a-c) / STEP-BY-STEP WORKED EXAMPLE

Select a timetable system for a school

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Need	Staff, rooms, accessibility	List must-have tasks and accommodations.
2. Options	Bespoke or existing package	Bespoke can fit workflow; existing package may deploy faster.
3. Security	Pupil data is sensitive	Check updates, access control, audit and compliance.
4. Support	Who maintains it?	Compare support commitments and long-term costs.
5. Decision	Explain choice against criteria	A defensible answer links features to school needs.

TRANSFER IT TO AN EXAM ANSWER

Open source and proprietary describe licensing/source availability. They are not synonyms for secure/insecure or free/paid.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.2(a-c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Questions are original teacher-written practice, not OCR past-paper questions. Work in black pen; check the next page only afterwards.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Evaluate bespoke and off-the-shelf software for a school timetabling system. Refer to at least three criteria.

QUESTION 2 | 4 MARKS

Explain why a backup utility and an open-source software licence solve different problems.

1.2.2(a-c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Bespoke can fit unusual school requirements but costs more to develop/test/support.
- Off-the-shelf deploys faster and can be cheaper but may require compromise.
- Justify with requirements such as accessibility, integration, security and ongoing support.
- A reasoned conclusion earns credit when anchored to stated needs.

Q2 / 4 MARKS

- Backup utility creates recoverable copies of files against loss.
- Open-source licence grants permission to inspect/modify/redistribute under stated conditions.
- One is software functionality; the other is a legal permission model.

COMMON MISCONCEPTION

A backup is not the same as a synchronised copy; deletion can propagate through sync, so recovery versions matter.

Green pen: Rewrite a generic preference as “This option is suitable because [feature] enables [specific user outcome].”

1.2.2(d,e) / BUILD THE MENTAL MODEL

A compiler checks structure before machine instructions run

Interpreters, compilers and assemblers translate different inputs or use different execution strategies; compilation itself has stages.

THREE PIECES OF THE IDEA

COMPILER

Translates a program (typically whole source unit) to target code before running it.

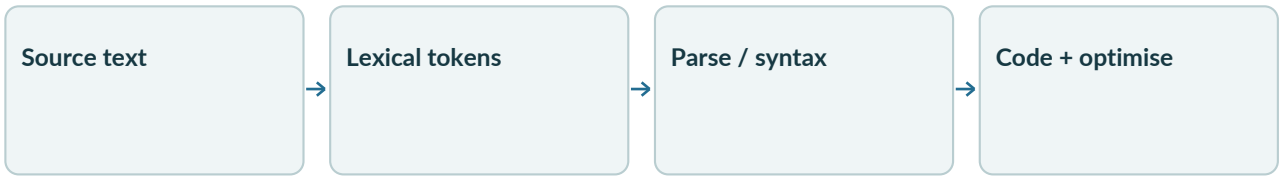
INTERPRETER

Executes through translation at run time; may report faults as reached.

ASSEMBLER

Maps assembly-language mnemonics and labels to machine-code representation.

MODEL / FOLLOW THE FLOW



Example: total = price + 2 becomes identifier, assignment, identifier, operator, integer tokens; then grammar and code checks.

EXAM-READY STATEMENT

A syntax error is found during analysis; a valid program can still contain semantic/logic errors.

WORDS THAT EARN MARKS

Lexical Tokens from text

Syntax Grammar checks

Code gen Produces target

Optimise Improve code

1.2.2(d,e) / STEP-BY-STEP WORKED EXAMPLE

Process a short assignment

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
Source	total = price + 2	Program contains identifiers, assignment, addition, literal.
Lexical	ID(total), =, ID(price), +, INT(2)	Scanner groups characters into tokens.
Syntax	assignment → ID = expression	Parser checks valid arrangement/grammar.
Code gen	LOAD price; ADD 2; STORE total	Illustrative low-level instructions, not literal LMC assembly.
Optimise	Reduce redundant loads where safe	Must preserve observable program meaning.

TRANSFER IT TO AN EXAM ANSWER

A syntax error is found during analysis; a valid program can still contain semantic/logic errors.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.2(d,e) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

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BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Describe lexical analysis, syntax analysis, code generation and optimisation for a compiler.

QUESTION 2 | 4 MARKS

Compare compiler, interpreter and assembler, with an appropriate input or use for each.

1.2.2(d,e) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Lexical analysis creates tokens and can discard irrelevant whitespace/comments.
- Syntax analysis checks token order against language grammar.
- Code generation produces target/intermediate instructions.
- Optimisation improves efficiency without changing intended behaviour.

Q2 / 4 MARKS

- Compiler translates high-level source to output code ahead of execution.
- Interpreter translates/executes at runtime (implementation details vary).
- Assembler converts symbolic assembly mnemonics to machine code.

COMMON MISCONCEPTION

Compiled does not mean bug-free. Translation can succeed while the algorithm still produces the wrong answer.

Green pen: Take one expression and underline which information belongs to lexical, syntax and code-generation stages.

1.2.2(f) / BUILD THE MENTAL MODEL

Compiled objects still need to become a running program

A compiler may create object code, but a linker resolves external references and a loader prepares the executable for execution.

THREE PIECES OF THE IDEA

LIBRARY

Reusable prewritten code providing functions, classes or services.

LINKER

Combines object modules and resolves symbols/references to libraries.

LOADER

Places executable in memory and prepares its process for running.

MODEL / FOLLOW THE FLOW



Two compiled source modules can call one library function; linking connects the references; loading starts the process.

EXAM-READY STATEMENT

Static and dynamic linking are different approaches: a library can be incorporated at build time or loaded/linked at run time.

WORDS THAT EARN MARKS

- **Object file** Compiled module
- **Linker** Resolves references
- **Symbol** Named reference
- **Loader** Prepares execution

1.2.2(f) / STEP-BY-STEP WORKED EXAMPLE

Program uses a shared graph library

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Compile	app.c → app.o	Object code may still refer to external drawGraph().
2. Reference	Unresolved drawGraph symbol	External routine implementation not in app.o.
3. Link	Link app.o to appropriate library	Resolve references; produce linked executable or dynamic references.
4. Load	OS loader maps code/data into memory	Arrange address space and required shared modules.
5. Run	Control transfers to entry point	Program uses library routine when required.

TRANSFER IT TO AN EXAM ANSWER

Static and dynamic linking are different approaches: a library can be incorporated at build time or loaded/linked at run time.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.2(f) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

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BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Explain what a linker does when a program calls a function stored in a library.

QUESTION 2 | 5 MARKS

Explain two advantages and one limitation of using libraries, and outline the loader's role.

1.2.2(f) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Compiled modules can contain unresolved external symbols.
- Linker matches symbols to module/library definitions.
- It combines code/metadata and creates appropriate addresses or relocations.
- Output is a linked executable or suitable dynamic-link structure.

Q2 / 5 MARKS

- Libraries provide tested, reusable functionality and reduce duplicated development.
- Updates/standardisation can improve maintainability.
- Version conflicts, dependencies or security flaws can be introduced.
- Loader maps executable and dependencies into memory and prepares execution.

COMMON MISCONCEPTION

A linker does not itself run the program. A loader does not perform the compiler's lexical analysis.

Green pen: Label where each of source, object, executable and running process exists in the pipeline.

1.2.3(a,b) / BUILD THE MENTAL MODEL

Choose a method by risk, certainty and feedback

A methodology structures change and validation; the best choice depends on stability, risk, user involvement and deadlines.

THREE PIECES OF THE IDEA

WATERFALL

Sequential staged work; useful with stable requirements and approval gates.

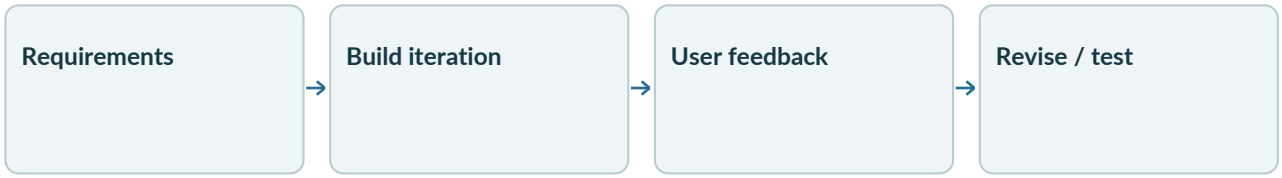
AGILE / XP

Incremental delivery; XP stresses practices such as pair programming and frequent tests.

SPIRAL / RAD

Spiral explicitly addresses risk across cycles; RAD prioritises quick prototypes and user feedback.

MODEL / FOLLOW THE FLOW



An iterative product improves after each usable increment. Waterfall may enforce sequential stage sign-off instead.

EXAM-READY STATEMENT

Agile is not “no documentation”; waterfall is not “no testing”. Explain when each trade-off matters.

WORDS THAT EARN MARKS

- **Iteration** Repeat build cycle
- **Prototype** Early model
- **Spiral** Risk-led cycle
- **XP** Agile practices

1.2.3(a,b) / STEP-BY-STEP WORKED EXAMPLE

Choose a method for two contrasting projects

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
A: payroll	Fixed legal calculation rules	Structured documentation and test stages are important.
Trade-off	Waterfall offers stage control	Late changes are costly; users see final product relatively late.
B: school app	Users uncertain about interface	RAD/agile prototypes can elicit precise feedback early.
Trade-off	Iterative approach enables changes	Scope and quality must still be controlled by tests.
High risk	Unproven technical integration	Spiral builds risk assessment into each cycle.
Team culture	Pair review and automated tests	XP practices can strengthen frequent releases.

TRANSFER IT TO AN EXAM ANSWER

Agile is not “no documentation”; waterfall is not “no testing”. Explain when each trade-off matters.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.3(a,b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Questions are original teacher-written practice, not OCR past-paper questions. Work in black pen; check the next page only afterwards.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Evaluate waterfall and agile approaches for a school reporting tool with changing teacher requirements.

QUESTION 2 | 5 MARKS

Explain why the spiral model may be preferable to RAD for safety-critical control software.

1.2.3(a,b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Waterfall offers up-front planning, staged review and predictable documentation.
- It may handle late changes poorly and delay user feedback.
- Agile enables incremental release and feedback when requirements evolve.
- It requires active user involvement and control of scope/test quality.
- Conclude using the actual school context.

Q2 / 5 MARKS

- Spiral includes explicit risk identification, assessment and mitigation each cycle.
- High-risk design decisions can be explored before major commitment.
- RAD rapidly develops prototypes but speed alone is not a safety guarantee.
- Safety still demands verification, documentation and controlled testing.

COMMON MISCONCEPTION

Extreme programming is a named agile approach; it is not simply writing code very quickly.

Green pen: Write one “suitable because” and one “limited because” statement for each method.

1.2.3(c) / BUILD THE MENTAL MODEL

Trace the condition before changing a variable

A trace table records state after clearly defined statements. Distinguish loop-entry tests from updates inside the body.

THREE PIECES OF THE IDEA

SPECIFY

State inputs, outputs and required behaviour before writing steps.

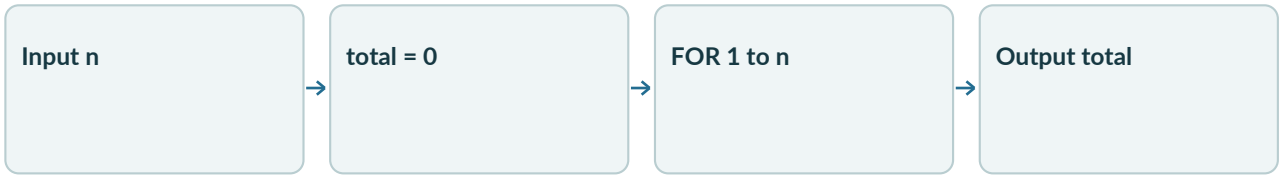
TRACE

Record each iteration's values and branch outcome systematically.

VERIFY

Test normal, boundary and invalid cases to expose faults.

MODEL / FOLLOW THE FLOW



Example algorithm accumulates 1+2+...+n; n=4 should output 10. The accumulator changes after each addition.

EXAM-READY STATEMENT

At each completed pass, total equals the sum of all integers already processed. This is an invariant useful for explanation.

WORDS THAT EARN MARKS

- **Trace** State by step
- **Boundary** Edge input
- **Invariant** Maintained truth
- **Dry run** Manual execution

1.2.3(c) / STEP-BY-STEP WORKED EXAMPLE

Dry-run a running total for n = 4

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
Initial	n = 4; total = 0	No iteration has run.
i = 1	total = 0 + 1 = 1	First pass.
i = 2	total = 1 + 2 = 3	Second pass.
i = 3	total = 3 + 3 = 6	Third pass.
i = 4	total = 6 + 4 = 10	Fourth pass.
After	Output 10	The loop has completed its inclusive upper bound.

TRANSFER IT TO AN EXAM ANSWER

At each completed pass, total equals the sum of all integers already processed. This is an invariant useful for explanation.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.3(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

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BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Trace total ← 0; FOR i FROM 1 TO 5: total ← total + i. Give total after every iteration and the output.

QUESTION 2 | 5 MARKS

Design an algorithm that inputs three integers and outputs the largest. State a boundary test.

1.2.3(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Totals after i=1,2,3,4,5 are 1,3,6,10,15.
- Final output is 15, assuming an inclusive loop 1..5.

Q2 / 5 MARKS

- Read a,b,c; set largest ← a; if b > largest then largest ← b; if c > largest then largest ← c; output largest.
- Test equal values, e.g. (3,3,3), and negatives, e.g. (−5,−2,−9), as relevant edge cases.
- A diagram or equivalent correct pseudocode may earn credit.

COMMON MISCONCEPTION

Updating total before recording it changes the meaning of a trace row. State exactly when values are measured.

Green pen: Add a trace table with columns for iteration number and accumulator after the loop body.

1.2.4(a,b) / BUILD THE MENTAL MODEL

Different styles organise the same computation

A paradigm is a programming approach. Procedural programs organise operations into procedures/functions with explicit control flow.

THREE PIECES OF THE IDEA

PROCEDURAL

Step-by-step algorithms, variables, loops, parameterised routines.

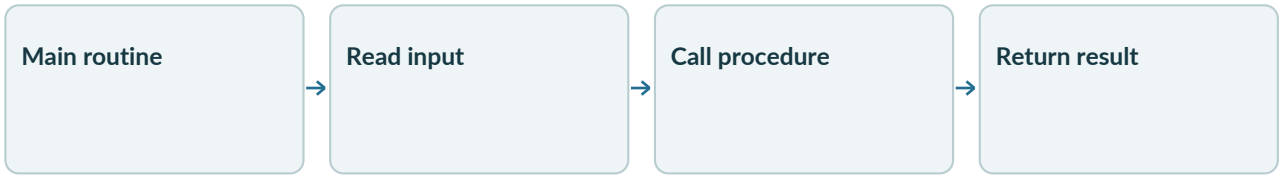
OBJECT-ORIENTED

Model objects containing state and behaviour, with encapsulation and reuse.

DECLARATIVE / FUNCTIONAL

Describe desired results or compose functions with fewer mutable side effects (different styles).

MODEL / FOLLOW THE FLOW



A procedural program decomposes a calculation into named routines. Data is passed by parameters and/or stored in state.

EXAM-READY STATEMENT

A procedure performs an action; a function returns a value in conventional terminology. Language implementations differ.

WORDS THAT EARN MARKS

Paradigm Programming style

Procedure Named action

Parameter Supplied value

State Current values

1.2.4(a,b) / STEP-BY-STEP WORKED EXAMPLE

Break a bill calculator into routines

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
Main	get unit price and quantity	Input handled before calculation.
Call	subtotal = cost(price,qty)	Function receives two arguments.
Inside	RETURN price × qty	Return value calculated, not printed inside function.
Process	vat = subtotal × 0.20	Separate stage can be tested independently.
Output	print subtotal + vat	Main routine handles presentation.

TRANSFER IT TO AN EXAM ANSWER

A procedure performs an action; a function returns a value in conventional terminology. Language implementations differ.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.4(a,b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Questions are original teacher-written practice, not OCR past-paper questions. Work in black pen; check the next page only afterwards.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Explain two advantages of procedural decomposition for a large program.

QUESTION 2 | 5 MARKS

Describe two differences between procedural and object-oriented approaches, with an appropriate example.

1.2.4(a,b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Named procedures/functions reduce duplication and support reuse.
- Individual routines can be tested and maintained separately.
- Parameters clarify data transferred between routines.
- Clear control flow supports reasoning and debugging.

Q2 / 5 MARKS

- Procedural code organises operations and data flow into routines.
- OOP groups attributes and methods in objects/classes.
- Procedural example: calculateTax function; OOP example: Invoice object with total() and stored line items.

COMMON MISCONCEPTION

Functions are not automatically object-oriented. Both procedural and OO programs can use functions and parameters.

Green pen: Replace a lengthy monolithic procedure with two named subroutines and state each input/output.

1.2.4(c,d) / BUILD THE MENTAL MODEL

Separate the instruction from the value it refers to

LMC illustrates an accumulator machine using mailboxes. Addressing modes determine how an operand is obtained.

THREE PIECES OF THE IDEA

LMC

LDA, STA, ADD, SUB, INP, OUT, BRA, BRZ, BRP, HLT and DAT model basic operations.

ADDRESS

Direct uses the addressed location; immediate embeds a value; indirect reads a pointer.

INDEXED

Effective address combines a base address with an index register or offset.

MODEL / FOLLOW THE FLOW

LDA 90 → ADD 91 → STA 92 → OUT / HLT

If mailbox 90=4 and 91=7, the accumulator reaches 11 and mailbox 92 stores 11. This example is valid LMC.

EXAM-READY STATEMENT

Immediate/indexed/indirect examples are generic pseudo-assembly, NOT instructions you may assume exist in standard OCR LMC.

WORDS THAT EARN MARKS

- **ACC** Accumulator
- **DAT** Data mailbox
- **Direct** Read location
- **Indirect** Follow pointer

1.2.4(c,d) / STEP-BY-STEP WORKED EXAMPLE

Trace a complete LMC sequence

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
Memory	90 DAT 4; 91 DAT 7; 92 DAT 0	These are data mailboxes, not literal operands.
LDA 90	ACC ← RAM[90] = 4	Load value, not the number 90.
ADD 91	ACC ← 4 + RAM[91] = 11	Arithmetic result retained in ACC.
STA 92	RAM[92] ← 11	Store result, ACC remains 11.
OUT	Output 11	Output current ACC.
HLT	Stop execution	No further instruction is executed.

TRANSFER IT TO AN EXAM ANSWER

Immediate/indexed/indirect examples are generic pseudo-assembly, NOT instructions you may assume exist in standard OCR LMC.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.4(c,d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

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BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Given 90 DAT 6, 91 DAT 2, 92 DAT 0, trace LDA 90, SUB 91, STA 92, OUT, HLT. Give ACC/output/mailbox 92.

QUESTION 2 | 6 MARKS

Explain immediate, direct, indirect and indexed addressing, using a simple example of each.

1.2.4(c,d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- After LDA 90: ACC = 6. After SUB 91: ACC = 4.
- STA 92 puts 4 in mailbox 92. OUT displays 4. HLT stops.

Q2 / 6 MARKS

- Immediate: operand is value in instruction, e.g. ADD #5 (generic).
- Direct: instruction gives location holding operand, e.g. ADD 90 means RAM[90].
- Indirect: instruction identifies a location containing an operand address/pointer.
- Indexed: effective address is base plus index/offset, e.g. ADD 90,X.
- Note only direct-style examples are in ordinary LMC syntax.

COMMON MISCONCEPTION

LMC has no standard LDI, indirect or indexed instruction in OCR's basic set. Do not write generic pseudo-assembly as an LMC answer.

Green pen: For each instruction, write "effective address" and "operand value" as two different entries.

1.2.4(e) / BUILD THE MENTAL MODEL

Encapsulation protects the state; polymorphism varies behaviour

A class defines structure and behaviour; an object is a runtime instance. Inheritance and polymorphism solve different design problems.

THREE PIECES OF THE IDEA

CLASS / OBJECT

A class is a blueprint; an instance holds attributes and exposes methods.

ENCAPSULATION

State and operations are grouped; access restrictions mediate valid changes.

INHERITANCE / POLYMORPHISM

Subclasses reuse/extend a base type; the same interface can select different behaviour.

MODEL / FOLLOW THE FLOW

Shape base class → **draw() method** → **Circle override** → **Square override**

Calling draw() on Shape references can dispatch to different implementations for Circle and Square objects.

EXAM-READY STATEMENT

Encapsulation means controlled access to object state, not merely placing public variables inside a class.

WORDS THAT EARN MARKS

- **Attribute** Stored property
- **Method** Object operation
- **Override** Replace method
- **Polymorphism** Same interface

1.2.4(e) / STEP-BY-STEP WORKED EXAMPLE

Model a safe bank account object

Follow each change in order. A model is only useful when you can explain both what changed and why.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
Class	Account(balance)	Define state and methods.
Private	balance not directly writable	Other code cannot bypass validation.
Method	withdraw(amount)	Validate amount positive and not greater than balance.
Instance	a = new Account(40)	Object a starts with balance 40.
Call	a.withdraw(15)	Balance becomes 25 after validation.
Extension	SavingsAccount overrides interest()	An inherited type can specialise behaviour.

TRANSFER IT TO AN EXAM ANSWER

Encapsulation means controlled access to object state, not merely placing public variables inside a class.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check calculations and intermediate values, not just the final result.

1.2.4(e) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

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BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Explain classes, objects, attributes, methods and encapsulation using a bank-account example.

QUESTION 2 | 6 MARKS

Explain how inheritance and polymorphism could be used for a hierarchy of shapes with draw() methods.

1.2.4(e) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Class is template defining data and operations.
- Account object is an instance with balance attribute and withdraw method.
- Encapsulation combines state/behaviour and controls access so invalid withdrawals are prevented.
- Methods can enforce data invariants.

Q2 / 6 MARKS

- Base class Shape defines shared interface/behaviour.
- Circle/Square inherit and can add or override methods.
- Polymorphic call to draw() selects implementation appropriate to runtime object.
- This enables common client code and easier extension.

COMMON MISCONCEPTION

Inheritance shares/reuses a type relationship; polymorphism selects different behaviour through a common interface.

Green pen: Draw a class and two instances; label which features belong to the class and which values belong to each object.

1.3 / TEACHER-REVIEW WORKING CHAPTER

Understand it. Model it. Try it. Check it.

A standalone revision upgrade for H446/01 section 1.3. Preserve and reconcile the 264-page published guide before any future release.

FOUR SPECIFICATION FAMILIES

1.3.1 COMPRESSION + SECURITY
Lossless/lossy, RLE, dictionary, encryption and hashing

1.3.2 DATABASES
Keys, 3NF, SQL, referential integrity and transactions

1.3.3 NETWORKS
TCP/IP, DNS, routing, switching and security

1.3.4 THE WEB
HTML, CSS, JavaScript, search and client/server

FOUR PASSES FOR EVERY TOPIC

01 UNDERSTAND
Build a precise conceptual picture.

02 MODEL
Follow a worked example or trace.

03 TRY
Answer original exam-style questions.

04 CHECK
Self-mark with separate indicative points.

RELEASE SAFETY

This is a teacher-review copy, not a replacement for the published H446 guide. Live lesson files, quizzes and page links are unchanged. Subject accuracy and mark points need teacher review.

Scope: 20 cycles x 4 pages plus 2 orientation pages = 82 pages. This is a candidate chapter to reconcile against the existing 1.3 content, not 78 pages simply appended to the current guide.

1.3 / SPECIFICATION CHECKLIST

Every 1.3 topic has a working model

Jump to any topic using the linked rows. Each cycle has an explanation, a worked model, two practice questions and separate mark points.

SPEC	LEARNING CYCLE	PAGES
1.3.1(a)	Lossy and lossless compression	3-6
1.3.1(b)	Run-length encoding	7-10
1.3.1(b)	Dictionary coding	11-14
1.3.1(c)	Symmetric and asymmetric encryption	15-18
1.3.1(d)	Hashing and integrity	19-22
1.3.2(a)	Relational data, keys and ER modelling	23-26
1.3.2(c)	Normalisation to third normal form	27-30
1.3.2(b)	Data capture, selection and exchange	31-34
1.3.2(d)	SQL SELECT and JOIN	35-38
1.3.2(d,e)	SQL changes and referential integrity	39-42
1.3.2(f)	Transactions, ACID and record locking	43-46
1.3.3(a,b)	Protocols, standards and the TCP/IP stack	47-50
1.3.3(b)	DNS, addressing and routing	51-54
1.3.3(b)	Packet and circuit switching	55-58
1.3.3(c)	Threats and layered network protection	59-62
1.3.3(d)	Network hardware and LAN/WAN	63-66
1.3.3(e)	Client-server and peer-to-peer	67-70
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SPECIFICATION BOUNDARY

Aligned to OCR H446 1.3.1-1.3.4. PageRank is taught conceptually, not as a required coded algorithm; relational normalisation is through 3NF.

Checked against: OCR H446 specification v2.7 (2024), plus OCR subject-content clarification (2020), sections 1.3.1-1.3.4. Questions and worked examples are new draft materials, not OCR past-paper content.

1.3.1(a) / BUILD THE MENTAL MODEL

Smaller file; choose what can be removed

Compression represents information using fewer bits. The method depends on whether every original bit must be recovered.

THREE PIECES OF THE IDEA

LOSSLESS

The reconstructed data is identical to the original: suitable for code, accounts and exact text.

LOSSY

Discard selected information: suitable where acceptable perceptual quality matters more than exact recovery.

TRADE-OFF

Consider file size, quality, processing cost, use and whether later editing will occur.

MODEL / FOLLOW THE FLOW

Original file → **Compressor** → **Stored bits** → **Decompressor**

Lossless returns the exact source. Lossy returns an approximation; repeated lossy saving can degrade quality.

EXAM-READY STATEMENT

Always justify the compression choice against the context; lossy is not automatically better because it is smaller.

WORDS THAT EARN MARKS

- **Lossless** Identical output
- **Lossy** Approximation
- **Ratio** Original ÷ compressed
- **Artefact** Visible or audible error

1.3.1(a) / STEP-BY-STEP WORKED EXAMPLE

Choose a method for three school files

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Source code	A program cannot tolerate missing characters	Use lossless compression so every character survives.
2. Photograph	Website thumbnail can trade some detail for size	A suitable lossy format may reduce transfer time.
3. Medical scan	Small features may matter diagnostically	Choose lossless if any data loss is unacceptable.
4. Calculate	12 MB compressed to 3 MB	Compression ratio = 12:3 = 4:1 (75% reduction).

TRANSFER IT TO AN EXAM ANSWER

Always justify the compression choice against the context; lossy is not automatically better because it is smaller.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.1(a) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

A school stores source code and photographs. Recommend a compression type for each and justify both.

QUESTION 2 | 4 MARKS

A file is reduced from 18 MB to 6 MB. Calculate the compression ratio and percentage reduction, and explain the result.

1.3.1(a) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Source code: lossless, because all characters/instructions must be exactly recovered.
- Photos: lossy may be appropriate if reduced size matters and small visual losses are acceptable.

Q2 / 4 MARKS

- Ratio = 18:6 = 3:1.
- Reduction = (18 - 6) / 18 × 100 = 66.7%.
- One third of the original size remains.

COMMON MISCONCEPTION

A compression ratio of 3:1 means the new file is one third the original, not that only 3% remains.

Green pen: State which property of the file makes lossless or lossy appropriate.

1.3.1(b) / BUILD THE MENTAL MODEL

Count the run; do not just delete repeats

Run-length encoding (RLE) replaces consecutive identical values with a count and the value. It is lossless if the representation can be decoded unambiguously.

THREE PIECES OF THE IDEA

ENCODE

Replace runs of a repeated symbol with a count and symbol.

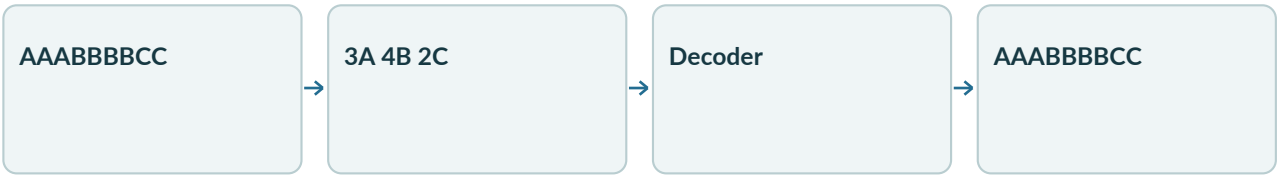
DECODE

Reconstruct each run exactly using the stored count.

LIMITATION

Short alternating runs may create extra data rather than reduce it.

MODEL / FOLLOW THE FLOW



In this teaching notation the encoded length is six tokens (3,A,4,B,2,C), compared with nine original characters.

EXAM-READY STATEMENT

RLE is effective when the source contains long runs of identical values, such as simple images with large solid areas.

WORDS THAT EARN MARKS

- **Run** Repeated sequence
- **Count** Run length
- **Lossless** Exact recovery
- **Overhead** Extra metadata

1.3.1(b) / STEP-BY-STEP WORKED EXAMPLE

Encode AAAAAABBBCC and compare an alternating string

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Identify runs	AAAAAA | BBB | CC	Runs have lengths 6, 3 and 2.
2. Encode	6A 3B 2C	Each stored pair gives count then symbol.
3. Decode	6A -> AAAAAA	All original information can be reconstructed.
4. Test failure	ABABAB -> 1A 1B 1A 1B 1A 1B	Encoding may be longer when repeats are scarce.
5. Consider format	Counts may need delimiters for multi-digit runs	Actual file size depends on how counts/symbols are represented.

TRANSFER IT TO AN EXAM ANSWER

RLE is effective when the source contains long runs of identical values, such as simple images with large solid areas.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.1(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Encode the symbols AAAABBBCCCAA using count-then-symbol RLE. Explain one reason the result may not always save space.

QUESTION 2 | 4 MARKS

A row of 12 white pixels is followed by one black pixel. Explain why RLE might work well for this image and how decoding restores it.

1.3.1(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Runs: 4A, 3B, 3C, 2A.
- Encoded as 4A3B3C2A in the stated notation.
- Every single-symbol run also needs a count, so alternating input can expand.

Q2 / 4 MARKS

- Store 12W 1B (with a known unambiguous format).
- Decoder outputs 12 white pixels then one black pixel.
- Long repeated runs yield fewer stored items than individual pixels.

COMMON MISCONCEPTION

RLE is not the same as removing duplicates anywhere in a file: only adjacent repeated symbols form a run.

Green pen: Check your result by decoding the sequence you created.

1.3.1(b) / BUILD THE MENTAL MODEL

One entry; reuse its shorter reference

Dictionary compression replaces recurring sequences with shorter references to entries in a shared dictionary.

THREE PIECES OF THE IDEA

BUILD

Identify repeating words, phrases or byte patterns and assign codes.

ENCODE

Write references to dictionary entries instead of repeating the full pattern.

DECODE

Use the same dictionary to rebuild the original data exactly.

MODEL / FOLLOW THE FLOW

Repeated phrases → **Dictionary entries** → **Index references** → **Exact original**

Simple illustrative example: the phrase COMPUTER repeats four times; a one-symbol token replaces later occurrences.

EXAM-READY STATEMENT

The coded message is not meaningful without an agreed dictionary; dictionary storage contributes to total compressed size.

WORDS THAT EARN MARKS

Dictionary Pattern-code map

Reference Short token

Lossless Exact recovery

Overhead Dictionary storage

1.3.1(b) / STEP-BY-STEP WORKED EXAMPLE

Substitute repeated colour names with a dictionary

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Source	RED BLUE RED BLUE RED	Repeated tokens are visible.
2. Dictionary	1 = RED; 2 = BLUE	Dictionary holds each full word once.
3. Encode	1 2 1 2 1	Every source token becomes a reference.
4. Decode	1 -> RED; 2 -> BLUE	Dictionary restores the exact token sequence.
5. Evaluate	Save size of codes minus dictionary cost	A large or infrequently reused dictionary may be inefficient.

TRANSFER IT TO AN EXAM ANSWER

The coded message is not meaningful without an agreed dictionary; dictionary storage contributes to total compressed size.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.1(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Create a two-entry dictionary for CAT DOG CAT DOG DOG and write the encoded message.

QUESTION 2 | 4 MARKS

Explain why a dictionary-based lossless method may perform poorly on a file where patterns seldom repeat.

1.3.1(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Example dictionary: A = CAT, B = DOG.
- Encoded data: A B A B B.
- Decoder uses each reference to restore the original order.

Q2 / 4 MARKS

- Few repeated patterns mean few substitutions.
- The dictionary itself takes storage space.
- References plus dictionary may be no smaller than the original.

COMMON MISCONCEPTION

A dictionary index is a compression reference, not a cryptographic secret or an encryption key.

Green pen: Explain how both sides access the dictionary and how this affects decompression.

1.3.1(c) / BUILD THE MENTAL MODEL

Which key is shared, and which stays private?

Encryption makes information unreadable to unauthorised parties. Symmetric and asymmetric schemes differ in their key use.

THREE PIECES OF THE IDEA

SYMMETRIC

The same shared secret key is used for encryption and decryption; generally efficient for bulk data.

ASYMMETRIC

Uses a public/private key pair; public encryption is decrypted with the matching private key.

HYBRID

Public-key methods can establish a shared session key; symmetric encryption then protects bulk traffic.

MODEL / FOLLOW THE FLOW

Public key step → **Session key** → **Symmetric cipher** → **Recipient decrypts**

A simplified secure connection often uses asymmetric techniques for key establishment/authentication and symmetric encryption for the data stream.

EXAM-READY STATEMENT

Avoid claiming that a recipient encrypts with their private key to keep a message confidential; that confuses confidentiality with signing.

WORDS THAT EARN MARKS

Plaintext	Readable data	**Ciphertext**	Encrypted data
Public key	Shared openly	**Private key**	Kept secret

1.3.1(c) / STEP-BY-STEP WORKED EXAMPLE

Alice sends a confidential file to Bob

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Obtain key	Alice obtains Bob's authentic public key	The identity of the key owner must be verified.
2. Protect	Alice encrypts a short secret for Bob	Only Bob's corresponding private key can recover it in this model.
3. Data	Shared secret protects the file symmetrically	Efficient bulk encryption uses the session key.
4. Receive	Bob recovers the secret and decrypts the file	Key secrecy protects confidentiality.
5. Check	Authentication is a separate concern	Encryption alone does not prove who sent a message.

TRANSFER IT TO AN EXAM ANSWER

Avoid claiming that a recipient encrypts with their private key to keep a message confidential; that confuses confidentiality with signing.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.1(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Compare symmetric and asymmetric encryption in terms of keys, speed and a realistic use case.

QUESTION 2 | 5 MARKS

Explain how a hybrid approach could protect a large file sent to a new recipient.

1.3.1(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Symmetric: same shared secret for encryption/decryption; efficient for bulk data.
- Asymmetric: public/private pair; suitable for securely establishing or transporting a session secret.
- Hybrid combines public-key key establishment with efficient symmetric data encryption.

Q2 / 5 MARKS

- Recipient has an authentic public key.
- Sender protects a session secret using a public-key mechanism.
- Recipient uses the corresponding private key to recover the secret.
- Both use the symmetric secret for bulk file encryption/decryption.

COMMON MISCONCEPTION

A public key can be public; the associated private key must not be sent to the other party.

Green pen: Label whose key is used at each stage and whether it is secret or public.

1.3.1(d) / BUILD THE MENTAL MODEL

A hash is a one-way fingerprint, not a cipher

A hash function maps input data to a fixed-size digest. Hashes help identify changes and support password verification with appropriate secure constructions.

THREE PIECES OF THE IDEA

DIGEST

The algorithm produces a fixed-length result from an arbitrary-length message.

INTEGRITY

A changed digest suggests the data is different; accidental changes can be detected.

PASSWORDS

Store a salted, deliberately slow password-derived hash rather than the plaintext password.

MODEL / FOLLOW THE FLOW

Input message → **Hash function** → **Digest value** → **Compare digests**

A hash is recomputed from the received data. Matching digests support an integrity check but do not independently authenticate the source.

EXAM-READY STATEMENT

Hash collisions are theoretically possible. A plain fast hash alone is not a recommended modern password-storage design.

WORDS THAT EARN MARKS

- **Hash** Fixed-length digest
- **Collision** Same digest
- **Salt** Unique random input
- **One-way** Hard to invert

1.3.1(d) / STEP-BY-STEP WORKED EXAMPLE

Check a file and store a password safely

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. File digest	Record an expected digest	A receiver calculates a digest of the downloaded file.
2. Compare	Received digest differs	This indicates a mismatch; redownload or investigate.
3. Password set	Generate a unique random salt	Use a suitable password hashing/KDF scheme to store a derived value.
4. Password login	Recompute using stored salt	Compare the derived result in constant-time code.
5. Remember	Digest is not reversible encryption	There is no decryption key for recovering the original input.

TRANSFER IT TO AN EXAM ANSWER

Hash collisions are theoretically possible. A plain fast hash alone is not a recommended modern password-storage design.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.1(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Explain two distinct uses of hashing and why hashing is not the same as encryption.

QUESTION 2 | 4 MARKS

A file digest changes after downloading. Explain what can be concluded and what cannot be proved using only the digest.

1.3.1(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Integrity check: compare stored and calculated digests for changes.
- Password verification: compare a secure salted password-derived hash.
- Hashing is designed to be one-way, unlike reversible encryption.

Q2 / 4 MARKS

- Different digests indicate the file data differs or an error occurred.
- Matching values alone do not prove the sender is authentic.
- Hashes can collide; protect the trusted expected digest.

COMMON MISCONCEPTION

Do not describe a hash as “encrypted data that the user can decrypt with a key”.

Green pen: Name the threat being addressed: accidental change, tampering, or password disclosure.

1.3.2(a) / BUILD THE MENTAL MODEL

Link facts using stable identifiers

A relational database organises data into tables; keys describe how records are uniquely identified and related.

THREE PIECES OF THE IDEA

PRIMARY KEY

Uniquely identifies each row in its own table; values cannot be null.

FOREIGN KEY

A field or field set referencing a key in another table.

SECONDARY KEY

A non-primary field used for searching; it need not be unique.

MODEL / FOLLOW THE FLOW

Customer 1 → **Many Orders** → **Many OrderLines** → **Product reference**

Customer(CustomerID PK) links to Order(CustomerID FK): one customer can have many orders. A line table bridges an order to a product.

EXAM-READY STATEMENT

A foreign key is not necessarily unique. It is supposed to repeat in a one-to-many relationship.

WORDS THAT EARN MARKS

- **Entity** Object of interest
- **PK** Unique row identifier
- **FK** Cross-table link
- **Cardinality** 1:1, 1:M, M:M

1.3.2(a) / STEP-BY-STEP WORKED EXAMPLE

Turn a many-to-many relationship into two one-to-many links

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Recognise	A student takes many courses	A course also has many students: M:M.
2. Make entities	Student(StudentID, Name)	StudentID is the primary key.
3. Course	Course(CourseID, Title)	CourseID is the primary key.
4. Bridge	Enrolment(StudentID, CourseID)	Both foreign keys; composite primary key may be the pair.
5. Explain	Student 1:M Enrolment M:1 Course	The bridge prevents repeated course columns.

TRANSFER IT TO AN EXAM ANSWER

A foreign key is not necessarily unique. It is supposed to repeat in a one-to-many relationship.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(a) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Design three tables for customers placing orders. Label primary and foreign keys and explain the relationships.

QUESTION 2 | 4 MARKS

Explain the purpose of a secondary key and why an email address may be unsuitable as a primary key.

1.3.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Customer(CustomerID PK, Name).
- Order(OrderID PK, CustomerID FK, OrderDate).
- OrderLine(LineID PK, OrderID FK, ProductID FK, Qty) with a Product table if product details are stored.
- One Customer can have many Orders; one Order many OrderLines.

Q2 / 4 MARKS

- Secondary key improves locating/selecting rows on a searched attribute.
- Email can change, and uniqueness/availability may not be guaranteed.
- A stable surrogate identifier is often more suitable as the PK.

COMMON MISCONCEPTION

A many-to-many relationship requires an associative table in a conventional relational design.

Green pen: Draw the crow-foot direction and identify the table that holds each foreign key.

1.3.2(c) / BUILD THE MENTAL MODEL

Remove partial and transitive dependency

Normalisation restructures tables so each fact is stored in an appropriate place, reducing update, insertion and deletion anomalies.

THREE PIECES OF THE IDEA

1NF

Atomic cells and a key; no repeating groups or multi-valued fields in a cell.

2NF

1NF plus every non-key attribute depends on the whole candidate key, not just part of a composite key.

3NF

2NF plus no transitive dependency of non-key attributes on a key via another non-key attribute.

MODEL / FOLLOW THE FLOW

Flat sales list → **1NF line records** → **2NF split by key** → **3NF isolate customer**

Worked decomposition below uses OrderID + ProductID as the original composite key.

EXAM-READY STATEMENT

Exam explanations need the dependency that causes the problem, not only a list of resulting tables.

WORDS THAT EARN MARKS

- **1NF** Atomic fields
- **Partial** Part of key
- **Transitive** Non-key via non-key
- **Anomaly** Data inconsistency

1.3.2(c) / STEP-BY-STEP WORKED EXAMPLE

Normalise sales lines in stages

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Raw row	OrderID, ProductID, OrderDate, CustomerID, CustomerName, ProductName, Qty	One atomic product per line; key (OrderID, ProductID).
2. Partial	OrderDate relies only on OrderID	ProductName relies only on ProductID: violates 2NF.
3. Into 2NF	Order(OrderID, OrderDate, CustomerID, CustomerName); Product(ProductID, ProductName); Line(OrderID, ProductID, Qty)	Remove those partial dependencies.
4. Transitive	OrderID -> CustomerID -> CustomerName	CustomerName depends on CustomerID, not directly on OrderID.
5. Into 3NF	Customer(CustomerID, CustomerName); Order(OrderID, OrderDate, CustomerID); Product; Line	CustomerID is an FK in Order; line has two FKs.

TRANSFER IT TO AN EXAM ANSWER

Exam explanations need the dependency that causes the problem, not only a list of resulting tables.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

A table Sale(OrderID, ItemID, ItemName, Qty, CustomerID, CustomerName) uses key (OrderID, ItemID). Describe one partial and one transitive dependency and suggest a 3NF structure.

QUESTION 2 | 4 MARKS

Explain how 3NF can reduce an update anomaly when a customer changes their name.

1.3.2(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- ItemName depends on ItemID only: partial dependency.
- CustomerName depends on CustomerID, itself identified through OrderID: transitive dependency.
- Use Order(OrderID,CustomerID), Customer(CustomerID,CustomerName), Item(ItemID,ItemName), SaleLine(OrderID,ItemID,Qty), with appropriate PKs/FKs.

Q2 / 4 MARKS

- Store the customer name once in Customer.
- Order rows reference CustomerID rather than repeat the name.
- One update changes the authoritative value for all orders.

COMMON MISCONCEPTION

2NF and 3NF are about functional dependencies; simply saying “no duplicates” does not show the reasoning.

Green pen: Underline each determinant and dependent attribute before deciding which tables to create.

1.3.2(b) / BUILD THE MENTAL MODEL

Collect, validate, query, export

Data may enter a database through forms, sensors, OCR and file imports; it must be selected and exchanged appropriately.

THREE PIECES OF THE IDEA

CAPTURE

Choose collection method and validate data at input; validation does not guarantee truth.

SELECT

Filter records using searches, predicates, query-by-example or SQL.

EXCHANGE

Move data as files or over networks using agreed structures such as CSV or JSON.

MODEL / FOLLOW THE FLOW



A joined workflow: a parent submits a form, a database stores valid entries, and an authorised report exports selected records.

EXAM-READY STATEMENT

Validation confirms that input meets specified rules; verification compares data against the intended source.

WORDS THAT EARN MARKS

Validation	Check rules	**Verification**	Check accuracy
QBE	Visual query design	**CSV**	Delimited text

1.3.2(b) / STEP-BY-STEP WORKED EXAMPLE

Safely import 120 student responses

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Capture	Form requires YearGroup as an integer	Use presence and range checks; collect only relevant data.
2. Verify	A valid postcode may still belong to the wrong person	Cross-check against a trusted source when appropriate.
3. Select	Return YearGroup = 12 and Consent = TRUE	Define a reproducible query filter.
4. Export	CSV: rows and fields; JSON: named structures	Choose format based on receiving system.
5. Protect	Control roles and sharing	Authorise access and transfer securely.

TRANSFER IT TO AN EXAM ANSWER

Validation confirms that input meets specified rules; verification compares data against the intended source.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Distinguish validation from verification in a school data-capture system, giving one example of each.

QUESTION 2 | 5 MARKS

Describe a suitable method to collect, select and exchange sensor data with a second application.

1.3.2(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Validation: rule checks, e.g. a permitted YearGroup range.
- Verification: compare value with a known accurate source, e.g. confirm ID against original record.

Q2 / 5 MARKS

- Capture via sensor or interface into structured records.
- Validate required types/ranges and handle missing readings.
- Query records by timestamp or threshold.
- Export in a supported structured format such as CSV/JSON.
- Use protected transfer appropriate to confidentiality.

COMMON MISCONCEPTION

A field passing a range check does not prove that the field's information is correct.

Green pen: Give the rule and the resulting action when invalid data is detected.

1.3.2(d) / BUILD THE MENTAL MODEL

Read the schema before writing the query

SQL retrieves related data by joining tables using matching keys and restricting the returned rows.

THREE PIECES OF THE IDEA

SELECT

Choose output fields. WHERE filters rows.

JOIN

Combine related tables through an explicit matching condition.

ORDER BY

Sort returned rows; use aggregation only when the task asks for summaries.

MODEL / FOLLOW THE FLOW

Student table → **StudentID match** → **Mark table** → **Filtered output**

For Student(StudentID,Name) and Mark(MarkID,StudentID,Score), join on Student.StudentID = Mark.StudentID.

EXAM-READY STATEMENT

SQL: SELECT Student.Name, Mark.Score FROM Student INNER JOIN Mark ON Student.StudentID = Mark.StudentID WHERE Mark.Score >= 70 ORDER BY Mark.Score DESC;

WORDS THAT EARN MARKS

- **SELECT** Columns
- **FROM** Source table
- **JOIN** Combine rows
- **WHERE** Filter rows

1.3.2(d) / STEP-BY-STEP WORKED EXAMPLE

Find learners scoring at least 70

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Schema	Student(StudentID, Name); Mark(MarkID, StudentID, Score)	StudentID in Mark is a foreign key.
2. Link	Student.StudentID = Mark.StudentID	Match each mark to its student.
3. Fields	SELECT Student.Name, Mark.Score	Return only the requested columns.
4. Tables	FROM Student INNER JOIN Mark ON ...	The join identifies matching rows.
5. Filter	WHERE Mark.Score >= 70 ORDER BY Mark.Score DESC	Filter before presenting sorted output.

TRANSFER IT TO AN EXAM ANSWER

SQL: SELECT Student.Name, Mark.Score FROM Student INNER JOIN Mark ON Student.StudentID = Mark.StudentID WHERE Mark.Score >= 70 ORDER BY Mark.Score DESC;

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Write SQL to display student names and scores at least 80 using Student(StudentID,Name) and Result(StudentID,Score).

QUESTION 2 | 4 MARKS

Explain what would go wrong if a student/result join omitted the condition matching StudentID.

1.3.2(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- SELECT Student.Name, Result.Score
- FROM Student INNER JOIN Result
- ON Student.StudentID = Result.StudentID
- WHERE Result.Score >= 80;

Q2 / 4 MARKS

- Without an ON condition or correct join predicate, unrelated rows may be combined.
- This may produce a Cartesian product or otherwise incorrect results.
- Scores could be attributed to the wrong student.

COMMON MISCONCEPTION

Do not join using Name if an appropriate stable StudentID key exists.

Green pen: Before writing SQL, circle the output fields, underline the join keys, then write WHERE.

1.3.2(d,e) / BUILD THE MENTAL MODEL

Modify records without breaking the relationships

INSERT, UPDATE and DELETE change data. Referential integrity restricts foreign keys to valid referenced rows (or permitted nulls).

THREE PIECES OF THE IDEA

INSERT

Add a record; provide required values and respect constraints.

UPDATE

Change specified records; a missing WHERE may affect every row.

DELETE

Remove specified rows; foreign-key relationships may block deletion.

MODEL / FOLLOW THE FLOW

Valid customer PK → **Order customer FK** → **FK constraint** → **Safe update/delete**

A new Order(CustomerID=99) is rejected if CustomerID 99 does not exist and that foreign-key value must reference Customer.

EXAM-READY STATEMENT

Referential integrity prevents dangling references, but the precise response to parent deletion depends on the database's FK policy.

WORDS THAT EARN MARKS

- **Constraint** Data rule
- **Orphan** Broken FK link
- **UPDATE** Modify rows
- **DELETE** Remove rows

1.3.2(d,e) / STEP-BY-STEP WORKED EXAMPLE

Protect customer and order data

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Schema	Customer(CustomerID PK); Orders(OrderID PK, CustomerID FK)	Foreign key references Customer.
2. Insert	INSERT INTO Customer(CustomerID) VALUES (99);	Create parent row before child.
3. Child	INSERT INTO Orders(OrderID,CustomerID) VALUES (501,99);	Now the reference is valid.
4. Update	UPDATE Orders SET CustomerID=99 WHERE OrderID=501;	WHERE limits the change to one order.
5. Delete	DELETE FROM Customer WHERE CustomerID=99;	May be blocked while Order 501 references customer 99 unless cascading rule is configured.

TRANSFER IT TO AN EXAM ANSWER

Referential integrity prevents dangling references, but the precise response to parent deletion depends on the database's FK policy.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(d,e) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

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BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Explain referential integrity using a Customer and Order table, including an invalid foreign-key example.

QUESTION 2 | 5 MARKS

Write an SQL UPDATE to set Quantity to 4 for OrderLine with OrderID=12 and ProductID=7. Explain why both keys matter.

1.3.2(d,e) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Order.CustomerID is a foreign key referring to Customer.CustomerID.
- A non-null reference such as 999 must match an existing Customer key.
- This avoids orphaned order records and inconsistent relationships.

Q2 / 5 MARKS

- UPDATE OrderLine
- SET Quantity = 4
- WHERE OrderID = 12 AND ProductID = 7;
- Both fields identify the intended row when their combination is the composite PK.

COMMON MISCONCEPTION

A foreign-key constraint does not mean every parent row must have a matching child.

Green pen: Say which row is the parent, which is the child and what the FK references.

1.3.2(f) / BUILD THE MENTAL MODEL

Both changes succeed, or neither does

A database transaction is a logical unit of work. ACID properties protect correctness when failures and concurrent users occur.

THREE PIECES OF THE IDEA

ATOMICITY

A transaction is all-or-nothing.

CONSISTENCY

Constraints and valid-state rules remain satisfied.

ISOLATION + DURABILITY

Concurrent transactions do not interfere incorrectly; committed results persist.

MODEL / FOLLOW THE FLOW

Begin transfer → **Debit account A** → **Credit account B** → **Commit or rollback**

One transfer must not debit money without corresponding credit, even if power fails mid-operation.

EXAM-READY STATEMENT

Locking can prevent another transaction changing a record until the first transaction completes, though it can reduce concurrency.

WORDS THAT EARN MARKS

- **Atomicity** All or nothing
- **Consistency** Rules hold
- **Isolation** Concurrent safety
- **Durability** Committed persists

1.3.2(f) / STEP-BY-STEP WORKED EXAMPLE

Transfer 20 between two account records

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Begin	BEGIN TRANSACTION;	Group operations as one unit.
2. Debit	UPDATE Account SET Balance=Balance-20 WHERE ID=1;	Account 1 is reduced.
3. Credit	UPDATE Account SET Balance=Balance+20 WHERE ID=2;	Account 2 gains 20.
4. Check	If any rule/check fails: ROLLBACK;	Atomicity restores initial consistent state.
5. Commit	Otherwise COMMIT;	Durability means committed changes survive a crash; locks/isolation avoid harmful interference.

TRANSFER IT TO AN EXAM ANSWER

Locking can prevent another transaction changing a record until the first transaction completes, though it can reduce concurrency.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(f) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

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BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Explain each ACID property for a bank transfer that moves 30 between two accounts.

QUESTION 2 | 5 MARKS

Two staff members update the last ticket at the same time. Explain how record locking can protect consistency and one disadvantage.

1.3.2(f) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Atomicity: debit and credit both happen or both roll back.
- Consistency: account constraints remain true before/after.
- Isolation: simultaneous transfers do not see inconsistent partial states.
- Durability: committed balances persist after restart.

Q2 / 5 MARKS

- Lock the seat/inventory record while a transaction reserves it.
- Second transaction waits or receives a conflict.
- First commits before the other proceeds.
- Disadvantage: waiting, reduced throughput or possible deadlock.

COMMON MISCONCEPTION

ACID Consistency is about preserved integrity constraints, not a guarantee that every field value is factually true.

Green pen: For a failed transfer, explicitly state which operations are undone and why.

1.3.3(a,b) / BUILD THE MENTAL MODEL

Each layer handles a different responsibility

Protocol layering separates application services, end-to-end transport, addressing and local delivery.

THREE PIECES OF THE IDEA

APPLICATION

HTTP, DNS and other protocols used by application processes.

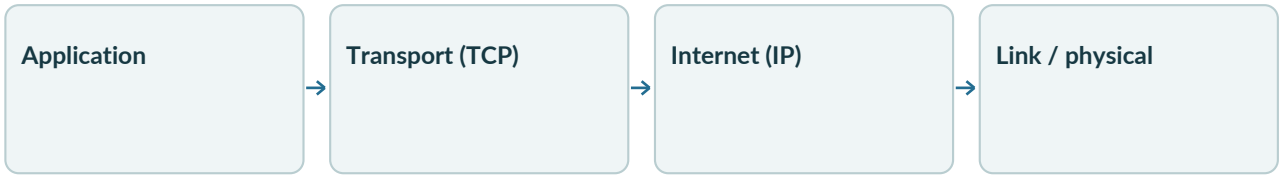
TRANSPORT + INTERNET

TCP provides ordered reliable streams; IP addresses and routes packets.

LINK

Local network frames and physical access such as Ethernet or Wi-Fi.

MODEL / FOLLOW THE FLOW



A four-layer TCP/IP teaching model shows encapsulation; some sources combine or split the bottom layers differently.

EXAM-READY STATEMENT

Protocol layering allows technologies to change at one layer while preserving well-defined interfaces to the others.

WORDS THAT EARN MARKS

- **Protocol** Agreed rules
- **Standard** Common specification
- **Layer** Responsibility
- **Packet** Network data unit

1.3.3(a,b) / STEP-BY-STEP WORKED EXAMPLE

Send a webpage request through the stack

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Application	Browser creates HTTP request	Application determines meaning of the message.
2. Transport	TCP segments stream and tracks delivery	Sequence/acknowledgement permits ordered reliable delivery in this example.
3. Internet	IP adds source/destination addressing	Routers forward towards destination network.
4. Link	NIC wraps packet for local network	Ethernet or Wi-Fi sends bits on next hop.
5. Receiver	Headers processed in reverse order	Application receives usable message content.

TRANSFER IT TO AN EXAM ANSWER

Protocol layering allows technologies to change at one layer while preserving well-defined interfaces to the others.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(a,b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Describe the roles of the application, transport, internet and link layers when a browser requests a webpage.

QUESTION 2 | 4 MARKS

Explain two benefits of protocols and common standards for devices made by different manufacturers.

1.3.3(a,b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Application: HTTP request format/semantics.
- Transport: TCP sequencing, acknowledgements and reliable delivery.
- Internet: IP addressing and route forwarding.
- Link: frames and local media delivery.

Q2 / 4 MARKS

- Devices use the same message formats and conventions.
- Interoperability avoids vendor-specific incompatible formats.
- Layers/components can be improved independently if standard interfaces are retained.

COMMON MISCONCEPTION

The protocol stack is not a literal list of four separate cables. Each layer supplies a service to the layer above.

Green pen: State one concrete responsibility and example per layer.

1.3.3(b) / BUILD THE MENTAL MODEL

The name is not the IP address

DNS resolves a domain name to DNS records including IP addresses. Routers use network addresses to forward packets.

THREE PIECES OF THE IDEA

DOMAIN

Human-readable name, for example school.example.

DNS

Distributed naming service returns records; local caches may speed repeated lookups.

IP + ROUTER

IP address identifies endpoints/interfaces; routers direct packets between networks.

MODEL / FOLLOW THE FLOW

Name entered → **DNS lookup** → **IP returned** → **Packets routed**

DNS maps names to records, not necessarily a permanent one-to-one server. A site may use multiple changing addresses.

EXAM-READY STATEMENT

DNS resolves names; a router forwards IP packets. Neither task alone guarantees encryption of the data being sent.

WORDS THAT EARN MARKS

- **DNS** Name resolution
- **IP** Network address
- **Router** Forwards packets
- **TTL** Cache lifetime

1.3.3(b) / STEP-BY-STEP WORKED EXAMPLE

Follow a request for a named service

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Name	Browser requests library.example	The URL host is a domain name.
2. Cache	Check cached DNS records	A valid cached answer may avoid a new lookup.
3. Resolve	DNS service returns address record	Result can be one of several possible IP addresses.
4. Connect	Client sends IP packets	Source/destination IPs support routing.
5. Route	Routers select next hops	Each router forwards based on its routing information.

TRANSFER IT TO AN EXAM ANSWER

DNS resolves names; a router forwards IP packets. Neither task alone guarantees encryption of the data being sent.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Explain what happens between entering a domain name and sending packets to a remote web server.

QUESTION 2 | 4 MARKS

Explain why DNS caching is useful and one reason a cached record might need refreshing.

1.3.3(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Client identifies hostname in the URL.
- Cached DNS answer may be used, otherwise resolver performs lookup.
- An address record is returned.
- Client sends IP packets addressed to the server.
- Routers forward packets towards destination.

Q2 / 4 MARKS

- Cache reduces lookup delay and query traffic.
- Records can change when services move or are load-balanced.
- TTL/expiry encourages refreshing stale values.

COMMON MISCONCEPTION

A hostname can map to several IP addresses; do not claim every website has exactly one fixed IP.

Green pen: Separate the naming question (DNS) from the delivery question (IP and routers).

1.3.3(b) / BUILD THE MENTAL MODEL

Shared routes versus reserved capacity

Packet switching divides a message into independently forwarded packets. Circuit switching reserves a route or capacity for a session.

THREE PIECES OF THE IDEA

PACKET

Shared network capacity; packets may take different paths and arrive out of order.

CIRCUIT

A dedicated path or reserved resources for the duration of communication.

CHOOSE

Bursty data favours efficient packet sharing; predictable reserved capacity may suit continuous real-time channels.

MODEL / FOLLOW THE FLOW

Message → **Packets** → **Shared network** → **Reassembly**

A file is split into packet payloads, routed over the internet, and reconstructed using higher-layer mechanisms.

EXAM-READY STATEMENT

Packet switching alone does not guarantee arrival in order; protocol mechanisms can provide that service.

WORDS THAT EARN MARKS

Packet	Routed unit	**Header**	Control details
Circuit	Reserved path	**Latency**	Transfer delay

1.3.3(b) / STEP-BY-STEP WORKED EXAMPLE

Send a 3 KB message as three 1 KB payloads

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Divide	P1, P2, P3 carry the data	Each packet gets addressing/metadata overhead.
2. Route	P1 via A; P2 via B; P3 via A	Shared network routes may differ.
3. Arrive	P2, P1, P3	Higher layers can reorder when required; packets may be lost.
4. Recover	TCP retransmits missing data in its stream	Reliability depends on protocols, not packet switching alone.
5. Compare	Reserved circuit stays allocated	Continuous predictable service but unused capacity can be wasted.

TRANSFER IT TO AN EXAM ANSWER

Packet switching alone does not guarantee arrival in order; protocol mechanisms can provide that service.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Explain why packet switching is appropriate for browsing but may suffer variable delay.

QUESTION 2 | 4 MARKS

Compare the benefits and disadvantages of packet switching and circuit switching for a fixed voice service.

1.3.3(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Traffic is bursty so many users share capacity.
- Packets are forwarded independently and can use available routes.
- Congestion, queueing and different paths introduce variable latency/jitter.
- Higher-layer protocols can provide reliability when needed.

Q2 / 4 MARKS

- Circuit reserves resources so delay/capacity can be predictable.
- Reserved capacity can be wasted during silence.
- Packet use is efficient but contention can cause delay/jitter.

COMMON MISCONCEPTION

Do not claim a circuit is established for ordinary IP packet switching.

Green pen: Link the network choice to occupancy, reliability and predictability requirements.

1.3.3(c) / BUILD THE MENTAL MODEL

One control cannot stop every threat

Security defences should match threats such as eavesdropping, malware, phishing, denial of service and SQL injection.

THREE PIECES OF THE IDEA

FIREWALL

Apply traffic rules; can block unwanted inbound or outbound connections.

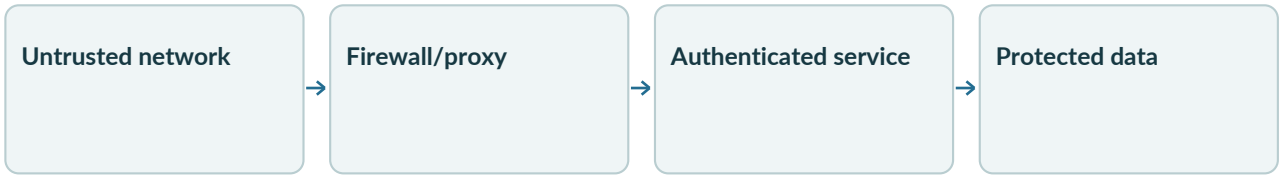
PROXY

Intermediary can filter requests, hide internal systems or cache content.

ENCRYPTION

Protects data confidentiality in transit when configured correctly; does not prevent all attacks.

MODEL / FOLLOW THE FLOW



Layered controls include user authentication, updates, backups and monitoring as well as perimeter devices.

EXAM-READY STATEMENT

A firewall filters traffic; it does not automatically remove malware already running or make application SQL safe.

WORDS THAT EARN MARKS

- **Phishing** Deceptive message
- **DoS** Disrupt service
- **TLS** Secured transport
- **Injection** Unsafe query input

1.3.3(c) / STEP-BY-STEP WORKED EXAMPLE

Protect a school results portal

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Threat	Attacker intercepts logins on public Wi-Fi	Use HTTPS/TLS so traffic is encrypted in transit.
2. Access	Attacker guesses a staff password	Require MFA and rate limiting; train staff.
3. Injection	User enters malicious SQL into a search box	Use parameterised queries and input validation.
4. Flood	Many requests exhaust server capacity	Rate limiting, filtering and DDoS protection may help.
5. Recover	Ransomware damages local files	Use tested backups and controlled permissions.

TRANSFER IT TO AN EXAM ANSWER

A firewall filters traffic; it does not automatically remove malware already running or make application SQL safe.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

A results portal suffers interception attempts and SQL injection. Recommend separate controls and explain each.

QUESTION 2 | 5 MARKS

Compare a firewall, proxy and encryption: state their distinct purposes and one limitation of each.

1.3.3(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Use TLS/HTTPS to protect data from passive interception.
- Use parameterised SQL to separate input values from query structure.
- Access controls/MFA further protect sensitive results.

Q2 / 5 MARKS

- Firewall controls traffic using rules but cannot stop all application attacks.
- Proxy intermediates/filters requests but is not inherently encryption.
- Encryption protects confidentiality but not availability or compromised endpoints.

COMMON MISCONCEPTION

Encryption does not protect against a legitimate user being fooled into entering credentials on a phishing website.

Green pen: Pair each named threat with a mechanism that blocks or reduces it.

1.3.3(d) / BUILD THE MENTAL MODEL

Choose the device by what it forwards

Devices operate at different levels: hosts need interfaces; local networks need switching; separate networks need routing.

THREE PIECES OF THE IDEA

NIC + SWITCH

NIC connects a host; a switch forwards frames within a LAN using MAC addresses.

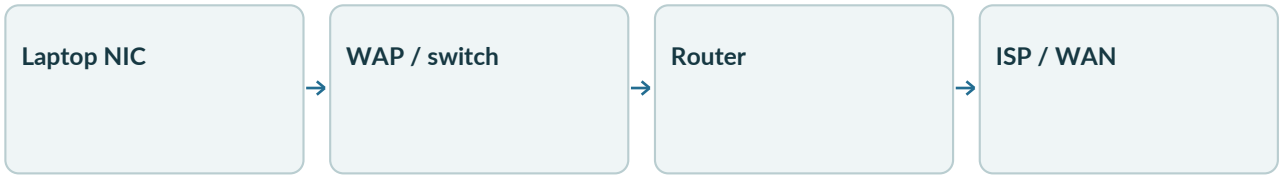
ROUTER

Forwards packets between IP networks using routing information.

WAP + MODEM

Wireless access point bridges Wi-Fi clients into a LAN; a modem adapts a link to an access technology.

MODEL / FOLLOW THE FLOW



A laptop joins Wi-Fi via an access point, local traffic uses switching, and a router sends traffic to remote networks.

EXAM-READY STATEMENT

A switch normally handles local Ethernet forwarding; a router connects distinct IP networks.

WORDS THAT EARN MARKS

- **NIC** Host interface
- **Switch** LAN frames
- **Router** Between networks
- **WAN** Wider network

1.3.3(d) / STEP-BY-STEP WORKED EXAMPLE

Design a school wired/wireless layout

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Devices	Desktop NICs connect using Ethernet	Wired endpoints plug into an access switch.
2. Wi-Fi	Mobile devices join the WAP	Wireless clients are bridged to the local network.
3. LAN	Switch sends local frames appropriately	MAC learning reduces unnecessary forwarding.
4. Exit	Router connects LAN to another IP network	Packets for outside subnets are forwarded.
5. WAN	ISP link reaches distant networks	The WAN connects geographically separated networks.

TRANSFER IT TO AN EXAM ANSWER

A switch normally handles local Ethernet forwarding; a router connects distinct IP networks.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

A school expands to two buildings with Wi-Fi and an internet connection. Name suitable network hardware and justify each.

QUESTION 2 | 4 MARKS

Explain the difference between LAN and WAN using ownership and geographic scope, with an example.

1.3.3(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- NICs provide host connections.
- Switches connect wired LAN endpoints.
- WAPs provide wireless access.
- Router forwards between school network and internet/WAN.
- Appropriate medium/links connect buildings.

Q2 / 4 MARKS

- LAN covers a local site/campus, often managed by one organisation.
- WAN spans larger geographic regions and connects LANs, often across provider links.
- Example: school LAN versus school-to-trust WAN.

COMMON MISCONCEPTION

A Wi-Fi access point is not necessarily the router, although home equipment often combines both functions.

Green pen: Name the required movement: host-to-LAN, LAN-to-LAN or LAN-to-WAN.

1.3.3(e) / BUILD THE MENTAL MODEL

Where does responsibility for a service sit?

Client-server networks rely on servers to provide services to requesting clients. Peer-to-peer networks let nodes provide services directly to one another.

THREE PIECES OF THE IDEA

CLIENT-SERVER

Centralised services, accounts, security policy and backup are easier to manage.

PEER-TO-PEER

Devices act as both service consumers and providers; may avoid a central server.

TRADE-OFF

Consider administration, resilience, expense, scalability and control of data.

MODEL / FOLLOW THE FLOW



In a school, a central file server controls access; on a simple peer-share network, computers share files directly.

EXAM-READY STATEMENT

Client-server is a logical relationship, not a rule that every network needs one physical central computer.

WORDS THAT EARN MARKS

Client	Requests service	**Server**	Provides service
Peer	Equal network node	**Centralise**	One control point

1.3.3(e) / STEP-BY-STEP WORKED EXAMPLE

Choose an architecture for two scenarios

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Exam files	100 users need audited access	Dedicated file server supports permissions and backup.
2. Updates	Teacher changes a shared document	Central server holds authoritative version.
3. Small group	Three laptops share temporary files	Peer-to-peer can work with little setup.
4. Availability	Server fails at exam time	Client-server needs reliable infrastructure/recovery.
5. Security	Peer machine is compromised	Distributed data sharing is harder to administer consistently.

TRANSFER IT TO AN EXAM ANSWER

Client-server is a logical relationship, not a rule that every network needs one physical central computer.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(e) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Recommend an architecture for a 500-user school needing controlled student file access and justify its limitations.

QUESTION 2 | 4 MARKS

Explain two differences between peer-to-peer and client-server networks with real implications.

1.3.3(e) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Use client-server for central authentication/access controls.
- Central backups, audit logs and consistent administration suit the scale.
- Costs include dedicated infrastructure and managing server failures.

Q2 / 4 MARKS

- Client-server requests go to a providing server; peer nodes both request and provide resources.
- Central management simplifies backups and access control.
- P2P can be simpler/cheaper but harder to secure consistently.

COMMON MISCONCEPTION

Peer-to-peer does not mean “there is no security”; access can be configured, but it is decentralised.

Green pen: Choose the architecture by citing user numbers, control needs and failure consequences.

1.3.4(a) / BUILD THE MENTAL MODEL

Structure, presentation, interaction

HTML defines document content and structure, CSS styles it and JavaScript can update content and respond to user input.

THREE PIECES OF THE IDEA

HTML

Elements such as headings, paragraphs, links and forms define the content structure.

CSS

Selectors and declarations control colour, spacing, layout and fonts.

JAVASCRIPT

Code can read inputs, respond to events and modify the DOM.

MODEL / FOLLOW THE FLOW

HTML element → **CSS selector** → **JS event** → **Updated webpage**

Concept: an HTML paragraph with id="msg" gets CSS style and is changed by a JavaScript event listener.

EXAM-READY STATEMENT

HTML does not execute the JavaScript just because a tag is present; the browser runs script code in the appropriate context.

WORDS THAT EARN MARKS

- **Tag** Markup syntax
- **Selector** CSS target
- **DOM** Page object tree
- **Event** User action

1.3.4(a) / STEP-BY-STEP WORKED EXAMPLE

Interpret a small page with a button

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. HTML	Ready	Creates a paragraph with identifier msg.
2. HTML	<button id="go">Start</button>	Creates an interactive button.
3. CSS	#msg { color: navy; }	Changes the paragraph's visual presentation.
4. JS	document.getElementById("go").onclick = function(){...}	Registers behaviour to run on clicking.
5. JS body	document.getElementById("msg").textContent = "Go";	Changes text shown in the existing paragraph.

TRANSFER IT TO AN EXAM ANSWER

HTML does not execute the JavaScript just because a tag is present; the browser runs script code in the appropriate context.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.4(a) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Explain how HTML, CSS and JavaScript work together to display and update a paragraph after clicking a button.

QUESTION 2 | 5 MARKS

A page has 0 . Give CSS to make text blue and one JavaScript statement to set the text to 10.

1.3.4(a) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- HTML paragraph/button defines structure and identifiers.
- CSS selects element and changes appearance.
- JavaScript attaches event handler to button.
- When clicked, handler updates DOM text content.

Q2 / 5 MARKS

- CSS: #score { color: blue; }
- JS: document.getElementById("score").textContent = "10";
- Identifier spelling/case must match the HTML.

COMMON MISCONCEPTION

CSS changes presentation, not the underlying meaning of an HTML element; JavaScript is not the same as Java.

Green pen: Point to the element, style rule, event and output separately.

1.3.4(b,c) / BUILD THE MENTAL MODEL

Discover pages; index content; rank importance

A search engine uses crawlers to discover pages, builds an index and then orders results using ranking signals. PageRank measures link-based importance.

THREE PIECES OF THE IDEA

CRAWL

Automated crawler follows links and discovers pages.

INDEX

Extract terms and metadata into a searchable structure.

RANK

Rank candidates using relevance and link-based authority; PageRank treats inbound links as weighted votes.

MODEL / FOLLOW THE FLOW

Web crawler → **Content index** → **Matching pages** → **Ranking order**

Page A receiving a link from an important page can earn more link importance than a link from an unimportant page.

EXAM-READY STATEMENT

PageRank is about the quantity and quality of inbound links, not a count of how many times a keyword appears.

WORDS THAT EARN MARKS

- **Crawler** Discovers pages
- **Index** Searchable entries
- **Inbound** Link pointing in
- **Damping** Random-jump effect

1.3.4(b,c) / STEP-BY-STEP WORKED EXAMPLE

Compare links in a tiny web graph

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Pages	A, B and C are web pages	Create directed arrows for hyperlinks.
2. Links	B -> A; C -> A; A -> B	A receives two inbound links; B receives one.
3. Quality	C is itself authoritative	A link from C may pass more weight than a weak page.
4. Share	A divides its outgoing rank across links	The share depends on the number of outgoing links.
5. Damping	Allow a small chance to jump to any page	Reduces traps/dead ends and supports stable scores.

TRANSFER IT TO AN EXAM ANSWER

PageRank is about the quantity and quality of inbound links, not a count of how many times a keyword appears.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.4(b,c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Outline how a search engine discovers, indexes and ranks a new page, including PageRank.

QUESTION 2 | 4 MARKS

Page X has ten inbound links from low-quality pages; Y has one from an authoritative page. Explain why Y may rank higher on PageRank.

1.3.4(b,c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Crawler follows links/discovers page.
- Indexer stores processed content and searchable terms.
- Query retrieves candidate documents.
- Ranking combines evidence; PageRank uses the importance of linking pages.

Q2 / 4 MARKS

- Inbound links act as weighted endorsements.
- Importance passed by a high-rank page can outweigh several weak links.
- Rank is also split across the linking page's outgoing links.

COMMON MISCONCEPTION

PageRank is not the same as the entire search-engine ranking algorithm; relevance and other signals also matter.

Green pen: When explaining a ranking, trace the direction of each hyperlink arrow.

1.3.4(d) / BUILD THE MENTAL MODEL

Run it close to the user or beside the data?

Client-side code executes in the browser; server-side code executes on the server. Most interactive sites use both.

THREE PIECES OF THE IDEA

CLIENT

Fast local feedback, page interaction and initial field validation.

SERVER

Handles trusted rules, database queries, sessions and private data.

SECURITY

Never trust browser validation alone: it can be bypassed. Recheck on the server.

MODEL / FOLLOW THE FLOW

Browser enters → **Client validates** → **Server verifies** → **Database query**

Shopping checkout: JS flags an empty postcode locally; server validates price/stock before writing an order.

EXAM-READY STATEMENT

Client-side checks are for convenience; essential security, permissions and consistency must be enforced server-side.

WORDS THAT EARN MARKS

Client Browser device

Server Remote service

Validation Rule check

Trust boundary Untrusted input

1.3.4(d) / STEP-BY-STEP WORKED EXAMPLE

Process an online order safely

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Form	Customer enters delivery address	HTML fields collect user input.
2. Client JS	Show missing postcode immediately	Reduces waiting; improves usability.
3. Submit	Browser sends request to server	Client data must be treated as untrusted.
4. Server	Check identity, stock and price	Authoritative rules execute at protected server.
5. Store	Server transaction saves the order	Database operation happens outside the browser.

TRANSFER IT TO AN EXAM ANSWER

Client-side checks are for convenience; essential security, permissions and consistency must be enforced server-side.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.4(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Compare client-side and server-side processing for an online shop, giving two tasks for each and a limitation.

QUESTION 2 | 5 MARKS

Explain why a JavaScript check that a price field is positive is not enough to secure an online purchase.

1.3.4(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Client JS can validate form format and update page instantly.
- Server checks stock, authorisation and writes database changes.
- Client can be manipulated or disabled; server has processing/network costs.

Q2 / 5 MARKS

- Client-side code/input can be modified or bypassed.
- An attacker can send a crafted request without using the original form.
- Server must validate price/permissions and perform authoritative calculations.
- Transaction checks and safe database access protect integrity.

COMMON MISCONCEPTION

JavaScript can run on a server too; “JavaScript” alone does not prove execution location. Say where the code runs.

Green pen: Label each stage browser or server and name the reason for that choice.

1.4 / TEACHER-REVIEW WORKING CHAPTER

Understand it. Model it. Try it. Check it.

Standalone revision upgrade for H446/01 section 1.4. It does not replace the current published 264-page guide or alter existing page links.

THREE SPECIFICATION FAMILIES

1.4.1 DATA TYPES + REPRESENTATION	Unsigned and signed binary, floating point, masks and Unicode
1.4.2 DATA STRUCTURES	Arrays, records, linked structures, trees, graphs and hashing
1.4.3 BOOLEAN ALGEBRA	Truth tables, K-maps, circuits, adders and flip-flops

FOUR PASSES FOR EVERY TOPIC

01 UNDERSTAND
Build a precise conceptual picture.

02 MODEL
Follow an explicit trace or diagram.

03 TRY
Try original exam-style questions.

04 CHECK
Self-mark with separate mark points.

RELEASE SAFETY

Teacher-review candidate only. Preserve and reconcile the published H446 guide before release. Live lessons, revision questions, page maps and bookmarks are not changed by this PDF.

Scope: 24 cycles x 4 pages plus 2 orientation pages = 98 pages. These are replacement/enhancement candidates, not pages to append to the published guide without editorial review.

1.4 / SPECIFICATION CHECKLIST

Every 1.4 topic has a working model

Select a linked row to jump to its Understand page. The following pages contain a model, exam-style practice and indicative answer points.

SPEC	LEARNING CYCLE	PAGES
1.4.1(a)	Primitive data types	3-6
1.4.1(b,e,f)	Unsigned binary and hexadecimal	7-10
1.4.1(c)	Negative binary values	11-14
1.4.1(d)	Binary addition and subtraction	15-18
1.4.1(g)	Normalised binary floating point	19-22
1.4.1(h)	Signed floating-point arithmetic	23-26
1.4.1(i)	Bitwise logic and shifts	27-30
1.4.1(j)	ASCII and Unicode	31-34
1.4.2(a)	Arrays, records, lists and tuples	35-38
1.4.2(b,c)	Stacks: push, pop and peek	39-42
1.4.2(b,c)	Queues and circular buffers	43-46
1.4.2(b,c)	Linked-list operations	47-50
1.4.2(b,c)	Trees and binary search trees	51-54
1.4.2(c)	Tree traversal and deletion	55-58
1.4.2(b,c)	Graphs and adjacency structures	59-62
1.4.2(b,c)	Hash tables and collisions	63-66
1.4.3(a,d)	Boolean problems and truth tables	67-70
1.4.3(d)	Logic gates and combined circuits	71-74
1.4.3(c)	Boolean laws and De Morgan	75-78
1.4.3(b)	Three-variable Karnaugh maps	79-82
1.4.3(b)	Four-variable K-maps	83-86
1.4.3(e)	Half adders	87-90
1.4.3(e)	Full adders	91-94
1.4.3(e)	D-type flip-flops	95-98

SPECIFICATION BOUNDARY

1.4.1(a-j), 1.4.2(a-c), and 1.4.3(a-e) from OCR H446 specification v2.7. Models state their representation/indexing conventions.

Source: OCR H446 A Level Computer Science specification v2.7 (2024), section 1.4; compared against current RHSC guide section map. All practice questions and explanations in this chapter are teacher-review drafts, not OCR past-paper material.

1.4.1(a) / BUILD THE MENTAL MODEL

Choose a representation that fits the value

An appropriate data type constrains the values and operations of a variable. Data types are not all stored as simple whole numbers.

THREE PIECES OF THE IDEA

INTEGER

Whole-number values, positive or negative, subject to a chosen range.

REAL / FLOAT

Approximates values with fractional parts using a limited number of bits.

OTHER PRIMITIVES

Boolean holds true/false; character is one symbol; string holds a sequence of characters.

MODEL / FOLLOW THE FLOW

Problem value → **Select data type** → **Check valid range** → **Choose operations**

Number of students: integer. Temperature 18.5: real. Permission granted: Boolean. Initial: character. Full surname: string.

EXAM-READY STATEMENT

Selection depends on the meaning and valid operations of the value, not merely whether quotation marks appear in pseudocode.

WORDS THAT EARN MARKS

Integer Whole number

Real Approximate fraction

Boolean True or false

Character Single symbol

1.4.1(a) / STEP-BY-STEP WORKED EXAMPLE

Declare variables for a library loan

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Count	copies = 12 / integer	You cannot have a fraction of a physical copy.
2. Price	fine = 1.25 / real	It includes a fractional component.
3. Flag	overdue = TRUE / Boolean	Only two logical states are needed.
4. Code	initial = "K" / character	Exactly one character is stored.
5. Text	title = "Data Structures" / string	A string is an ordered sequence of characters.

TRANSFER IT TO AN EXAM ANSWER

Selection depends on the meaning and valid operations of the value, not merely whether quotation marks appear in pseudocode.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.1(a) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Choose a type for pupilCount, temperature, hasPaid, surname and firstInitial. Justify two choices.

QUESTION 2 | 4 MARKS

Explain why a real/floating-point value may be inappropriate for comparing financial totals exactly.

1.4.1(a) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Integer, real, Boolean, string and character respectively.
- Count uses whole numbers; hasPaid has two possible states.

Q2 / 4 MARKS

- Many decimal fractions cannot be represented exactly in binary floating point.
- Small representation/rounding errors can occur.
- A suitable fixed-point/integer-minor-unit representation can preserve exact pence.

COMMON MISCONCEPTION

A floating-point value can be approximate even when it appears with only one or two decimal places.

Green pen: Always attach the example value to the reason for selecting the type.

1.4.1(b,e,f) / BUILD THE MENTAL MODEL

Move between bases without losing place value

Binary uses powers of 2; hexadecimal uses powers of 16. One hex digit maps exactly to four binary bits.

THREE PIECES OF THE IDEA

BINARY

Read bits from right to left as 1, 2, 4, 8, 16, 32 and so on.

HEXADECIMAL

Digits 0-9 and A-F represent values 0-15.

NIBBLE

Group a binary word from the right into four-bit groups, padding on the left if needed.

MODEL / FOLLOW THE FLOW

Denary 173 → **Binary 10101101** → **Nibbles 1010 1101** → **Hex AD**

173 = 128 + 32 + 8 + 4 + 1; binary 10101101; 1010 is A and 1101 is D.

EXAM-READY STATEMENT

Write out place values for denary conversions. Use exact nibble mapping for binary/hex conversions.

WORDS THAT EARN MARKS

- **Base** Radix
- **Bit** Binary digit
- **Nibble** 4 bits
- **Place value** Power of the base

1.4.1(b,e,f) / STEP-BY-STEP WORKED EXAMPLE

Convert 0x3C to binary and denary

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Hex digits	3 C	C is 12 in denary.
2. Nibble 1	3 becomes 0011	Each hex digit is exactly four bits.
3. Nibble 2	C becomes 1100	12 = 8 + 4.
4. Join	00111100	Maintain eight positions.
5. Denary	3 x 16 + 12 = 60	Also 32 + 16 + 8 + 4 = 60.

TRANSFER IT TO AN EXAM ANSWER

Write out place values for denary conversions. Use exact nibble mapping for binary/hex conversions.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.1(b,e,f) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Convert binary 11010110 to hexadecimal and denary. Show working.

QUESTION 2 | 4 MARKS

Convert denary 157 into 8-bit binary and hexadecimal. Show the place values.

1.4.1(b,e,f) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- 1101 0110 = D6.
- 128 + 64 + 16 + 4 + 2 = 214.

Q2 / 4 MARKS

- 157 = 128 + 16 + 8 + 4 + 1.
- 10011101 = 9D hexadecimal.

COMMON MISCONCEPTION

Do not map individual bits to hex digits; each hex digit replaces exactly four bits.

Green pen: Write the nibble grouping line explicitly before converting.

1.4.1(c) / BUILD THE MENTAL MODEL

Sign-and-magnitude is not two’s complement

Two different signed representations can give different bit patterns for the same negative value. State the width of the word.

THREE PIECES OF THE IDEA

SIGN / MAGNITUDE

Most significant bit gives sign; remaining bits are the magnitude.

TWO'S COMPLEMENT

Invert an 8-bit positive representation and add 1; one zero pattern exists.

RANGE

8-bit two's complement: -128 to +127. Sign/magnitude: -127 to +127 with two zeros.

MODEL / FOLLOW THE FLOW

+13 = 00001101 → **Invert 11110010** → **Add 1 = 11110011** → **Result = -13**

In 8 bits, sign/magnitude -13 is 10001101, but two's-complement -13 is 11110011.

EXAM-READY STATEMENT

Specify word size, representation and the conversion method; two's complement is not simply setting the top bit.

WORDS THAT EARN MARKS

- **MSB** Sign position
- **Magnitude** Absolute value
- **Invert** Flip all bits
- **Overflow** Out of range

1.4.1(c) / STEP-BY-STEP WORKED EXAMPLE

Represent -23 in both 8-bit conventions

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Positive	+23 = 00010111	16 + 4 + 2 + 1.
2. Sign/mag	10010111	Set sign bit; retain 7-bit magnitude.
3. Invert	11101000	Flip all eight bits of +23.
4. Add one	11101001	This encodes -23 in two's complement.
5. Check	256 - 233 = 23	Unsigned value 233 represents -23 in 8 bits.

TRANSFER IT TO AN EXAM ANSWER

Specify word size, representation and the conversion method; two's complement is not simply setting the top bit.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.1(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Write -42 in 8-bit sign-and-magnitude and 8-bit two's complement. Show the invert/add-one step.

QUESTION 2 | 4 MARKS

State the range of 8-bit two's-complement values and explain why there is only one zero.

1.4.1(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- +42 = 00101010. Sign/magnitude = 10101010.
- Invert -> 11010101; add 1 -> 11010110.

Q2 / 4 MARKS

- -128 through +127 inclusive.
- Zero is 00000000; inversion and adding one wraps zero back to the same pattern.
- There is no separate negative-zero representation.

COMMON MISCONCEPTION

Setting the sign bit in a two's-complement number generally changes its magnitude.

Green pen: Check the decoded value using negative MSB weight or invert/add-one.

1.4.1(d) / BUILD THE MENTAL MODEL

Use fixed width and detect overflow

Binary arithmetic uses carries. In fixed-width two's complement, an out-of-range result wraps; overflow must be recognised.

THREE PIECES OF THE IDEA

ADDITION

0+0=0, 0+1=1, 1+1=10, 1+1+1=11 in binary.

SUBTRACTION

A - B can be performed as A plus the two's complement of B.

OVERFLOW

Two same-sign operands giving the opposite-sign result indicates signed overflow.

MODEL / FOLLOW THE FLOW

13 = 00001101 → **-5 = 11111011** → **Add = 00001000** → **Answer +8**

The ninth carry bit is discarded in an eight-bit register; 00001000 equals +8.

EXAM-READY STATEMENT

A carry out is not the same test as signed overflow. Check operand signs and result sign.

WORDS THAT EARN MARKS

Carry	Column overflow	**Width**	Number of bits
Wrap	Discard excess bit	**Signed**	Two's complement

1.4.1(d) / STEP-BY-STEP WORKED EXAMPLE

Calculate 19 - 7 as 8-bit addition

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. A	19 = 00010011	An eight-bit positive operand.
2. +7	00000111	Write out the subtrahend.
3. Negate	11111001	Invert 00000111 -> 11111000; add 1.
4. Add	00010011 + 11111001 = 1 00001100	Discard the ninth carry in fixed width.
5. Decode	00001100 = +12	19 - 7 = 12; no signed overflow.

TRANSFER IT TO AN EXAM ANSWER

A carry out is not the same test as signed overflow. Check operand signs and result sign.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.1(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Use eight-bit two's complement to calculate 14 - 9, displaying the bit patterns.

QUESTION 2 | 4 MARKS

Explain why 01111111 + 00000001 causes signed overflow in eight bits.

1.4.1(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- 14 = 00001110; +9 = 00001001; -9 = 11110111.
- 00001110 + 11110111 = 1 00000101 -> 00000101 = +5.

Q2 / 4 MARKS

- 01111111 is +127 and 00000001 is +1.
- Expected +128 is outside [-128,127].
- Stored result 10000000 decodes as -128: two positives produce a negative.

COMMON MISCONCEPTION

A carry from the most significant bit alone does not determine signed overflow.

Green pen: Check expected mathematical result against range and signs.

1.4.1(g) / BUILD THE MENTAL MODEL

Treat mantissa as a signed binary fraction

For these worked examples the format is an 8-bit two's-complement fractional mantissa followed by a 4-bit signed exponent.

THREE PIECES OF THE IDEA

VALUE

Number = mantissa x 2 raised to exponent; the binary point sits immediately after the sign bit.

NORMALISE

Positive mantissa begins 01; negative mantissa begins 10 (except zero).

PRECISION

Mantissa length limits precision; exponent range limits scale. Truncation/rounding can lose bits.

WORKED VISUAL / NOT TO SCALE

SIGN + FRACTION (8 BITS)

0	1	1	0	1	0	0	0

EXPONENT 4 BITS

0	0	1	1

0.1101000 x 2^3 = 0.8125 x 8 = 6.5

EXAM-READY STATEMENT

Before any calculation, write the mantissa/exponent widths and the location of the binary point.

WORDS THAT EARN MARKS

- **Mantissa** Significant bits
- **Exponent** Power of two
- **Normalised** Leading 01 or 10
- **Precision** Representable detail

1.4.1(g) / STEP-BY-STEP WORKED EXAMPLE

Encode +6.5 and -6.5 in the 8+4 scheme

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Convert	6.5 = 110.1 binary	110 base 2 = 6; .1 base 2 = 0.5.
2. Normalise	+0.1101 x 2^3	Positive starts 01 when the sign bit is included.
3. Encode +	01101000 0011	8-bit mantissa, 4-bit exponent +3.
4. Encode -	10011000 0011	Two's-complement mantissa of -0.8125.
5. Check	-104/128 x 8 = -6.5	Negative mantissa begins 10.

TRANSFER IT TO AN EXAM ANSWER

Before any calculation, write the mantissa/exponent widths and the location of the binary point.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.1(g) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

In an 8-bit signed fractional mantissa and 4-bit signed exponent format, represent +5 using a normalised mantissa.

QUESTION 2 | 4 MARKS

Explain the difference between the precision and range of floating-point representations.

1.4.1(g) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- 5 = 101 binary = 0.101 x 2^3.
- Mantissa 01010000, exponent 0011; value 0.625 x 8 = 5.

Q2 / 4 MARKS

- More mantissa bits provide more significant precision.
- More exponent bits enable a wider range of magnitudes.
- Finite bits still cause rounding/representation error.

COMMON MISCONCEPTION

Do not interpret the mantissa as an ordinary unsigned integer. It is a signed binary fraction.

Green pen: State the convention and calculate the mantissa decimal fraction to check.

1.4.1(h) / BUILD THE MENTAL MODEL

Align exponent, operate on mantissas, normalise

To add or subtract binary floating-point values, align exponents first, then operate on signed mantissas and normalise.

THREE PIECES OF THE IDEA

ALIGN

Shift the mantissa of the smaller exponent right until exponents match.

OPERATE

Add signed mantissas; for subtraction, add a negated mantissa.

RENORMALISE

Shift mantissa and change exponent in opposite directions; lost low bits can reduce precision.

MODEL / FOLLOW THE FLOW

2 = .5 x 2^2 → **-1 = -.5 x 2^1** → **Align -1 to -.25 x 2^2** → **Result .25 x 2^2 = 1**

Intermediate 8-bit mantissa 00100000 is not normalised; shift left to 01000000 and reduce exponent from 2 to 1.

EXAM-READY STATEMENT

Keep mantissa and exponent changes together. When adding positive values, reserve an extra bit if a mantissa sum overflows.

WORDS THAT EARN MARKS

- **Align** Same exponent
- **Shift** Move binary point
- **Renormalise** Restore 01 or 10
- **Rounding** Precision loss

1.4.1(h) / STEP-BY-STEP WORKED EXAMPLE

Use 8+4 signed-fraction format to compute 2 - 1

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Encode +2	01000000 0010	0.5 x 2^2 = 2.
2. Encode -1	11000000 0001	-0.5 x 2^1 = -1.
3. Align -1	11100000 0010	-0.25 x 2^2 = -1.
4. Add	00100000 0010	0.25 x 2^2 = 1; mantissa starts 00.
5. Normalise	01000000 0001	Shift left one and decrease exponent to 1.

TRANSFER IT TO AN EXAM ANSWER

Keep mantissa and exponent changes together. When adding positive values, reserve an extra bit if a mantissa sum overflows.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.1(h) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Show the aligned mantissas and normalised result for +3 plus -1 using the stated 8+4 signed-fraction format.

QUESTION 2 | 4 MARKS

Explain why aligning exponents before addition can reduce precision.

1.4.1(h) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- +3: 01100000 0010 (0.75 x 4).
- -1: 11000000 0001; align to 11100000 0010 (-0.25 x 4).
- Add = 01000000 0010: 0.5 x 4 = +2, normalised.

Q2 / 4 MARKS

- Right-shifting the smaller-exponent mantissa discards low-order bits if width is fixed.
- Once discarded, those bits cannot be recovered by normalisation.

COMMON MISCONCEPTION

A binary string that starts 00 or 11 is not a normalised nonzero mantissa in this convention.

Green pen: Write each exponent beside its mantissa; verify the final denary value.

1.4.1(i) / BUILD THE MENTAL MODEL

A mask edits selected bit positions

Bitwise AND, OR and XOR operate independently on corresponding bits. Logical shifts move bits and fill newly empty positions with zeros.

THREE PIECES OF THE IDEA

AND MASK

AND with 1 preserves a bit; AND with 0 clears it.

OR / XOR

OR with 1 sets a bit. XOR with 1 toggles a bit.

SHIFTS

Logical left/right shifts move positions; bits shifted off the fixed width are lost.

MODEL / FOLLOW THE FLOW

Value 10110110 → **AND mask 00001111** → **Output 00000110** → **Keep low nibble**

A mask must be aligned to the same bit positions as the value being modified.

EXAM-READY STATEMENT

For fixed-width logical shifts, label direction, inserted zeros, discarded bits and chosen bit width.

WORDS THAT EARN MARKS

- **AND** Both bits 1
- **OR** At least one 1
- **XOR** Exactly one 1
- **Mask** Bit selector

1.4.1(i) / STEP-BY-STEP WORKED EXAMPLE

Modify the 8-bit value 10110110

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Clear upper	AND 00001111 -> 00000110	Zeros clear the high nibble.
2. Set bit 0	OR 00000001 -> 10110111	One forces the low bit on.
3. Toggle all	XOR 11111111 -> 01001001	XOR with one flips each bit.
4. Shift right	10110110 logical >> 2 = 00101101	Insert zeros at the left.
5. Check width	Discard bits shifted beyond 8 places	Do not extend the register indefinitely.

TRANSFER IT TO AN EXAM ANSWER

For fixed-width logical shifts, label direction, inserted zeros, discarded bits and chosen bit width.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.1(i) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Starting at 11010101, apply AND 11110000 then OR 00000011. Show both intermediate words.

QUESTION 2 | 4 MARKS

Explain how XOR can toggle a flag bit without changing other bits. Give an 8-bit example.

1.4.1(i) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- 11010101 AND 11110000 = 11010000.
- 11010000 OR 00000011 = 11010011.

Q2 / 4 MARKS

- XOR with 1 toggles; XOR with 0 leaves unchanged.
- For 10110000, XOR 00000100 gives 10110100 (bit 2 changes).

COMMON MISCONCEPTION

A logical right shift is not always division by 2 for signed two's-complement values; signed/arithmetic shifts need separate treatment.

Green pen: Number the bit positions from the right before selecting a mask.

1.4.1(j) / BUILD THE MENTAL MODEL

Character and encoding are different concepts

A character set assigns identities to symbols. An encoding determines the bytes used to store those identities.

THREE PIECES OF THE IDEA

ASCII

Original ASCII uses seven bits and represents 128 character codes.

UNICODE

Assigns code points to characters across many writing systems and symbols.

UTF-8

A variable-length Unicode encoding; ASCII-range characters use one byte, others may require more.

MODEL / FOLLOW THE FLOW

Letter A → **Code point U+0041** → **UTF-8 byte 41** → **Decode to A**

Unicode code point U+20AC (euro sign) is stored in UTF-8 as three bytes E2 82 AC.

EXAM-READY STATEMENT

Do not equate Unicode with a fixed 16-bit or 32-bit encoding; Unicode characters can be encoded using UTF-8, UTF-16 or UTF-32.

WORDS THAT EARN MARKS

- **Code point** Character identifier
- **ASCII** 7-bit set
- **UTF-8** Variable-length bytes
- **Glyph** Rendered symbol

1.4.1(j) / STEP-BY-STEP WORKED EXAMPLE

Compare two characters sent in UTF-8

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. A	U+0041	An ASCII-range Unicode code point.
2. A bytes	41 hex	One UTF-8 byte.
3. Euro	U+20AC	Outside original 7-bit ASCII.
4. Euro bytes	E2 82 AC	Three UTF-8 bytes.
5. Interpret	One character need not equal one byte	Storage size depends on encoding, not character count alone.

TRANSFER IT TO AN EXAM ANSWER

Do not equate Unicode with a fixed 16-bit or 32-bit encoding; Unicode characters can be encoded using UTF-8, UTF-16 or UTF-32.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.1(j) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Explain why the text file containing “A” and “€” may need different numbers of bytes per character.

QUESTION 2 | 4 MARKS

Distinguish a Unicode code point from UTF-8, giving an example.

1.4.1(j) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- A uses one byte in UTF-8 because it is in the ASCII range.
- Euro sign U+20AC requires three UTF-8 bytes (E2 82 AC).
- UTF-8 is variable-length.

Q2 / 4 MARKS

- Code point identifies a character, e.g. U+0041.
- UTF-8 encodes that identifier as bytes, e.g. 0x41.
- Same Unicode character may have another byte sequence in UTF-16.

COMMON MISCONCEPTION

Unicode is not the name of a single fixed-width bit pattern.

Green pen: Always name the encoding when giving a byte count.

1.4.2(a) / BUILD THE MENTAL MODEL

Choose a structure by shape and behaviour

These structures differ in whether they have fixed positions, named fields, changing length or immutable groups.

THREE PIECES OF THE IDEA

ARRAY

Indexed elements of a common type; OCR includes one-, two- and three-dimensional arrays.

RECORD

Named fields of potentially different types; e.g. Student.Name and Student.Age.

LIST / TUPLE

A list is an ordered, commonly mutable collection; a tuple is an ordered fixed group, often immutable.

MODEL / FOLLOW THE FLOW

Sensor[day] → **Grid[row,col]** → **Voxel[x,y,z]** → **Record fields**

A 3D array index has three coordinates. A record is better when each item holds different named attributes.

EXAM-READY STATEMENT

Array indices locate elements; record field names identify attributes. A 3D array is not the same as three separate arrays.

WORDS THAT EARN MARKS

- **Index** Position
- **Dimension** Independent index
- **Field** Named attribute
- **Tuple** Fixed ordered group

1.4.2(a) / STEP-BY-STEP WORKED EXAMPLE

Represent six measurements and one student

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Array	temp[0..5]	A 1D sequence of six same-type readings.
2. Matrix	seat[1..3,1..4]	A 2D grid with 3 rows and 4 columns.
3. Volume	pixel[0..1,0..2,0..3]	A 3D array with 2 x 3 x 4 = 24 elements.
4. Record	student = (name, year, present)	Fields may have string, integer, Boolean types.
5. List	tasks.add(newTask)	The number of tasks can grow dynamically.

TRANSFER IT TO AN EXAM ANSWER

Array indices locate elements; record field names identify attributes. A 3D array is not the same as three separate arrays.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.2(a) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

A game stores blocks in a 4 x 3 x 2 region. Choose a data structure, state its capacity and give an access expression.

QUESTION 2 | 4 MARKS

Compare a record with an array for storing a pupil's name, year and attendance flag.

1.4.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- A three-dimensional array is suitable.
- Capacity = 4 x 3 x 2 = 24 elements.
- For zero-based indices, e.g. block[3,2,1] is a valid access.

Q2 / 4 MARKS

- Record has named fields and allows different data types.
- A conventional array has indexed elements of a single type.
- Use Student.Name, Student.Year, Student.Present.

COMMON MISCONCEPTION

Do not infer zero-based indexing unless a question defines it; state your assumption.

Green pen: Draw the dimensions and label the index order before calculating capacity.

1.4.2(b,c) / BUILD THE MENTAL MODEL

Last in, first out means top changes first

A stack permits insertion and removal at the top. A bounded array stack must detect full and empty states.

THREE PIECES OF THE IDEA

PUSH

Check capacity; increment top pointer and store a new item.

POP

Check nonempty; return top value and decrement top pointer.

PEEK

Return top value without removing it; do not change top pointer.

MODEL / FOLLOW THE FLOW

Empty top=-1 → **Push A top=0** → **Push B top=1** → **Pop gives B top=0**

An array-backed example uses top = -1 for an empty stack and zero-based slots.

EXAM-READY STATEMENT

Always name the stack convention (top at last occupied index) and preserve the return order under LIFO.

WORDS THAT EARN MARKS

- **LIFO** Last in first out
- **Top** Last occupied index
- **Overflow** Push onto full
- **Underflow** Pop from empty

1.4.2(b,c) / STEP-BY-STEP WORKED EXAMPLE

Trace operations on a capacity-three stack

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
0. Empty	[_, _, _], top=-1	The stack has no items.
1. PUSH A	[A, _, _], top=0	Store in the first slot.
2. PUSH B	[A, B, _], top=1	B is now the top element.
3. PUSH C	[A, B, C], top=2	Stack is full.
4. POP	returns C; top=1	Next POP returns B; pushing D is now possible.

TRANSFER IT TO AN EXAM ANSWER

Always name the stack convention (top at last occupied index) and preserve the return order under LIFO.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.2(b,c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Trace PUSH(4), PUSH(9), POP(), PUSH(7), PEEK(). Give returned values and final stack.

QUESTION 2 | 4 MARKS

Explain both overflow and underflow for an array stack of capacity two.

1.4.2(b,c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- POP returns 9.
- Final stack bottom -> top: [4,7].
- PEEK returns 7 and does not remove it.

Q2 / 4 MARKS

- Overflow occurs when PUSH is attempted on a full stack.
- Underflow occurs when POP is attempted on an empty stack.
- Bounds checks prevent invalid array access.

COMMON MISCONCEPTION

PEEK does not remove data. POP changes the size even if the physical memory slot has not been cleared.

Green pen: List returned values on a separate line from the stored stack.

1.4.2(b,c) / BUILD THE MENTAL MODEL

First in, first out uses front and rear

An array queue can reuse freed slots by wrapping indices modulo capacity. Track count to distinguish full from empty.

THREE PIECES OF THE IDEA

ENQUEUE

Add at the rear after checking capacity.

DEQUEUE

Remove from the front after checking nonempty.

CIRCULAR

Update indices modulo array capacity; count disambiguates full versus empty.

MODEL / FOLLOW THE FLOW

front = 0 → **Enqueue A,B,C** → **Dequeue A** → **Enqueue D at wrap**

In this convention front is the index of the next item to remove; rear is the index of the next free slot.

EXAM-READY STATEMENT

Do not treat front and rear as universally defined; state whether rear means final occupied or next free index.

WORDS THAT EARN MARKS

FIFO	First in first out	**Front**	Next removed
Rear	Next inserted	**MOD**	Wrap index

1.4.2(b,c) / STEP-BY-STEP WORKED EXAMPLE

Circular queue of capacity three

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
0. Empty	front=0 rear=0 count=0	Use rear as the next insertion index.
1. Enqueue A	slot 0=A; rear=1 count=1	Front remains 0.
2. Enqueue B,C	slots 1=B,2=C; rear=0 count=3	Queue full; rear wraps (2+1) MOD 3.
3. Dequeue	returns A; front=1 count=2	The freed slot is zero.
4. Enqueue D	slot 0=D; rear=1 count=3	Logical order is B,C,D although physical slots are D,B,C.

TRANSFER IT TO AN EXAM ANSWER

Do not treat front and rear as universally defined; state whether rear means final occupied or next free index.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.2(b,c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Starting empty with capacity three, enqueue 1,2,3; dequeue once; enqueue 4. State physical slots and logical order.

QUESTION 2 | 4 MARKS

Explain why front = rear alone cannot distinguish a circular queue's full and empty conditions.

1.4.2(b,c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Slots become [4,2,3] using next-free-rear convention.
- Logical FIFO order is 2,3,4.
- front=1, rear=1, count=3.

Q2 / 4 MARKS

- Indices can be equal after all items are removed or after all slots are filled.
- Maintain count, a separate full flag, or reserve a spare slot.

COMMON MISCONCEPTION

Circular does not mean items move in memory. Pointers wrap and freed slots are reused.

Green pen: Record both physical array and logical order after each operation.

1.4.2(b,c) / BUILD THE MENTAL MODEL

Change links without shifting every node

A singly linked list stores a value plus a reference to the next node. The head references the first node.

THREE PIECES OF THE IDEA

NODE

A data field plus a next pointer/reference.

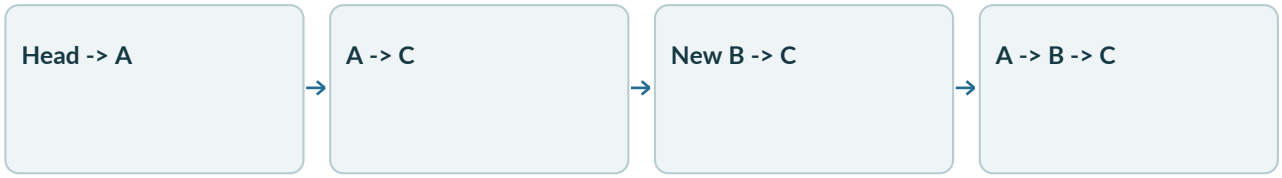
INSERT

Connect new node to successor before reconnecting predecessor or head.

DELETE

Bypass the removed node and update head if required; reclaim storage as appropriate.

MODEL / FOLLOW THE FLOW



To insert between A and C, set B.next to C before setting A.next to B.

EXAM-READY STATEMENT

Pointers change, not the positions of all other stored values. Handle head/tail and empty-list cases explicitly.

WORDS THAT EARN MARKS

- **Head** First-node reference
- **Null** No successor
- **Pointer** Node reference
- **Traversal** Follow links

1.4.2(b,c) / STEP-BY-STEP WORKED EXAMPLE

Insert B into A -> C, then delete A

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Before	head -> A -> C -> NULL	A.next refers to C.
2. Make B	B.next = C	Preserve access to C before changing A.
3. Reconnect	A.next = B	Now A -> B -> C.
4. Remove A	head = A.next	New head becomes B.
5. After	head -> B -> C -> NULL	A can be released if not referenced elsewhere.

TRANSFER IT TO AN EXAM ANSWER

Pointers change, not the positions of all other stored values. Handle head/tail and empty-list cases explicitly.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.2(b,c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Describe insertion of new node X immediately after node P in a singly linked list. Include statement order.

QUESTION 2 | 5 MARKS

Explain how to remove the first node from a singly linked list and identify two edge cases.

1.4.2(b,c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Set X.next = P.next.
- Set P.next = X.
- Maintain any tail pointer if P was originally the tail.

Q2 / 5 MARKS

- Check whether list is empty before deletion.
- Save old head if required, set head = head.next.
- If list becomes empty update tail; reclaim removed node according to implementation.

COMMON MISCONCEPTION

Reversing the assignment order during insertion can make the remainder of the list unreachable.

Green pen: Sketch head and next arrows before and after the operation.

1.4.2(b,c) / BUILD THE MENTAL MODEL

A search tree enforces an ordering invariant

A tree has a root and parent-child edges without cycles. In a binary search tree (BST), left subtree keys are smaller and right subtree keys larger under the chosen duplicate policy.

THREE PIECES OF THE IDEA

ROOT / LEAF

Root has no parent. A leaf has no children.

BINARY TREE

Each node has no more than two children, but the keys need not be sorted.

BST

Left keys < parent < right keys if duplicate keys are disallowed.

MODEL / FOLLOW THE FLOW

Insert 8 → **Insert 3 left** → **Insert 11 right** → **Search 11 right**

A balanced BST can be efficient, but a badly ordered insertion sequence can produce a chain with linear search time.

EXAM-READY STATEMENT

BST ordering is recursive: every key in the whole left subtree must be smaller, not only the immediate left child.

WORDS THAT EARN MARKS

- **Root** Top node
- **Leaf** No children
- **Height** Longest path
- **BST** Ordered keys

1.4.2(b,c) / STEP-BY-STEP WORKED EXAMPLE

Build and search for a key

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Insert 8	root=8	First item becomes root.
2. Insert 3	3 < 8 -> left child	Compare at every visited node.
3. Insert 11	11 > 8 -> right child	11 becomes right child.
4. Insert 5	5 < 8; 5 > 3	5 is right child of node 3.
5. Search 5	8 -> 3 -> 5	At each step choose left or right by comparison.

TRANSFER IT TO AN EXAM ANSWER

BST ordering is recursive: every key in the whole left subtree must be smaller, not only the immediate left child.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.2(b,c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Insert 10, 6, 14, 8, 4 into an empty BST. Describe the edges and the search path to 8.

QUESTION 2 | 4 MARKS

Explain why inserting sorted values into an ordinary unbalanced BST may slow searches.

1.4.2(b,c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Root 10; left 6, right 14; node 6 has left 4 and right 8.
- Search path 10 -> 6 -> 8.

Q2 / 4 MARKS

- Sorted order can make each new node the right or left child of the last.
- Height grows to n rather than approximately log2(n).
- Search can then require O(n) comparisons.

COMMON MISCONCEPTION

A binary tree is not necessarily a BST. Ordering is an additional invariant.

Green pen: Write the comparator direction along each search arrow.

1.4.2(c) / BUILD THE MENTAL MODEL

In-order on a BST yields sorted keys

Traversal order depends on whether a node is processed before, between or after its subtrees.

THREE PIECES OF THE IDEA

PRE-ORDER

Node, left subtree, right subtree.

IN-ORDER

Left subtree, node, right subtree; sorted keys for a BST.

POST-ORDER

Left subtree, right subtree, node.

MODEL / FOLLOW THE FLOW

Root 8 → **Left 3 with right 5** → **Right 11** → **Visit in order**

For BST with root 8, left 3 (right child 5) and right 11: in-order outputs 3,5,8,11.

EXAM-READY STATEMENT

When deleting a BST node with two children, replace it consistently by an in-order predecessor or successor, then remove the moved node.

WORDS THAT EARN MARKS

Traversal Visit each node

In-order Left-node-right

Successor Next larger key

Deletion Preserve ordering

1.4.2(c) / STEP-BY-STEP WORKED EXAMPLE

Traverse and delete node 8 from a BST

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Tree	Root 8; left 3 (right 5); right 11	All keys follow BST order.
2. In-order	3, 5, 8, 11	Visit left, node, right.
3. Pre-order	8, 3, 5, 11	Visit node before subtrees.
4. Post-order	5, 3, 11, 8	Visit node after subtrees.
5. Delete root	Replace 8 with in-order successor 11	11 has no left subtree here; reconnect remaining nodes.

TRANSFER IT TO AN EXAM ANSWER

When deleting a BST node with two children, replace it consistently by an in-order predecessor or successor, then remove the moved node.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.2(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

For BST root 6, children 2 and 9, with 2 having right child 4, give in-, pre- and post-order traversals.

QUESTION 2 | 4 MARKS

Outline deletion of a BST node that has two children without breaking the ordering property.

1.4.2(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- In-order: 2,4,6,9.
- Pre-order: 6,2,4,9.
- Post-order: 4,2,9,6.

Q2 / 4 MARKS

- Find in-order successor (minimum of right subtree) or predecessor (maximum left).
- Replace target key/value with that key/value.
- Delete the successor/predecessor from its original location and reconnect its child if needed.

COMMON MISCONCEPTION

Pre-order and post-order results are not generally sorted even for a BST.

Green pen: Write the left-node-right pattern above each traversal before tracing.

1.4.2(b,c) / BUILD THE MENTAL MODEL

An edge may have a direction and a weight

A graph contains vertices and edges. Choose a representation suited to whether the graph is dense or sparse and whether direction matters.

THREE PIECES OF THE IDEA

UNDIRECTED

An edge A-B permits travel in either direction.

DIRECTED

A -> B is an ordered relationship; reverse edge is separate.

REPRESENTATION

Adjacency lists store neighbours; adjacency matrices use a cell for each ordered pair.

MODEL / FOLLOW THE FLOW

Vertices A,B,C → **Edges A->B, B->C** → **Adjacency list** → **Visit each once**

Directed list: A:[B], B:[C], C:[]; its adjacency matrix is asymmetric because edges are one-way.

EXAM-READY STATEMENT

Search orders depend on specified adjacency order. State that convention before writing a single valid traversal.

WORDS THAT EARN MARKS

- **Vertex** Graph node
- **Edge** Connection
- **Directed** One-way relation
- **Adjacency** Neighbours

1.4.2(b,c) / STEP-BY-STEP WORKED EXAMPLE

Trace BFS and DFS on a small graph

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Graph	A-B, A-C, B-D, C-E	Edges are undirected; neighbours processed alphabetically.
2. BFS start	Queue [A], visited {A}	Dequeue A; enqueue B and C.
3. BFS order	A, B, C, D, E	Breadth-first explores each level with a queue.
4. DFS start	A -> B -> D, backtrack	Depth-first follows an unvisited path.
5. DFS order	A, B, D, C, E	Assumes neighbours visited alphabetically and marked on discovery.

TRANSFER IT TO AN EXAM ANSWER

Search orders depend on specified adjacency order. State that convention before writing a single valid traversal.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.2(b,c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

For undirected edges A-B, A-C, B-D, C-E, give BFS and DFS orders from A using alphabetical neighbours.

QUESTION 2 | 4 MARKS

Compare adjacency lists with adjacency matrices for a sparse directed network.

1.4.2(b,c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- BFS: A,B,C,D,E.
- DFS: A,B,D,C,E with the specified neighbour ordering.
- BFS uses a queue; DFS uses recursion or a stack.

Q2 / 4 MARKS

- Lists store present outgoing edges and can be space-efficient when sparse.
- Matrix uses V x V positions and gives constant-time direct edge lookup.
- Direction means matrix A,B and B,A may differ.

COMMON MISCONCEPTION

BFS and DFS are not one universal ordering unless the neighbour selection rule is provided.

Green pen: Mark vertices as visited to avoid infinite loops in cyclic graphs.

1.4.2(b,c) / BUILD THE MENTAL MODEL

A good hash chooses a slot; collisions need a policy

A hash table converts a key to an array index. Two distinct keys can map to the same slot; the table must handle that collision.

THREE PIECES OF THE IDEA

HASH FUNCTION

Computes an index, e.g. key MOD capacity for simple integer examples.

COLLISION

Two keys receive the same initial index.

RESOLUTION

Chaining stores multiple items per bucket; open addressing probes another slot.

MODEL / FOLLOW THE FLOW

Key 17 -> 3 → **Key 24 -> 3** → **Probe next free slot** → **Search same sequence**

With seven slots, 17 MOD 7 = 3 and 24 MOD 7 = 3; collision handled using linear probing to slot 4.

EXAM-READY STATEMENT

Deletion in open addressing must preserve the probe chain, commonly with a deleted marker or careful rehashing.

WORDS THAT EARN MARKS

- **Hash** Key-to-index
- **Bucket** Slot or chain
- **Probe** Alternative slot
- **Tombstone** Deleted marker

1.4.2(b,c) / STEP-BY-STEP WORKED EXAMPLE

Insert/search/delete in capacity-7 open-address table

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. h(k)	k MOD 7	Slots indexed 0 to 6.
2. Insert 10	10 MOD 7 = 3 -> slot 3	Slot was empty.
3. Insert 17	17 MOD 7 = 3, probe slot 4	Collision; store at first free slot.
4. Search 17	Check slot 3 then slot 4	Use identical probe sequence.
5. Delete 10	Mark slot 3 DELETED, not EMPTY	Otherwise a later search for 17 could stop too early.

TRANSFER IT TO AN EXAM ANSWER

Deletion in open addressing must preserve the probe chain, commonly with a deleted marker or careful rehashing.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.2(b,c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

With hash key MOD 5 and linear probing, insert 7,12,17 into an empty table. Give the slots used.

QUESTION 2 | 4 MARKS

Explain why replacing a deleted collided entry with an EMPTY marker can cause failed lookups.

1.4.2(b,c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- 7 MOD 5 = 2 -> slot 2.
- 12 MOD 5 = 2 -> probe slot 3.
- 17 MOD 5 = 2 -> probe slots 3 then 4.

Q2 / 4 MARKS

- A search stops on an actually empty slot.
- Later colliding records may be beyond the deleted index.
- Mark DELETED (tombstone) or correctly rehash to preserve probe sequence.

COMMON MISCONCEPTION

Hashing is not encryption, and a hash table is not necessarily sorted.

Green pen: Write the table index under each attempt and keep the search probe order identical to insertion.

1.4.3(a,d) / BUILD THE MENTAL MODEL

Turn a written rule into a testable expression

Boolean expressions encode conditions using true/false inputs. A truth table checks the output for every combination.

THREE PIECES OF THE IDEA

AND

All specified inputs must be true.

OR

At least one input must be true (inclusive OR).

NOT

Reverse a truth value; combine brackets to control the scope.

MODEL / FOLLOW THE FLOW

Door rule → **A = staff** → **B = valid code** → **F = A AND B**

A door opens only if a staff credential and valid code are both present: F = A AND B.

EXAM-READY STATEMENT

A truth table with n Boolean inputs has 2^n rows, and inclusive OR is true if either or both operands are true.

WORDS THAT EARN MARKS

- **Input** Condition
- **Output** Decision
- **Truth table** All combinations
- **Expression** Logic rule

1.4.3(a,d) / STEP-BY-STEP WORKED EXAMPLE

Build a three-input access rule

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Define	A=staff, B=code, C=override	Name every variable and whether TRUE means present.
2. Write	F=(A AND B) OR C	Override opens independently; otherwise both conditions needed.
3. Example	A=1 B=0 C=0 -> F=0	Staff without code and no override: locked.
4. Example	A=0 B=0 C=1 -> F=1	Override alone grants access.
5. Check	Three inputs require 2^3=8 rows	Do not omit input combinations.

TRANSFER IT TO AN EXAM ANSWER

A truth table with n Boolean inputs has 2^n rows, and inclusive OR is true if either or both operands are true.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.3(a,d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

A light comes on if motion M is detected AND it is dark D, OR the manual switch S is on. Write and explain the Boolean expression.

QUESTION 2 | 4 MARKS

For F=(A AND B) OR C, calculate outputs for 000, 101, 110 and 111 (A,B,C order).

1.4.3(a,d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- F=(M AND D) OR S.
- Motion and darkness must both be true for automatic activation.
- S=TRUE switches it on independently.

Q2 / 4 MARKS

- 000 -> 0.
- 101 -> 1 (C=1).
- 110 -> 1 (A AND B=1).
- 111 -> 1.

COMMON MISCONCEPTION

OR is inclusive, not exclusive, unless XOR is explicitly specified.

Green pen: State what each variable represents before drawing a table.

1.4.3(d) / BUILD THE MENTAL MODEL

Read the gate at every stage

Logic gates implement Boolean operations; intermediate wire values make a multi-gate circuit easier to follow.

THREE PIECES OF THE IDEA

BASIC

AND, OR and NOT form building blocks; NAND and NOR negate AND/OR outputs.

XOR

TRUE when the two inputs differ. XNOR is TRUE when they match.

CIRCUIT

Mark intermediate outputs so the final truth table follows the circuit, not a guess.

MODEL / FOLLOW THE FLOW

Input A,B → **AND gate X** → **NOT gate Y** → **OR with C -> F**

For the example F=NOT(A AND B) OR C, the intermediate output is X=A AND B, then Y=NOT X.

EXAM-READY STATEMENT

For multi-stage circuits, identify intermediate wires and show their values. NAND = NOT(AND), NOR = NOT(OR).

WORDS THAT EARN MARKS

- **NAND** NOT AND
- **NOR** NOT OR
- **XOR** Different bits
- **XNOR** Same bits

1.4.3(d) / STEP-BY-STEP WORKED EXAMPLE

Trace F=NOT(A AND B) OR C

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Inputs	A=1, B=1, C=0	Three specified Boolean inputs.
2. AND	X=1 AND 1 = 1	Output X of first gate.
3. NOT	Y=NOT 1 = 0	Invert X.
4. OR	F=0 OR 0 = 0	Evaluate final stage.
5. Alternative	A=1 B=0 C=0 -> F=1	AND becomes 0; NOT becomes 1.

TRANSFER IT TO AN EXAM ANSWER

For multi-stage circuits, identify intermediate wires and show their values. NAND = NOT(AND), NOR = NOT(OR).

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.3(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Evaluate F=(A XOR B) AND (NOT C) for A=1, B=0, C=0, showing intermediate results.

QUESTION 2 | 5 MARKS

Describe the truth table for a two-input NAND gate and explain why its output is false only once.

1.4.3(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- A XOR B = 1 because inputs differ.
- NOT C = 1.
- F=1 AND 1 = 1.

Q2 / 5 MARKS

- 00 -> 1, 01 -> 1, 10 -> 1, 11 -> 0.
- AND is TRUE only when both inputs are TRUE.
- NAND negates the AND output, so only 11 gives FALSE.

COMMON MISCONCEPTION

An XOR gate is not the same as an inclusive OR gate when both inputs are 1.

Green pen: Give the intermediate values in the same order as the circuit.

1.4.3(c) / BUILD THE MENTAL MODEL

Simplify by stating the law used

Boolean algebra has identities that preserve output across all input combinations. De Morgan swaps AND/OR when negating a bracket.

THREE PIECES OF THE IDEA

DE MORGAN

NOT(A AND B) = (NOT A) OR (NOT B); NOT(A OR B) = (NOT A) AND (NOT B).

DOUBLE NEGATION

NOT(NOT A) = A.

REARRANGEMENT

Commutative, associative and distributive laws permit valid reorderings and expansions.

MODEL / FOLLOW THE FLOW

Start NOT(A AND B) → **De Morgan** → **(NOT A) OR (NOT B)** → **Verify by table**

Negation applies to each input AND changes the operator between them.

EXAM-READY STATEMENT

Name each law. NOT(A AND B) becomes NOT A OR NOT B, not NOT A AND NOT B.

WORDS THAT EARN MARKS

- **De Morgan** Negate + swap
- **Distribute** Factor or expand
- **Associate** Regroup
- **Commute** Reorder

1.4.3(c) / STEP-BY-STEP WORKED EXAMPLE

Simplify (A AND B) OR (A AND NOT B)

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Start	(A AND B) OR (A AND NOT B)	Identify common A.
2. Distribute	A AND (B OR NOT B)	Factor A using distributive law.
3. Complement	B OR NOT B = 1	Excluded-middle / complement law.
4. Identity	A AND 1 = A	One AND A is A.
5. Check	B=0 or B=1 gives result A	Verify using a two-input truth table.

TRANSFER IT TO AN EXAM ANSWER

Name each law. NOT(A AND B) becomes NOT A OR NOT B, not NOT A AND NOT B.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.3(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Simplify (P AND Q) OR (P AND NOT Q), naming each law used.

QUESTION 2 | 5 MARKS

Use De Morgan to transform NOT(X OR (Y AND Z)); show the negations and operators.

1.4.3(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Factor P: P AND (Q OR NOT Q).
- Complement law: Q OR NOT Q = 1.
- Identity law: P AND 1 = P.

Q2 / 5 MARKS

- NOT X AND NOT(Y AND Z).
- Apply De Morgan within bracket: NOT X AND (NOT Y OR NOT Z).
- Both operator changes must be shown.

COMMON MISCONCEPTION

De Morgan changes the operator as well as distributing NOT.

Green pen: Underline the bracket that is being negated at each step.

1.4.3(b) / BUILD THE MENTAL MODEL

Adjacent ones let a changing variable disappear

A Karnaugh map (K-map) places truth-table outputs in Gray-code order. Group adjacent ones in rectangles of sizes 1, 2, 4 or 8.

THREE PIECES OF THE IDEA

ORDER

For two-bit axis labels use 00, 01, 11, 10.

GROUP

Choose largest available power-of-two groups, allowing overlap and wrap-around.

EXPRESSION

A variable disappearing within a group is omitted from that group term.

WORKED VISUAL / NOT TO SCALE

A / BC	00	01	11	10
0	0	0	1	1
1	0	1	1	0

TWO PAIRS

Group row A=0: m2,m3. Group row A=1: m5,m7.

F = (NOT A AND B) OR (A AND C)

EXAM-READY STATEMENT

If a variable takes both 0 and 1 inside a group, it cancels from the product term.

WORDS THAT EARN MARKS

- **Minterm** Input row with 1
- **Gray code** One bit changes
- **Group** Power of two
- **Implicant** Simplified product

1.4.3(b) / STEP-BY-STEP WORKED EXAMPLE

Simplify F=1 for minterms 2, 3, 5 and 7

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Labels	A rows 0,1; BC columns 00,01,11,10	Each adjacent cell differs in one input.
2. Put ones	m2,m3,m5,m7 are 1	All other input rows have F=0.
3. First group	m2 and m3	A=0 and B=1 constant; C varies.
4. Second group	m5 and m7	A=1 and C=1 constant; B varies.
5. Result	F=(NOT A AND B) OR (A AND C)	Combine group products by OR.

TRANSFER IT TO AN EXAM ANSWER

If a variable takes both 0 and 1 inside a group, it cancels from the product term.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.3(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

For three inputs A,B,C, F=1 on minterms 0,1,2,3. Give a simplified expression and explain the group.

QUESTION 2 | 5 MARKS

Explain why K-map columns are not in ordinary binary order 00,01,10,11.

1.4.3(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- All four rows have A=0 while B and C vary.
- One four-cell group gives F=NOT A.

Q2 / 5 MARKS

- Gray-code order 00,01,11,10 keeps adjacent labels different by one bit.
- This allows neighbouring ones to differ in only one variable, which can then be removed.

COMMON MISCONCEPTION

A group of three cells is invalid; group sizes must be powers of two.

Green pen: Highlight the variables constant within the group, not those that change.

1.4.3(b) / BUILD THE MENTAL MODEL

Wrap-around groups include all four corners

In a 4-variable map, each axis is in Gray-code order. Opposite edges touch; the four corner cells can form one 4-cell group.

THREE PIECES OF THE IDEA

FOUR AXES BITS

Use AB across columns and CD down rows, both ordered 00,01,11,10.

WRAP

First and last rows touch; first and last columns touch.

OVERLAP

One cell may belong to multiple groups if this allows fewer/larger terms.

WORKED VISUAL / NOT TO SCALE

CD / AB	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	0	0	0	0
10	1	0	0	1

FOUR CORNERS

One group wraps across **both** pairs of edges.

B = 0 and D = 0 remain fixed.

F = NOT B AND NOT D

EXAM-READY STATEMENT

Always consider corner and opposite-edge adjacency before settling for multiple small groups.

WORDS THAT EARN MARKS

- **K-map** Gray-code grid
- **Wrap** Opposite edge
- **Constant** Keep variable
- **Overlap** Reuse a 1

1.4.3(b) / STEP-BY-STEP WORKED EXAMPLE

Simplify the four-corner pattern

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Grid	AB: 00 01 11 10; CD same order	Use Gray-coded axes.
2. Ones	(AB,CD): (00,00),(10,00),(00,10),(10,10)	These are the four corners.
3. Group	One 4-cell rectangle wraps both edges	All four positions are adjacent through wrapping.
4. Constants	B=0 and D=0	A and C vary, so they disappear.
5. Result	F=(NOT B) AND (NOT D)	Equivalent to NOT(B OR D).

TRANSFER IT TO AN EXAM ANSWER

Always consider corner and opposite-edge adjacency before settling for multiple small groups.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.3(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

A 4-variable K-map has ones only in the four corners using AB columns and CD rows. Give the simplest expression and justify it.

QUESTION 2 | 5 MARKS

A second group has all cells with A=1 and C=1. Give its term and combine it by OR with the four-corner expression.

1.4.3(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Four corners form one 4-cell wrap-around group.
- B=0 and D=0 are constant.
- F=NOT B AND NOT D.

Q2 / 5 MARKS

- All A=1 and C=1 cells give A AND C.
- Combined F=(NOT B AND NOT D) OR (A AND C).
- Overlapping cells may be included in more than one group.

COMMON MISCONCEPTION

Grid edges are not walls. Do not create four singleton groups when one corner group exists.

Green pen: Write down the axis order and mark wrap connections on both pairs of edges.

1.4.3(e) / BUILD THE MENTAL MODEL

Add two input bits without a carry-in

A half adder adds two one-bit inputs. It produces a one-bit sum and a carry-out.

THREE PIECES OF THE IDEA

SUM

S = A XOR B; the sum is one when inputs differ.

CARRY

C = A AND B; carry is one only for 1+1.

LIMITATION

A half adder has no carry-in from a lower-order bit.

WORKED VISUAL / NOT TO SCALE



Input 1 + 1 yields sum 0 and carry 1, i.e. binary 10.

EXAM-READY STATEMENT

Draw two gates fed by both A and B: XOR to sum, AND to carry. A half adder cannot accept Cin.

WORDS THAT EARN MARKS

- **XOR** Different inputs
- **Carry** Extra binary place
- **Half adder** No Cin
- **Truth table** Four input rows

1.4.3(e) / STEP-BY-STEP WORKED EXAMPLE

Trace all half-adder input combinations

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
0+0	A=0 B=0 -> S=0 C=0	No ones at either output.
0+1	A=0 B=1 -> S=1 C=0	Different bits, no carry.
1+0	A=1 B=0 -> S=1 C=0	Same sum as the previous row.
1+1	A=1 B=1 -> S=0 C=1	Binary 10 represents decimal 2.

TRANSFER IT TO AN EXAM ANSWER

Draw two gates fed by both A and B: XOR to sum, AND to carry. A half adder cannot accept Cin.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.3(e) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Give the half-adder outputs for A=1, B=1, and name the gates that produce each output.

QUESTION 2 | 4 MARKS

Explain why a half adder alone is insufficient for a middle bit in multi-bit addition.

1.4.3(e) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Sum S=0 using XOR.
- Carry C=1 using AND.
- Output 10 represents decimal 2.

Q2 / 4 MARKS

- A middle column must add A, B and carry-in from the previous column.
- A half adder accepts only A and B; a full adder supports three inputs.

COMMON MISCONCEPTION

The carry output is not the same wire as the sum output.

Green pen: Check the 1+1 row: sum 0 and carry 1.

1.4.3(e) / BUILD THE MENTAL MODEL

A full adder accepts three one-bit inputs

A full adder adds A, B and Cin. It produces S and Cout; two half adders and an OR gate form a standard implementation.

THREE PIECES OF THE IDEA

FIRST HALF ADDER

S1 = A XOR B; C1 = A AND B.

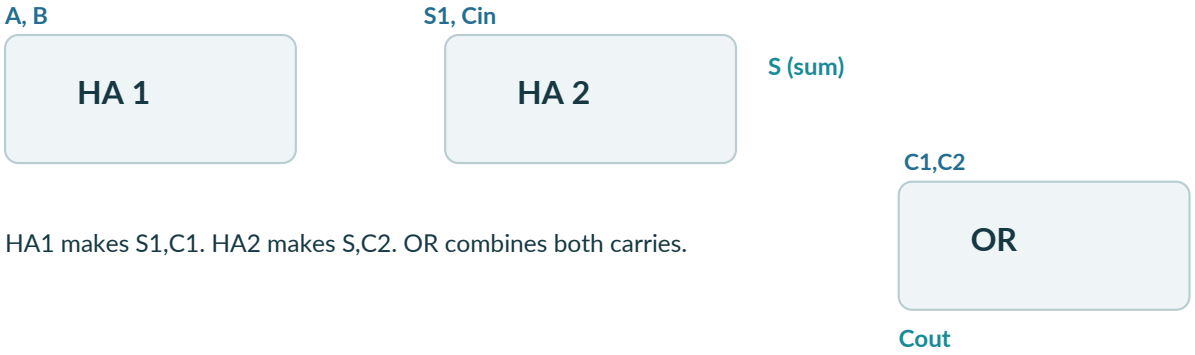
SECOND HALF ADDER

S = S1 XOR Cin; C2 = S1 AND Cin.

COMBINE CARRIES

Cout = C1 OR C2; equivalently AB OR (A XOR B)Cin.

WORKED VISUAL / NOT TO SCALE



HA1 makes S1,C1. HA2 makes S,C2. OR combines both carries.

EXAM-READY STATEMENT

For a full adder, S = A XOR B XOR Cin and Cout = (A AND B) OR (Cin AND (A XOR B)).

WORDS THAT EARN MARKS

Cin	Incoming carry	**Cout**	Outgoing carry
HA	Half adder	**XOR**	Sum parity

1.4.3(e) / STEP-BY-STEP WORKED EXAMPLE

Trace A=1, B=0, Cin=1

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. First sum	S1 = 1 XOR 0 = 1	First half adder sum.
2. First carry	C1 = 1 AND 0 = 0	First half adder carry.
3. Second sum	S = 1 XOR 1 = 0	Second half adder sum.
4. Second carry	C2 = 1 AND 1 = 1	Second half adder carry.
5. Combine	Cout = 0 OR 1 = 1	Binary output CoutS = 10 (decimal 2).

TRANSFER IT TO AN EXAM ANSWER

For a full adder, S = A XOR B XOR Cin and Cout = (A AND B) OR (Cin AND (A XOR B)).

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.3(e) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

A full adder receives A=1, B=1, Cin=1. Calculate S and Cout through two half adders.

QUESTION 2 | 5 MARKS

Explain how two half adders plus one OR gate produce a full-adder circuit.

1.4.3(e) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- HA1: S1=0, C1=1.
- HA2 with Cin=1: S=1, C2=0.
- Cout=1 OR 0=1, output 11 (decimal 3).

Q2 / 5 MARKS

- First half adder receives A,B -> S1 and C1.
- Second receives S1,Cin -> S and C2.
- OR combines C1,C2 to Cout.

COMMON MISCONCEPTION

Cout is not just A AND B because Cin can produce a carry when one of A/B is 1.

Green pen: Label both intermediate carry outputs before drawing the final OR gate.

1.4.3(e) / BUILD THE MENTAL MODEL

Store the input bit at the active clock edge

A D-type flip-flop stores one bit. In a rising-edge-triggered version, the input D is sampled at the clock's rising edge and output Q holds until the next rising edge.

THREE PIECES OF THE IDEA

DATA

D is the bit offered for storage.

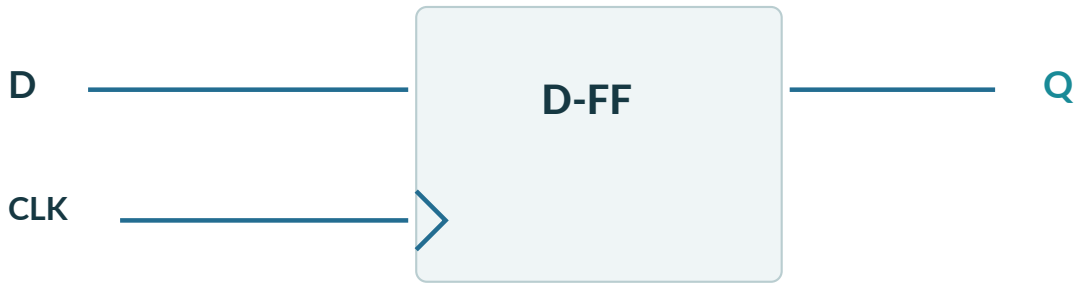
CLOCK

An edge-triggered symbol uses a small triangle at CLK; a rising edge causes sampling.

OUTPUT

Q is the stored bit. The complementary output, if shown, is NOT Q.

WORKED VISUAL / NOT TO SCALE



Rising clock edge: Q copies D. Between edges: Q holds.

EXAM-READY STATEMENT

A D-type flip-flop is storage, not a gate that continuously copies D. A collection of flip-flops can form a register.

WORDS THAT EARN MARKS

D	Data input	**CLK**	Clock input
Q	Stored output	**Edge**	Sampling instant

1.4.3(e) / STEP-BY-STEP WORKED EXAMPLE

Trace an edge-triggered register bit

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
0. Initial	Q=0	Known stored state.
1. Between edges	D changes to 1; Q stays 0	No active clock edge.
2. Rising edge 1	D=1 -> Q=1	Sample the present input.
3. Between edges	D changes to 0; Q stays 1	Still holding the previous value.
4. Rising edge 2	D=0 -> Q=0	Update on this clock edge.

TRANSFER IT TO AN EXAM ANSWER

A D-type flip-flop is storage, not a gate that continuously copies D. A collection of flip-flops can form a register.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.4.3(e) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

A rising-edge D-flop starts Q=0. D=1 at edge 1, D=0 at edge 2, and D=1 at edge 3. State Q after each edge.

QUESTION 2 | 4 MARKS

Explain why a D-flop output need not change immediately when its data input changes.

1.4.3(e) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- After edge 1 Q=1; after edge 2 Q=0; after edge 3 Q=1.
- Each active edge captures D at that moment; between edges Q holds.

Q2 / 4 MARKS

- For an edge-triggered device the state updates only on the configured clock transition.
- Between active edges, stored Q stays constant even if D changes.

COMMON MISCONCEPTION

A D-type flip-flop is not a toggle flip-flop: it copies D on the active edge.

Green pen: Mark the rising clock edges before tracing Q.

1.5 / TEACHER-REVIEW WORKING CHAPTER

Understand it. Model it. Try it. Check it.

Standalone H446/01 revision upgrade for OCR 1.5. Existing live 264-page guide and all lesson and revision-quiz page links are unchanged.

TWO SPECIFICATION FAMILIES

1.5.1 COMPUTING-RELATED LEGISLATION
Data protection, computer misuse, copyright and investigatory powers

1.5.2 MORAL AND ETHICAL ISSUES
Nine specified contexts: workforce, AI, environment, expression, culture and privacy

FOUR PASSES FOR EVERY TOPIC

01 UNDERSTAND
Learn precise concepts and stakeholder context.

02 MODEL
Follow a decision trace or scenario.

03 TRY
Answer original exam-style prompts.

04 CHECK
Self-mark, correct and explain.

CRITICAL LEGAL UPDATE

The OCR specification names the Data Protection Act 1998 and RIPA 2000. Those are historical specification references. Current UK law includes UK GDPR / DPA 2018 as amended in 2025, and later investigatory-powers legislation. Always distinguish an exam listing from present law.

58 pages: 14 learning cycles x 4 pages, plus this cover and linked contents. These are replacement/enhancement drafts, NOT 58 extra pages automatically added to the published guide.

1.5 / SPECIFICATION CHECKLIST

Every named context is covered

Use a linked topic row to jump directly to its Understand page. Then follow Model, Try and Check.

SPEC	TOPIC	PAGES
1.5.1(a)	Data protection: old and current law	3-6
1.5.1(a)	Data principles, rights and controls	7-10
1.5.1(b)	Computer Misuse Act 1990	11-14
1.5.1(c)	Copyright, designs and patents	15-18
1.5.1(d)	Surveillance and investigatory powers	19-22
1.5.2(a)	Computers in the workforce	23-26
1.5.2(a)	Automated decision making	27-30
1.5.2(a)	Artificial intelligence	31-34
1.5.2(a)	Environmental effects	35-38
1.5.2(a)	Censorship and the Internet	39-42
1.5.2(a)	Monitoring behaviour	43-46
1.5.2(a)	Analysing personal information	47-50
1.5.2(a)	Piracy and offensive communications	51-54
1.5.2(a)	Layout, colour and character sets	55-58

KEY EXAM DISCIPLINE

A-level extended responses should identify stakeholders, apply the scenario, weigh plausible benefits and harms, and reach a justified judgement. Avoid a generic “pros and cons” list.

Sources checked (30 Oct 2026): OCR H446 v2.7; ICO on UK GDPR, DPA 2018 and DUAA 2025; CPS/NCA on CMA; GOV.UK copyright; legislation.gov.uk on investigatory powers.

Source links: OCR specification | ICO data law updates | CPS computer misuse | GOV.UK copyright

1.5.1(a) / BUILD THE MENTAL MODEL

Know the named OCR Act and the modern legal framework

OCR H446 v2.7 still lists the Data Protection Act 1998. Do not describe the 1998 Act as the current UK data-protection framework: today the UK GDPR and Data Protection Act 2018 apply, as amended by the Data (Use and Access) Act 2025.

THREE PIECES OF THE IDEA

OCR LISTING

DPA 1998 is explicitly named in specification 1.5.1(a); it is historical legislation.

CURRENT FRAMEWORK

The UK GDPR and DPA 2018 govern personal-data processing in the UK.

2025 UPDATE

The Data (Use and Access) Act 2025 amended, but did not replace, that framework.

VISUAL EXPLANATION / FOLLOW THE RELATIONSHIP

1998

OCR names

Historic Act

2018

Core law

DPA + UK GDPR

2025

Updates

DUAA amendments

The 2025 changes AMEND; they do not replace the UK GDPR and DPA 2018.

EXAM-READY STATEMENT

For a real organisation in 2026, refer to the UK GDPR and DPA 2018 as amended, while recognising the DPA 1998 in the OCR specification.

WORDS THAT EARN MARKS

- **Personal data** — Identifiable person
- **Processing** — Use of personal data
- **Controller** — Decides purposes
- **Amendment** — Changes existing law

1.5.1(a) / STEP-BY-STEP SCENARIO

A revision answer about a new school attendance app

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Identify	Attendance records + names	Those records can identify individual pupils.
2. Specification	DPA 1998 named in H446	Recognise the historic Act without treating it as current.
3. Current law	UK GDPR + DPA 2018	Modern processing must have a lawful basis and meet principles.
4. Updates	DUAA 2025 amendments	Changes some rules and processes; does not repeal the framework.
5. Apply	Minimise, secure, retain appropriately	Only collect required data and restrict access.

EXAM TRANSFER

For a real organisation in 2026, refer to the UK GDPR and DPA 2018 as amended, while recognising the DPA 1998 in the OCR specification.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.1(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

The H446 specification names the Data Protection Act 1998. Explain why a new school app should not rely on that Act alone as its description of current UK data-protection law.

QUESTION 2 | 4 MARKS

Identify two safeguards a school should use when processing pupil attendance data and explain why each is appropriate.

1.5.1(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- The DPA 1998 is named historically in the OCR content.
- The current core framework is UK GDPR and DPA 2018, amended by DUAA 2025.
- A current school app must comply with the applicable present-day obligations.

Q2 / 4 MARKS

- Limit collection to necessary attendance fields; reduces exposure.
- Apply appropriate access controls; restricts unauthorised viewing.
- Set justified retention periods; prevents keeping data longer than needed.

COMMON MISCONCEPTION

The 2025 Act amended existing rules; it did not replace UK GDPR or the DPA 2018.

Green pen: Keep one phrase for the exam listing and another for modern law; do not merge the dates.

1.5.1(a) / BUILD THE MENTAL MODEL

Make a data flow lawful, fair and secure

Personal data creates responsibilities through its collection, use, disclosure and deletion. A system may be technically secure but still breach data-protection principles if its collection or use is unjustified.

THREE PIECES OF THE IDEA

PRINCIPLES

Lawful, fair, transparent, limited in purpose and quantity, accurate, secure and not kept unnecessarily.

RIGHTS

People may have rights to information, access, rectification, erasure and other protections; rights have conditions or exceptions.

ACCOUNTABILITY

A controller should demonstrate compliance through governance, assessment and appropriate safeguards.

VISUAL MODEL / TRACK THE DECISION

Collect necessary data → **Explain purpose and lawful basis** → **Limit access and retention** → **Support rights and accountability**

Minimisation is not the same as encryption: collecting fewer fields and securing stored fields solve different risks.

EXAM-READY STATEMENT

Identify the exact data, the purpose, a lawful basis and a proportional control. Do not assume that collecting data is always unlawful or that consent is always required.

WORDS THAT EARN MARKS

- **Lawful basis** — Legal ground to process
- **Minimisation** — Only required data
- **Retention** — Justified storage period
- **Transparency** — Explain the processing

1.5.1(a) / STEP-BY-STEP SCENARIO

Redesign a sports-club membership form

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Audit	Name, contact, shoe size, health details	Some fields may not be necessary; health data needs additional safeguards.
2. Purpose	Contact for fixtures	Explain why contact details are needed.
3. Reduce	Do not collect shoe size by default	Data minimisation limits unnecessary personal data.
4. Protect	Role-based permissions + encryption	Reduce exposure and unauthorised access.
5. Review	Retention, accuracy and requests	Delete when no longer needed; provide appropriate rights processes.

EXAM TRANSFER

Identify the exact data, the purpose, a lawful basis and a proportional control. Do not assume that collecting data is always unlawful or that consent is always required.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.1(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

A club collects names, dietary details and home addresses for an online newsletter. Evaluate two privacy issues and propose specific changes.

QUESTION 2 | 4 MARKS

Explain the difference between data minimisation and data security using one practical example of each.

1.5.1(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Question whether each field is necessary for the newsletter purpose.
- Dietary details may be particularly sensitive and unjustified for this purpose.
- Collect only needed contact details, explain the use and set retention.
- Use limited access or encryption to protect retained records.

Q2 / 4 MARKS

- Minimisation reduces how much data is collected, e.g. no date of birth if unnecessary.
- Security protects retained data, e.g. access permissions and encryption.

COMMON MISCONCEPTION

Consent is one possible lawful basis, not a universal requirement for every use of personal data.

Green pen: Connect every safeguard to an identified risk instead of giving a generic security list.

1.5.1(b) / BUILD THE MENTAL MODEL

Authorisation matters more than technical skill

The Computer Misuse Act 1990 targets unauthorised access and other unauthorised acts affecting computer systems. Relevant offences differ according to access, intention and resulting or risked harm.

THREE PIECES OF THE IDEA

SECTION 1

Knowingly attempting to secure unauthorised access to computer material.

SECTION 2

Unauthorised access with intent to commit or facilitate a further serious offence.

SECTION 3

Unauthorised acts intending, or reckless as to, impairment of a system, access or data.

VISUAL EXPLANATION / FOLLOW THE RELATIONSHIP

S1

Access

Knowingly unauthorised

S2

Purpose

Further offence intended

S3

Impact

Intended/reckless impairment

EXAM-READY STATEMENT

Describe the unauthorised action, the relevant intention and its effect. There are additional modern provisions beyond the three core introductory sections.

WORDS THAT EARN MARKS

Unauthorised Without permission

Section 1 Unauthorised access

Section 2 Further offence intent

Section 3 Impairment / recklessness

1.5.1(b) / STEP-BY-STEP SCENARIO

Classify three cybersecurity scenarios

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Student	Uses guessed credentials on a classmate account	Likely unauthorised access under s1.
2. Criminal	Accesses account intending to steal funds	May involve s2 because of further-offence intent.
3. Disruption	Launches traffic to take a site offline	May involve s3 because the act impairs availability.
4. Permission	Pen-tests own authorised lab	Authorisation and agreed scope change the legal analysis.
5. Extension	Severe damage to critical systems	Other modern CMA provisions such as s3ZA may be relevant.

EXAM TRANSFER

Describe the unauthorised action, the relevant intention and its effect. There are additional modern provisions beyond the three core introductory sections.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.1(b) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

A student guesses a password, reads an account and later launches a denial-of-service attack against a school service. Explain which CMA concerns arise.

QUESTION 2 | 4 MARKS

Explain why a security researcher needs explicit scope and permission even when they intend to improve security.

1.5.1(b) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Guessing/logging in without permission can constitute unauthorised access (s1).
- The traffic attack may impair service availability and can engage s3.
- The circumstances, knowledge and intention determine offences.
- Explain real impact, e.g. legitimate users unable to access service.

Q2 / 4 MARKS

- Without permission, accessing systems can be unauthorised.
- Authorisation defines which systems and tests are permitted.
- A helpful motive alone is not equivalent to permission.

COMMON MISCONCEPTION

Hacking is not one single offence: the precise conduct and required mental element matter.

Green pen: In a response, label the section and prove it with a scenario-specific detail.

1.5.1(c) / BUILD THE MENTAL MODEL

Code, images and music can have separate permissions

The Copyright, Designs and Patents Act 1988 protects many creative works including computer programs. Copyright generally arises automatically; permitted use depends on ownership, licences and applicable exceptions.

THREE PIECES OF THE IDEA

PROTECTED WORK

Source code, artwork, recorded audio and written content can be protected independently.

LICENCE

Permission sets conditions of use, modification or redistribution; read the actual terms.

EXCEPTIONS

Some uses are permitted by law, but educational use does not mean unlimited copying or posting online.

VISUAL MODEL / TRACK THE DECISION

Identify original asset → **Find owner and licence** → **Check proposed reuse** → **Credit and comply / seek permission**

A working program on a public website is not automatically public-domain software; source availability alone is not a licence.

EXAM-READY STATEMENT

Distinguish legal permission from ethical credit. Crediting a creator alone does not automatically grant rights to copy their work.

WORDS THAT EARN MARKS

- **Copyright** Automatic protection
- **Licence** Usage permission
- **Attribution** Credit creator
- **Infringement** Use beyond rights

1.5.1(c) / STEP-BY-STEP SCENARIO

Publish a student game with four external assets

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Source	Student-written game code	Identify ownership and any collaborators.
2. Image	Screenshot from a commercial game	Check permission or an applicable exception; do not assume free reuse.
3. Music	Track found on a streaming service	Streaming access does not grant redistribution rights.
4. Library	Open-source graphics library	Read and comply with the relevant software licence.
5. Publish	Credits + licences + replacement assets	Use permitted material and keep attribution where required.

EXAM TRANSFER

Distinguish legal permission from ethical credit. Crediting a creator alone does not automatically grant rights to copy their work.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.1(c) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

A programmer uses public GitHub code and a commercial music track in a published app. Discuss legal and ethical responsibilities.

QUESTION 2 | 4 MARKS

Explain why an open-source software licence is not the same thing as work being free of copyright.

1.5.1(c) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Publicly visible code may still have copyright; check its licence and comply with terms.
- Commercial music requires permission or another legal basis for use.
- Licences may impose attribution or redistribution conditions.
- The developer should document permitted sources or replace assets.

Q2 / 4 MARKS

- Copyright protection still applies to open-source work.
- Its licence grants specific rights under stated conditions.
- A user must comply with those conditions when copying/modifying/distributing.

COMMON MISCONCEPTION

Attribution is important but does not alone authorise reproduction or distribution.

Green pen: Name the asset and the permission needed rather than writing only “copyright applies”.

1.5.1(d) / BUILD THE MENTAL MODEL

Separate public safety powers from privacy protections

OCR explicitly names the Regulation of Investigatory Powers Act 2000 (RIPA). Investigatory powers have since evolved, including under the Investigatory Powers Act 2016. These laws address particular investigative activities and provide restrictions and safeguards.

THREE PIECES OF THE IDEA

RIPA 2000

OCR-named legislation associated with specified interception and surveillance powers and regulatory controls.

IPA 2016

Modern framework for certain interception, communications-data and equipment-interference powers.

PROPORTIONALITY

Consider legitimate aim, necessity, intrusiveness and applicable approval/oversight safeguards.

VISUAL MODEL / TRACK THE DECISION

Identify legitimate aim → **Specify what data is sought** → **Check lawful authority and safeguards** → **Balance against privacy impact**

Do not claim an organisation has a blanket right to monitor all communications or that a generic warrant rule covers every investigatory power.

EXAM-READY STATEMENT

Explain both how lawful investigatory powers may support public safety and how safeguards protect privacy and civil liberties.

WORDS THAT EARN MARKS

- **Interception** Access during transit
- **Communications data** Metadata about contact
- **Oversight** Independent review
- **Proportionate** No more than justified

1.5.1(d) / STEP-BY-STEP SCENARIO

A police investigation requests communications information

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Need	Find connections during a serious investigation	A defined purpose matters.
2. Scope	Numbers/timestamps vs message content	Different types of information raise different intrusions.
3. Authority	Apply the relevant statutory process	Permissions depend on the precise power and circumstances.
4. Safeguards	Authorisation, necessity and oversight	Legislation limits how powers may be exercised.
5. Balance	Privacy of uninvolved people	Use a proportionate, targeted approach where required.

EXAM TRANSFER

Explain both how lawful investigatory powers may support public safety and how safeguards protect privacy and civil liberties.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.1(d) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Evaluate a proposal to collect communications records while investigating serious crime. Discuss a benefit and two risks or safeguards.

QUESTION 2 | 4 MARKS

Explain why RIPA 2000 should be distinguished from later UK investigatory-powers law in a present-day answer.

1.5.1(d) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Records may help establish networks or timing relevant to an investigation.
- Broad collection risks intruding on uninvolved people.
- Applicable authorisation, necessity, proportionality and oversight protect rights.
- Different kinds of communications information can be governed differently.

Q2 / 4 MARKS

- RIPA 2000 is named in the OCR specification.
- Later legislation, including the Investigatory Powers Act 2016, governs or amended parts of the modern framework.
- Specific powers require checking the current applicable law.

COMMON MISCONCEPTION

"The government can read everything" and "every request needs the same warrant" are both unjustified blanket statements.

Green pen: Qualify claims about lawful authority: it depends on the power, purpose and process.

1.5.2(a) / BUILD THE MENTAL MODEL

Automation changes tasks, not only job totals

Computing can raise productivity, create roles, and remove or transform tasks. Ethical evaluation must recognise uneven effects on workers, businesses and service users.

THREE PIECES OF THE IDEA

OPPORTUNITY

Automate repetitive work, improve accuracy and free time for complex or human-centred tasks.

RISK

Displacement, deskilling, surveillance and unequal access to training can harm groups.

RESPONSE

Reskilling, human oversight, consultation and redesigning roles can share benefits more fairly.

VISUAL MODEL / TRACK THE DECISION

Automate routine task → **Change job requirements** → **Assess affected groups** → **Support fair transition**

A warehouse robot may reduce repetitive lifting while changing staffing, training needs and worker safety responsibilities.

EXAM-READY STATEMENT

A strong discussion weighs an outcome for named stakeholders and proposes realistic mitigation, rather than asserting all automation is good or bad.

WORDS THAT EARN MARKS

Automation	Machine-performed tasks	**Deskilling**	Reduced skill use
Reskilling	Learning new tasks	**Stakeholder**	Affected group

1.5.2(a) / STEP-BY-STEP SCENARIO

A supermarket introduces automated checkout

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Customer	Potentially shorter queue	Depends on usability, reliability and staffing.
2. Business	Routine transactions need fewer staff minutes	Savings may be reinvested or reduce employee hours.
3. Staff	More exception-handling and customer support	Roles can change, not simply disappear.
4. Equity	Some customers need assistance	Accessibility and human support are important.
5. Response	Training + assisted checkout	Balances efficiency and inclusion.

EXAM TRANSFER

A strong discussion weighs an outcome for named stakeholders and proposes realistic mitigation, rather than asserting all automation is good or bad.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.2(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Evaluate the impact of introducing automated checkouts on employees, customers and the retailer.

QUESTION 2 | 4 MARKS

Explain how reskilling can change the ethical impact of workplace automation.

1.5.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Retailer may lower routine staffing cost and gain speed.
- Employees may face role change or reduced hours, creating unequal impact.
- Customers may gain convenience but some need accessible human assistance.
- Training and staffed alternatives mitigate disadvantages.

Q2 / 4 MARKS

- Reskilling prepares workers for new technical or support tasks.
- It reduces barriers to taking redesigned roles.
- It does not guarantee every displaced worker can remain employed.

COMMON MISCONCEPTION

Automation does not necessarily mean zero jobs; the nature and distribution of work matter.

Green pen: Use at least two stakeholders and link benefits/costs to each.

1.5.2(a) / BUILD THE MENTAL MODEL

A prediction is not automatically a fair decision

Algorithms can assess applications at scale but encode assumptions about data, thresholds and acceptable errors. People affected by significant decisions may need safeguards and opportunities to challenge outcomes.

THREE PIECES OF THE IDEA

ADVANTAGE

Fast and consistent rule application; transparent rules can reduce arbitrary differences.

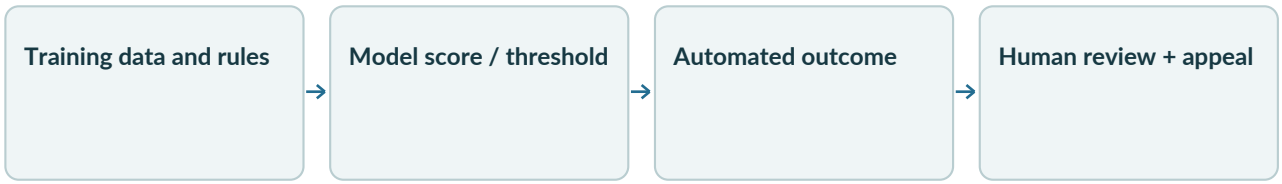
RISK

Proxy variables, incomplete records and rigid cut-offs can produce unfair exclusions.

SAFEGUARD

Audit outcomes, explain factors where appropriate, support correction and human challenge.

VISUAL MODEL / TRACK THE DECISION



A student with a score just below a threshold may have very different circumstances from another with a slightly higher score.

EXAM-READY STATEMENT

Evaluate input quality, decision logic, impact and safeguards. Current UK rules for significant automated decisions are nuanced and were amended by the 2025 Act.

WORDS THAT EARN MARKS

Threshold Decision boundary

Bias Systematic unfairness

Explainability Can reasons be given?

Appeal Challenge an outcome

1.5.2(a) / STEP-BY-STEP SCENARIO

Review a bursary eligibility algorithm

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Inputs	Attendance and family income	Check whether fields are accurate and appropriate.
2. Rule	Reject applications with attendance < 90%	A rigid boundary may misrepresent illness-related absence.
3. Output	Automatic refusal	High-stakes decision affects a named student.
4. Check	Compare outcomes across groups	Look for disparate or unfair impact.
5. Remedy	Correction + human review	Enable contextual evidence and a challenge route.

EXAM TRANSFER

Evaluate input quality, decision logic, impact and safeguards. Current UK rules for significant automated decisions are nuanced and were amended by the 2025 Act.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.2(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

A college uses a scoring algorithm to award financial support. Evaluate its risks and possible safeguards.

QUESTION 2 | 4 MARKS

Explain why a uniform numeric threshold might still produce an unfair outcome.

1.5.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Fast and consistent processing can help large application volumes.
- Biased/incomplete data may disadvantage some students.
- A threshold can ignore context such as illness or caring responsibilities.
- Audits and a meaningful review/challenge route can mitigate risks.

Q2 / 4 MARKS

- Equal treatment by a fixed rule is not the same as fair outcomes.
- Relevant contextual differences can be missing from the input data.
- A hard threshold can change the result for very similar cases.

COMMON MISCONCEPTION

“The computer is objective” is not a justification; designers chose the data, labels, rules and thresholds.

Green pen: Name one test you would run to discover an unfair outcome.

1.5.2(a) / BUILD THE MENTAL MODEL

Assess accuracy, bias, accountability and trust

AI systems can recognise patterns and generate content, but outputs may be incorrect, unrepresentative or difficult to explain. Their deployment creates social and ethical questions beyond raw model accuracy.

THREE PIECES OF THE IDEA

BENEFIT

Assist diagnosis, translation, accessibility, forecasting or routine document work.

RISK

Bias, hallucination, privacy leakage, manipulation and overreliance.

GOVERNANCE

Use representative evaluation, human oversight, disclosure and restrictions appropriate to stakes.

VISUAL EXPLANATION / FOLLOW THE RELATIONSHIP



Test error rates by group and severity. Recheck after deployment; performance can drift.

EXAM-READY STATEMENT

Match the strength of oversight to the consequences of error; do not say all AI applications have equal risk.

WORDS THAT EARN MARKS

Training data	Learning examples	**Hallucination**	Unsupported output
False negative	Missed real case	**Human oversight**	Person can intervene

1.5.2(a) / STEP-BY-STEP SCENARIO

Evaluate AI-assisted health triage

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Benefit	Prioritise urgent appointments	May reduce waiting time for some users.
2. Error	Urgent case incorrectly labelled low risk	False negative creates potential harm.
3. Bias	Group underrepresented in training	Model performance may vary across groups.
4. Control	Clinician review of high-impact outputs	A human can challenge or reverse an outcome.
5. Monitor	Track errors and complaints over time	Deployment conditions and data may change.

EXAM TRANSFER

Match the strength of oversight to the consequences of error; do not say all AI applications have equal risk.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.2(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Discuss the opportunities and risks of AI triage in a hospital. Recommend safeguards.

QUESTION 2 | 4 MARKS

Explain why 95% overall model accuracy does not establish that all groups are treated fairly.

1.5.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Faster triage may help allocate staff and reduce delay.
- Incorrect urgent-case classification can harm a patient.
- Training-data imbalance can cause unequal error rates.
- Review, subgroup testing and continuous monitoring mitigate harm.

Q2 / 4 MARKS

- Overall accuracy may conceal very different subgroup error rates.
- Compare false-positive and false-negative rates across relevant groups.
- Representative evaluation is needed to test fairness.

COMMON MISCONCEPTION

A confident AI-generated explanation may still be factually wrong or ungrounded.

Green pen: Use a concrete failure mode and name the person accountable for correcting it.

1.5.2(a) / BUILD THE MENTAL MODEL

Assess the whole digital life cycle

Computing can reduce some travel and resource use while consuming energy, water and materials. Life-cycle assessment considers extraction, manufacturing, operation, repair and disposal.

THREE PIECES OF THE IDEA

PRODUCTION

Mining, component manufacture and shipping use energy and raw materials.

OPERATION

Data centres and devices consume electricity; cooling may use water depending on design.

END OF LIFE

Reuse, repair and appropriate recycling can reduce demand for virgin materials and e-waste.

VISUAL EXPLANATION / FOLLOW THE RELATIONSHIP

MAKE — RUN — REPAIR — REUSE

Compare embodied impacts + operational impacts + end-of-life effects.

EXAM-READY STATEMENT

A balanced answer states the environmental mechanism, who benefits, what is displaced and when a claimed saving might not occur.

WORDS THAT EARN MARKS

- **E-waste** Discarded electronics
- **Embodied carbon** Production emissions
- **Life cycle** Whole-product impacts
- **Rebound effect** Extra use offsets savings

1.5.2(a) / STEP-BY-STEP SCENARIO

Compare replacing devices every year with every four years

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Manufacture	Each new device needs material + energy	Frequent replacement increases production impacts.
2. In use	A newer device may be more efficient	Operational benefits depend on electricity consumption and use.
3. Rebound	Faster device encourages more activity	Efficiency gains can be partly offset.
4. Repair	Battery and screen replacement	May extend life without replacing whole device.
5. Decision	Use life-cycle evidence	Compare total impacts for the actual scenario.

EXAM TRANSFER

A balanced answer states the environmental mechanism, who benefits, what is displaced and when a claimed saving might not occur.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.2(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Evaluate whether replacing every school laptop with a more efficient model is environmentally beneficial.

QUESTION 2 | 4 MARKS

Explain how remote meetings can reduce emissions yet still have environmental costs.

1.5.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Newer devices may consume less electricity in use.
- Manufacture, transport and disposal impose environmental costs.
- Repair or longer life may reduce material demand.
- Compare complete life-cycle impacts before recommending replacement.

Q2 / 4 MARKS

- Fewer journeys can reduce travel emissions.
- Devices, networks and data centres still consume energy.
- Net impact depends on what travel is replaced and how technology is used.

COMMON MISCONCEPTION

The phrase “digital is paperless” does not mean an activity has zero environmental footprint.

Green pen: Use production, use and disposal in every high-mark environmental evaluation.

1.5.2(a) / BUILD THE MENTAL MODEL

Freedom of expression and harm prevention can conflict

Online moderation can reduce illegal or harmful material but may remove legitimate discussion, restrict minority voices or create inconsistent enforcement. Judgements depend on context and rule design.

THREE PIECES OF THE IDEA

SAFETY

Moderation can limit harassment, threats and illegal material.

FREEDOM

Overbroad filters risk suppressing lawful discussion, satire or public-interest reporting.

PROCESS

Clear rules, proportionate action, transparency and appeals can improve accountability.

VISUAL MODEL / TRACK THE DECISION

Set rules and reasons → **Flag content** → **Human / contextual review** → **Appeal and audit**

An automated block on a keyword cannot reliably understand every quotation, dialect, reclaimed term or satire.

EXAM-READY STATEMENT

Evaluate both the harms of leaving content visible and the harms of removing lawful speech; avoid an absolute stance without evidence.

WORDS THAT EARN MARKS

- **Censorship** Suppression of expression
- **Moderation** Content rule enforcement
- **False positive** Legitimate item blocked
- **Appeal** Challenge removal

1.5.2(a) / STEP-BY-STEP SCENARIO

A platform automatically removes posts containing a term

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Aim	Protect users from abuse	A legitimate safety interest.
2. Filter	Match a prohibited word	Cheap and fast, but lacks context.
3. Error	Historical quotation incorrectly removed	False positive affects legitimate speech.
4. Review	Appeal to contextual human review	Can correct mistakes but costs time.
5. Publish	Explain rules and outcomes	Lets people judge policy consistency.

EXAM TRANSFER

Evaluate both the harms of leaving content visible and the harms of removing lawful speech; avoid an absolute stance without evidence.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.2(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Evaluate an online platform using automated keyword filters to moderate conversations between students.

QUESTION 2 | 4 MARKS

Explain why a transparent appeals process is important when content is removed automatically.

1.5.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Filters can act quickly on common harmful patterns.
- May wrongly block legitimate context such as quotations or satire.
- Human review and clear rules help reduce false positives.
- Provide appeal process and proportionate responses.

Q2 / 4 MARKS

- Users can challenge mistakes and supply context.
- Corrections reduce wrongful exclusion and promote accountability.
- An appeal process does not guarantee a correct first decision.

COMMON MISCONCEPTION

A rule that targets harmful abuse can still be implemented in an unfair or overbroad way.

Green pen: Identify the error type and the stakeholder before proposing a moderation policy.

1.5.2(a) / BUILD THE MENTAL MODEL

Being able to monitor does not justify every use

Workplaces, schools and services may log activity for security, performance or welfare. Pervasive monitoring can harm privacy, trust, autonomy and relationships between people.

THREE PIECES OF THE IDEA

LEGITIMATE USE

Security alerts, fraud detection, safeguarding and troubleshooting may need limited monitoring.

POTENTIAL HARM

Continuous surveillance can chill behaviour, misclassify harmless activity and reduce trust.

GOOD DESIGN

Define purpose, limit data and retention, inform users, restrict access and review proportionality.

VISUAL MODEL / TRACK THE DECISION

Purpose → **Collect only needed events** → **Secure / limit access** → **Review necessity**

A system logging failed logins is different from recording every private message and keystroke.

EXAM-READY STATEMENT

The best answer compares a less intrusive method with a more intrusive alternative and judges whether the extra data is necessary.

WORDS THAT EARN MARKS

Telemetry Usage measurements

Chilling effect Behaviour changes

Proportionality Appropriate to aim

Retention Storage duration

1.5.2(a) / STEP-BY-STEP SCENARIO

Consider staff-laptop monitoring

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Aim	Investigate repeated account lockouts	Security-focused purpose.
2. Option A	Log failed sign-ins	Targeted low-volume data may suffice.
3. Option B	Capture all typed messages	Much more intrusive; likely excessive for the aim.
4. Controls	Access limits and retention policy	Lower privacy and misuse risks.
5. Evaluate	Ask staff and review outcomes	Trust and accuracy matter alongside security.

EXAM TRANSFER

The best answer compares a less intrusive method with a more intrusive alternative and judges whether the extra data is necessary.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.2(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Evaluate an employer recording employees' screen activity throughout the day.

QUESTION 2 | 4 MARKS

Distinguish targeted security logs from continuous surveillance, with one ethical issue for each.

1.5.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Monitoring may identify misuse or assist support teams.
- Continuous recording may invade privacy and erode trust.
- Activity metrics can misrepresent actual work quality.
- Targeted logs, transparency and retention limits reduce harm.

Q2 / 4 MARKS

- Targeted logs capture events relevant to a stated security issue.
- Continuous surveillance captures much more personal behaviour.
- Even necessary logs need controls; continuous capture demands stronger justification.

COMMON MISCONCEPTION

A manager owning the devices does not automatically make every kind of monitoring ethically or legally justified.

Green pen: Use the purpose and data collected to judge proportionality.

1.5.2(a) / BUILD THE MENTAL MODEL

Useful insight can become unfair profiling

Personal-data analysis can improve services or detect problems, but combining datasets can reveal sensitive patterns and affect people who never expected the new use.

THREE PIECES OF THE IDEA

BENEFIT

Find service gaps, detect fraud or identify individuals needing support.

RISK

Re-identification, secondary uses, bias and opaque profiling.

CONTROL

Purpose limitation, minimisation, fairness checks and appropriate aggregation or anonymisation.

VISUAL MODEL / TRACK THE DECISION

Collect records → **Join datasets** → **Infer / predict pattern** → **Assess impact and safeguards**

Removing names is not automatically anonymisation if other attributes can identify a person when combined.

EXAM-READY STATEMENT

Explain how a new inference is generated, why it may harm a particular group and which specific technical or governance control can reduce harm.

WORDS THAT EARN MARKS

- **Profiling** Inferring characteristics
- **Re-identification** Identity recovered
- **Aggregation** Group-level summary
- **Secondary use** New purpose for data

1.5.2(a) / STEP-BY-STEP SCENARIO

Analyse travel-card records and shopping data

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Original	Journey data for fare payments	People expect the data to support travel.
2. Join	Link location + purchases	New combined data reveals richer personal patterns.
3. Risk	Unique routes reveal a person	An “anonymous” record may be re-identifiable.
4. Use	Plan services from grouped totals	Aggregated trends may achieve the purpose with less intrusion.
5. Review	Assess purpose and appropriate lawful basis	Extra use needs its own justification and safeguards.

EXAM TRANSFER

Explain how a new inference is generated, why it may harm a particular group and which specific technical or governance control can reduce harm.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.2(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Evaluate a council combining library, transport and benefits data to target services.

QUESTION 2 | 4 MARKS

Explain how removing names from a dataset may still leave a privacy risk.

1.5.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Joined data may identify underserved groups and improve resource allocation.
- Linking detailed records can expose sensitive or unexpected patterns.
- Risk of wrong inference or disproportionate impact on groups.
- Use justified purpose, access restriction, aggregation and bias testing.

Q2 / 4 MARKS

- Combinations such as unique location/time patterns may still identify someone.
- External datasets can enable re-identification.
- Aggregating or appropriately anonymising data may reduce this risk.

COMMON MISCONCEPTION

Pseudonymised data can still be personal data; it is not automatically fully anonymous.

Green pen: Name what can be inferred that could not easily be seen from one dataset alone.

1.5.2(a) / BUILD THE MENTAL MODEL

Separate legal restrictions, social harm and platform policy

Digital copying can be nearly costless, and online communication can spread quickly. Evaluate rights and harms separately: copyright infringement, abusive communications and content-policy breaches are not interchangeable.

THREE PIECES OF THE IDEA

PIRACY

Unauthorised copying/distribution can damage creators' revenue and may breach copyright.

OFFENSIVE MESSAGES

Targeted harassment or threatening communications can cause distress and exclusion.

BALANCE

Enforcement should distinguish unlawful conduct from unpopular but lawful expression.

VISUAL MODEL / TRACK THE DECISION

Identify the actual conduct → **Determine rights / harms** → **Apply proportionate response** → **Offer education / appeal**

A platform removing a post under its terms of service is not itself proof that the speaker committed a criminal offence.

EXAM-READY STATEMENT

In an extended answer identify the type of harm, responsible stakeholder and remedy; do not merge every objectionable online act into one law.

WORDS THAT EARN MARKS

- **Piracy** Unauthorised distribution
- **Harassment** Targeted abuse
- **Terms of service** Platform contract
- **Takedown** Removal request

1.5.2(a) / STEP-BY-STEP SCENARIO

Respond to two unrelated digital incidents

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Incident A	Student uploads a paid video course	Potential infringement of the owner's copyright.
2. Mitigation A	Licence or legitimate access	Creators can be paid and users have clear permissions.
3. Incident B	Repeated targeted threatening messages	Potential serious harm; relevant law depends on context.
4. Mitigation B	Report, preserve evidence, protect users	Provide a proportionate safety response.
5. Distinguish	Two distinct issues	Do not call all offensive speech “copyright piracy”.

EXAM TRANSFER

In an extended answer identify the type of harm, responsible stakeholder and remedy; do not merge every objectionable online act into one law.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.2(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Discuss the effects of illegal file sharing and repeated offensive messages on users, creators and platforms.

QUESTION 2 | 4 MARKS

Explain why a platform's content removal does not necessarily prove a criminal offence.

1.5.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Creators may lose income or control when work is distributed without permission.
- Users may face harmful, threatening or distressing communication.
- Platforms need fair reporting, review and enforcement processes.
- Education, licensing and proportionate moderation address different harms.

Q2 / 4 MARKS

- Platform rules can prohibit conduct that is still lawful.
- Criminal liability depends on the elements of the relevant offence and circumstances.
- A takedown or moderation decision is not a court determination.

COMMON MISCONCEPTION

“Offensive” is not by itself a precise universal legal category; specifics and context matter.

Green pen: Keep copyright consequences and communications harms in separate parts of the answer.

1.5.2(a) / BUILD THE MENTAL MODEL

Design for different languages, cultures and users

Interfaces are experienced through language, reading direction, symbols, layout and colour. A design that works for one audience may confuse or exclude another; localisation goes beyond translation.

THREE PIECES OF THE IDEA

LAYOUT

Right-to-left languages may need mirrored navigation and text alignment; layouts should be tested.

COLOUR

Colour meanings differ across cultures and cannot be the only signal for meaning or state.

CHARACTERS

Unicode-capable storage and fonts support many scripts; rendering and input methods also matter.

VISUAL EXPLANATION / FOLLOW THE RELATIONSHIP

LEFT-TO-RIGHT

Home > Results

One reading convention

RIGHT-TO-LEFT

< Results Home

Another reading convention

EXAM-READY STATEMENT

Distinguish cultural localisation (beliefs, conventions, reading patterns) from accessibility, while explaining how the two can overlap.

WORDS THAT EARN MARKS

Localisation Adapt to locale

Unicode Character code points

RTL Right-to-left layout

Accessibility Inclusive usability

1.5.2(a) / STEP-BY-STEP SCENARIO

Localise an international competition registration page

Track what happens and why. Apply each point to the precise situation rather than repeating a memorised list.

SCENARIO / DECISION TABLE

STEP	ACTION / EVIDENCE	REASON / INTERPRETATION
1. Language	Offer translated text	Reduce comprehension barriers.
2. Layout	Support RTL where appropriate	Navigation flow can change with writing direction.
3. Form	Accept Unicode names	Avoid rejecting valid non-ASCII characters.
4. State	Success icon + written confirmation	Colour alone is ambiguous or inaccessible.
5. Validate	Test with users from target contexts	Reveal assumptions and cultural mismatches.

EXAM TRANSFER

Distinguish cultural localisation (beliefs, conventions, reading patterns) from accessibility, while explaining how the two can overlap.

Exam habit: point + scenario-specific evidence + consequence + considered judgement. Check any claim of law is accurate for its time and context.

1.5.2(a) / INDEPENDENT EXAM PRACTICE

Apply the idea without the model

Original teacher-written exam practice (not OCR past-paper material). Work in BLACK pen first; check the next page.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Evaluate how an international sports-event website should handle layout, colour and character sets for a diverse audience.

QUESTION 2 | 4 MARKS

Explain why Unicode support alone does not guarantee a well-localised interface.

1.5.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Self-mark, correct and explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Some users read RTL and need suitable alignment/navigation conventions.
- Colour meaning varies; pair colour with text/icons.
- Unicode is needed for many names and scripts but fonts/input must also work.
- Test translated interfaces with target audiences.

Q2 / 4 MARKS

- Unicode represents character identities, not page layout, translation or local conventions.
- Fonts, directional rendering and input methods must work.
- Content and visual signals still need cultural evaluation.

COMMON MISCONCEPTION

“Use Unicode” is not a complete answer to cultural UI design; layout and colour are distinct concerns.

Green pen: Give one concrete language, colour or layout decision and link it to a user group.

Additional recall

Memory banks from the earlier guide

The following seven pages reproduce the recall material at the end of the archived 21 October 2026 H446 Paper 1 revision guide. They are retained for teaching value, not silently rewritten.

What students should do

1. Cover the scientific definition, recall the analogy, then explain the idea accurately.

2. Compare your explanation with the precise technical wording.

3. Use these pages as additional retrieval practice, not as a substitute for topic examples.

Editorial status: these are original pages from an earlier edition. A page-by-page comparison with the current published 264-page edition is still required before release.

H446 MEMORY BANK

Read the hook, then cover the precise wording and try to explain the concept without the analogy.

Processor / 1.1.1 / CPU as head chef

NOW BE PRECISE

PC = next instruction address; MAR = memory address; MDR = data/instruction; CIR = fetched instruction.

Careful: A chef cannot model processor pipelining or precise register transfers.

Binary / 1.4.1 / Two’s complement = invert + one

NOW BE PRECISE

For a fixed width: flip all bits, then add 1 to form a negative value.

Careful: Check representable range and overflow; this is not sign-magnitude.

Data structures / 1.4.2 / Plates & queues

NOW BE PRECISE

Stack = LIFO; queue = FIFO.

Careful: An abstract data type defines operations independent of implementation.

Boolean / 1.4.3 / K-map = joined-edge map

NOW BE PRECISE

Opposite edges are adjacent. Circle power-of-two groups of 1s.

Careful: Do not group diagonals; draw only one source for each logic input.

MEMORY BANK / continued

Operating systems / 1.2.1 / Airport controller

NOW BE PRECISE

Scheduler allocates CPU time; ready processes wait for service.

Careful: Scheduling is a defined algorithm, not intuition.

Applications / 1.2.2 / Translator + linker

NOW BE PRECISE

Lexing → parsing → code generation → linking/loading.

Careful: Implementations vary; compilation does not guarantee correctness.

Data exchange / 1.3.1 / ZIP, lock, fingerprint

NOW BE PRECISE

Compression reduces representation; encryption protects confidentiality; hashing produces a digest.

Careful: Hashes have collisions and are not decrypted.

Databases / 1.3.2 / Library card number

NOW BE PRECISE

Unique primary key identifies one row; foreign key links related rows.

Careful: Human names may repeat; FK values can repeat.

REVISION ROUTINE

1. Recall the hook without looking. 2. Explain in precise H446 language. 3. Trace or draw an example. 4. Attempt a fresh exam question.

MEMORY BANK

Cover the precise meaning and explain each memory hook aloud. The analogy helps you remember; the technical explanation earns the marks.

WATERFALL	Staircase
EXAM PRECISION	Distinct phases with defined outputs; late change can mean rework.
AGILE	Short laps
EXAM PRECISION	Iterative increments incorporate frequent stakeholder feedback.
XP	Engineering pit crew
EXAM PRECISION	An Agile approach emphasising testing, pair work and integration.
SPIRAL	Risk radar
EXAM PRECISION	Revisit risk identification and evaluation in each cycle.
RAD	Prototype workshop
EXAM PRECISION	Rapid prototypes help users refine unclear requirements.
LAW / ETHICS	Permitted / justified
EXAM PRECISION	Legislation establishes duties and rights; ethics weighs fairness and impact.
AI DECISIONS	Human check
EXAM PRECISION	Model outputs require reliable data, oversight and explainability.

Memory bank · additional recall

Cover the precise wording, recall the memory hook, then explain the computer-science meaning aloud.

Recipe cards	Procedural Procedures/functions organise operations; data is supplied as arguments or shared state.
Vending machine	OOP Objects combine attributes and methods. Encapsulation regulates state changes.
Numbered mailboxes	LMC ACC is working storage; STA/LDA transfer data to and from addressed mailboxes.
Follow the pointer	Indirect address First access an address held in memory, then use that address for the operand.
Offset on a map	Indexed address Calculate effective address from base plus index or displacement.

QUICK RECALL / PAPER 1

Memory bank — data and network models

C-S-M-E

Capture the input; **Select** the records; **Manage** the stored data; **Exchange** it with another system.

Keyword	Remember it precisely
Validation	Check an input against rules or constraints.
Verification	Check captured data against a source.
Foreign key	Field referencing a key in a related table.
CSV	Row-and-column exchange format.
JSON	Structured objects/arrays, including nested data.
Client-server	Managed services supplied to clients.
Peer-to-peer	Peers may be both resource clients and providers.
Topology	Physical/logical connection layout; not the service model.

TWO EXAM TRAPS

SQL SELECT does not change data. **Client-server** does not imply exactly one server, and **peer-to-peer** does not require a physical mesh.

Five one-minute retrieval prompts

- Give one appropriate data-capture method for a barcode-labelled library book.
- Explain validation and verification using the same captured identifier.
- Explain when JSON might be preferable to CSV.
- Give one benefit and one cost of central network management.
- Explain why a star network may support either client-server or peer-to-peer.

Quick recall / memory management and databases

Cover the precise explanation, recall the memory image, then explain it in Computer Science vocabulary.

Remember	H446 concept	Exam-accurate meaning
Hotel room key	Paging	Page table maps fixed-size virtual pages to RAM frames.
Moving boxes	Virtual memory	Absent pages can be retrieved from backing storage.
Fire alarm	Interrupt / ISR	Save context, service the event, restore and resume.
Library card	Referential integrity	Foreign keys reference existing parent keys.
One fact, one home	3NF	Remove partial and transitive dependencies.
Find the rows	SQL JOIN / WHERE	JOIN matches keys; WHERE filters.
All or nothing	Atomicity	Commit all dependent changes or roll back.
One ticket left	Isolation / locking	Control concurrent work to avoid conflicting updates.

A 30-second check

Can you explain page vs frame, interrupt vs ISR, primary vs foreign key, and DELETE vs DROP TABLE without looking at your notes?

Memory bank: systems and applications

Cover the precise definition. Recall the hook, then explain the real Computer Science.

Memory hook	Specification term	Use exam language
Airport control desk	OS purpose	Allocate CPU, memory, files, I/O and secure access.
Five questions	OS types	Machine / device / tasks / users / deadline.
Opening the building	BIOS / UEFI	Initialise platform and begin boot sequence.
Device translator	Driver	OS-specific request mapped to device operations.
Machine made of software	Virtual machine	Guest OS on virtual hardware OR bytecode runtime.
User’s job	Application	Choose functions that meet a concrete user need.
Backstage crew	Utility	Maintain, protect, compress or recover system data.
Join, then start	Linker / loader	Resolve modules, then prepare executable to run.

Five 30-second retrieval prompts

- Why is a real-time system not merely a fast system?
- What separate roles do UEFI firmware and the boot loader play?
- Why can a missing driver stop a connected printer working?
- How does a guest OS VM differ from a bytecode VM?
- What is the distinction between linking and loading?